Where Should Science Funding Go?

Estimating Marginal Returns to Public Funding Across Research Fields*

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Abstract

Science funding is shrinking in many countries, while commercial innovation increasingly relies on publicly funded science. I study the optimal allocation of science funding across fields. I estimate marginal returns to funding in 19 biomedical fields using a micro-founded structural model where scientists compete for grants. These scientists differ in productivity and shift their research focus towards better-funded fields. Once funded, they produce knowledge by combining grant funding with existing knowledge from their own and other fields, giving rise to knowledge spillovers. Marginal returns vary widely across fields: an additional $1 million yields 2.2 publications in Nursing Research and 14.8 in Human Genome. Knowledge spillovers account for most of this gap. Upstream fields, such as General Medicine and Human Genome, generate far larger spillovers than specialised downstream fields, such as Nursing Research or Deafness and Communication Disorders. Scientists do move towards better-funded fields, but their replacements are equally productive once funded, so this reallocation costs little in equilibrium. I construct three welfare-relevant measures of scientific output: academic impact, commercial potential and disease-burden reduction. Marginal-return rankings barely change across the three measures. I combine each measure with the estimated marginal returns from the model to compare the current allocation with two counterfactual allocations: allocating funding in proportion to disease burden, as the medical community often advocates, and in proportion to marginal returns, as the model suggests. Across the three measures, allocation based on the model estimates yields 39–54% more output than the current allocation and 15.1–26.4% more than disease-burden alignment.

Keywords: economics of science; public research; research policy; public research grants; sectoral planning

JEL codes: O30; O32; O38; O51; O25; H4; E13; L52

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1. Introduction

Science drives technological change and economic growth. Yet scientific progress appears to be slowing (Bloom et al., 2020; Holst et al., 2026; Jones, 2009; Jones, 2010; Park, Leahey and Funk, 2023). At the same time, firms have retreated from upstream basic science and now rely more on publicly funded science to support their innovations (Arora, Belenzon and Patacconi, 2018; Fleming et al., 2019; Poege et al., 2019). Bell Labs was progressively dismantled after the break-up of AT&T; DuPont closed its central research laboratory in 2016. Governments now support a larger share of the scientific base just as public science budgets tighten in many countries.[1] In such a context, it is crucial that governments allocate their science budgets efficiently.

Once policymakers have set the total science budget, they must divide it across competing fields. What share should go to cancer research? What about infectious disease? Should mental health be prioritised? This paper asks how science funding should be allocated across fields. I study this question in the United States, where Congress sets each year the budget of the National Institutes of Health (NIH) and its distribution across NIH institutes (ICs). Each institute funds research on a body system or a class of disease, such as the National Eye Institute or the National Cancer Institute. For the purpose of this paper, each NIH institute is a scientific field and each scientific field is an NIH institute. I use the terms IC, institute and field interchangeably.

The NIH budget stood at $48.5 billion in 2025 (more than half of all federal non-defence R&D spending and more than the total R&D spending of all but ten countries). Given its scale, reallocating the NIH budget could substantially accelerate progress in some fields while greatly slowing it in others. It is therefore unsurprising that the allocation of funding across institutes is fiercely debated and attracts intense lobbying in Congress (Hegde and Mowery, 2008; Hegde and Sampat, 2015; Sampat, 2012). Yet Congress makes this choice every year without estimates of how much science an additional dollar buys in each field.

The most prominent benchmark in this debate is disease burden. In 1997, Congress asked the Institute of Medicine to review how the NIH sets priorities, and the Institute recommended giving greater weight to disease burden (Gillum et al., 2011; Gross, Anderson and Powe, 1999). The medical and health-policy literature has since measured how far NIH funding departs from this benchmark. AIDS and diabetes appear overfunded relative to their burden, whereas chronic obstructive pulmonary disease and depression appear underfunded (Gillum et al., 2011; Gross, Anderson and Powe,

[1] In the United States, the proposed 2026 budget included a 40% cut to the National Institutes of Health (NIH). In Europe, changes to European Research Council grant allocation rules in 2014 reduced biomedical research funding by 20% (Adda and Ottaviani, 2024), while in the UK, the 2025–2026 UK Research and Innovation budget fell by roughly £300 million.

1999). Disease burden explains about a third of the variation in disease-level funding, and funding tracks its own level a decade earlier more closely than changes in burden (Ballreich et al., 2021; Gillum et al., 2011). These studies treat disease burden as the benchmark for appropriate funding. Yet burden alignment ignores how much science a dollar buys in each field.

Economic theory offers a different rule: invest the marginal dollar where its marginal return is highest. Despite the stakes, economists have paid little attention to this allocation problem, perhaps because of its apparent simplicity. Applying this principle in practice is much more difficult. At least two key challenges arise. First, the marginal return to funding is unobservable. Output per dollar is a natural proxy, but it ignores two cross-field externalities. Scientific fields form a network of knowledge exchange: research produced in one field becomes an input into research in another. Scientists also move towards fields where competition for funding is weaker, so a budget increase in one field can draw talent away from others and lower their output. A field-by-field accounting misses both externalities. Second, the social value of scientific output is difficult to define. Any mapping from output into welfare is imperfect: what welfare change should be attributed to Alexander Fleming's penicillin paper?

Motivated by these challenges, I develop and estimate a tractable micro-founded model of scientists' field and production choices that captures both externalities. The model specifies (i) how additional funds raise output in each NIH institute, (ii) how scientists redirect their grant applications when the funding allocation across fields changes, and (iii) how research produced in one field creates knowledge spillovers for itself and for other fields. The resulting marginal-return estimates are anchored in scientists' revealed preferences and can therefore support policy counterfactuals.

The model works as follows. Each year scientists draw field-specific research ideas and choose whether to apply for funding, and at which institute, given expected success rates. Institutes evaluate the applications and fund the best ones down to a threshold that clears their budget. Funded scientists combine grant money with existing knowledge to produce new knowledge. An institute that receives more funding becomes more attractive and draws additional scientists. The resulting output gain depends on the productivity of the incoming researchers and on the research they would otherwise have produced elsewhere. These funded scientists produce knowledge that may, in turn, be used as an input into research in other fields, generating cross-field knowledge spillovers.

The model is agnostic about how scientific output is measured: it estimates the marginal return to funding without assigning a social value to that output. To make counterfactual policy evaluation welfare-relevant, I construct three measures that aim to capture some of this social value. I measure the social value of scientific output in terms of (i) its academic impact, (ii) its commercial potential, and (iii) the burden of the

disease it addresses. Combining each measure with the model's marginal returns, I compare the 2018 NIH funding allocation with counterfactual allocations, including one that aligns funding with each field's disease burden, an allocation rule frequently championed in the biomedical literature.

I estimate the model by GMM using a panel of 106,664 unique investigator–grant pairs over 2005–2018, constructed by linking NIH R01 grant data to academic publications from OpenAlex and investigator histories from the Dimensions database. Two broad sets of moments identify the model: application volumes and scientist flows across fields. The model fits empirical moments well, including in out-of-sample years 2002–2004 and 2019–2022. Five findings emerge. Each answers a question that a policymaker dividing the NIH budget would ask.

What does an additional dollar buy in each field? First, marginal returns vary widely across fields. An additional $1 million yields 2.2 publications in Nursing Research but 14.8 in Human Genome, a factor of 6.8. General Medicine (11.0) and Environmental Health (8.3) also rank near the top; Aging (3.1) sits near the bottom. Where the marginal dollar goes therefore matters.

Why do returns differ so much? There are baseline differences in how much science $1 million produce, but this differences are greatly accentuated by knowledge spillovers. Upstream or general-purpose institutes whose research feeds many other fields (Human Genome, General Medicine, Environmental Health, and Biomedical Imaging and Bioengineering) generate large cross-field spillovers and are among the most productive. Knowledge spillovers account for 71.2 per cent of Human Genome's return, most of it accruing in other fields. In contrast, specialised institutes with narrower citation footprints (Nursing Research or Deafness and Communication Disorders) sit at the bottom of the return distribution. A field-by-field accounting would therefore greatly understate the return to upstream institutes.

Does funding one field simply drain talent from the others? It does draw talent, but at little cost. Scientists move towards better-funded fields: a budget increase in one institute draws applicants away from others. But those who replace them are equally productive once funded, so reallocation leaves marginal returns almost unchanged. Funding a field thus imposes a small cost on other fields through talent and confers a large benefit on them through knowledge.

Do these return estimates depend entirely on what the planner values and on how scientific output is measured? Marginal-return estimates and the resulting performance of counterfactual allocations are surprisingly stable across output measures. Marginal returns are measured in units of scientific output, and the choice of units embeds a welfare judgement. A planner who prioritises reducing heart-disease burden might assign greater weight to a mediocre paper on heart disease than to a cancer paper in a prestigious journal; a planner who prioritises academic impact might make the

opposite choice. I use the number of publications as the base measure of scientific output, then weight each publication by its commercial potential, academic impact, and disease burden. If field rankings shifted greatly with these weights, any optimal allocation would be optimal only under a narrowly defined welfare function. The ranking of fields barely moves across measures of scientific output. The most productive fields remain so across measures, as do the least productive fields. Rankings also vary little in the middle of the distribution.[2] This stability is an artefact of the similarity between measures, but only to a degree. The academic-impact and commercial-potential measures are strongly correlated (0.81) at the publication level, but neither is strongly correlated with disease-burden weights (0.04 with academic-impact and 0.11 with commercial-potential). Measures that disagree on individual grants nevertheless agree on fields. The ranking therefore reflects how each field converts funding and knowledge into publications, a feature that all three measures share. Across these three objectives, the case for reallocating funding does not hinge on a particular measure.

Should funding then follow disease burden, as the medical literature often argues? It should not. Allocating a budget increment in proportion to disease burden does improve substantially on the current allocation, increasing scientific output by 21.9 to 25.7 per cent. But allocating it in proportion to marginal returns performs better still: depending on how scientific output is measured, it produces between 15.1 and 26.4 per cent more output than the disease-burden allocation, and 39 to 54 per cent more output than the current allocation. This holds even when output is measured by disease burden alone. A planner who cares only about disease burden should therefore allocate funding according to marginal returns rather than disease-burden alignment. Although disease-burden alignment improves on the current allocation, the current allocation performs worse than any alternative based on marginal returns or disease burden.

Are these findings robust? Several checks suggest that they are. First, the estimates do not depend on the estimation window. Holding the parameters estimated on 2005–2018 fixed, the model predicts application volumes in 2002–2004 and 2019–2022 with mean absolute percentage errors of 12.3 and 17.2 per cent, and bilateral scientist-flow shares with an R^2 of 0.964 (Section 6.1). Second, the disease-burden results do not depend on missing data or on imperfect matching between grants and diseases. Only 45 per cent of grants link directly to a disease-burden category: 33.4 per cent carry no disease tag and 21.6 per cent carry tags that do not match the burden data. I impute the missing data with machine learning. I fine-tune BiomedBERT, a pretrained language model, to assign disease categories from grant titles and abstracts. Each grant then receives the burden of its predicted diseases. The re-estimated marginal returns

[2]The most notable exception is Heart, Lung, and Blood, which ranks third on disease burden but fourteenth on academic impact. This gap reflects the burden of cardiovascular disease in the United States.

correlate at 0.986 with the baseline with missing data, and Human Genome and Nursing Research remain first and last (Appendix K.6). Third, the policy results do not depend on the most productive field. Excluding Human Genome narrows the gains, but each marginal-return rule still outperforms disease-burden alignment, by 7.4 to 10.5 per cent (Appendix L.3). Finally, the data support the model's interpretation of scientist flows. Before switching institutes, scientists' grants are already about 10 per cent closer to their destination in a text-embedding space than the grants of matched stayers, consistent with switches reflecting reallocation at the margin between fields (Appendix B).

This paper contributes first to the medical and health-policy literature on the allocation of funds across NIH institutes. These studies typically assume that institutes addressing diseases with a greater social burden should receive proportionally more funds and then discuss how to measure disease burden and how to connect funding levels to disease burden in the data (Ballreich et al., 2021; Gao et al., 2025; Gillum et al., 2011; Gross, Anderson and Powe, 1999; Nimgaonkar et al., 2022; Sampat, Buterbaugh and Perl, 2013). Early economic work also links disease-level NIH funding to pharmaceutical innovation and health outcomes (Lichtenberg, 2001).[3] These studies typically ignore differences in productivity between fields and instead focus on differences in the social value of research (is curing cancer more pressing than preventing infectious disease?) I complement these studies by focussing on the field differences in the marginal return to funding: the rate at which additional dollars translate into additional scientific output. These differences are large and persist across various measures of scientific output.

The paper sits between two programmes of research on the public funding of science and innovation. The first concerns the design and effectiveness of public scientific funding at the micro level. For example, Boudreau et al. (2016), Lanei et al. (2022), and Li (2017) show how study-section composition determines which proposals are funded; Li and Agha (2015) assesses whether peer review selects productive science; Adda and Ottaviani (2024) and Carson, Zivin and Shrader (2023) highlight the role of application grading schemes; Howell et al. (2025) and Myers (2020) analyse the trade-off between untargeted and targeted grants; Azoulay, Graff Zivin and Manso (2011) compares person- and project-based funding; and Jacob and Lefgren (2011) estimates the effect of receiving an NIH grant. Carnehl, Ottaviani and Preusser (2025) provide a useful overview of this literature. Together, these studies ask how to allocate funds among competing individual projects. Instead, I ask how to allocate funds across broad lines of research.

The second programme concerns the public funding of science and technology for the economy as a whole. In particular, this paper adds to the growing literature estimating the return on public R&D funding (Akcigit, Hanley and Serrano-Velarde,

[3] In addition to disease burden, concerns over racial (Taffe and Gilpin, 2021), gender (Mirin, 2021), and age (Lauer and Roychowdhury, 2021; Qiu, 2023) have also been raised.

2021; Azoulay et al., 2019; Babina et al., 2023; Dyevre, n.d.; Fieldhouse and Mertens, 2025; Gross and Sampat, 2023; Kantor and Whalley, 2025; Myers and Lanahan, 2022; Toole, 2012; Yin et al., 2022). While these studies focus on the return on public R&D funding in general, I focus on the cross-field differences in the return to funding.

The paper also contributes to two literatures on the mechanisms of knowledge production. First, its spillover mechanism builds on the tradition of mapping knowledge flows with citation networks (Acemoglu, Akcigit and Kerr, 2016; Adams, 1990; Jaffe, Trajtenberg and Fogarty, 2000; Liu and Ma, 2021). It documents large heterogeneity in knowledge spillovers across scientific fields and shows that this heterogeneity drives the optimal funding allocation. I am perhaps closest to Liu and Ma (2021), who find that some industries produce far more knowledge spillovers than others and argue that industrial policy should prioritise them. Second, the paper shows how scientists redirect their research focus in response to funding conditions. It thereby contributes to the literature on scientists' labour supply and topic choice (Foster, Rzhetsky and Evans, 2015; Hill et al., 2025; Myers, 2020; Rzhetsky et al., 2015).

The allocation problem I study concerns 'picking winners' and actively choosing which fields to support. By studying this question in biomedical research, I contribute to the fast-growing literature on mission-oriented innovation and modern industrial policy (Gross and Sampat, 2023; Juhász, 2018; Kantor and Whalley, 2025; Lane, 2025; Lashkaripour and Lugovskyy, 2023; Liu, 2019; Mazzucato, 2018; Sampat, 2012). See Juhász, Lane and Rodrik (2024) for a review. A quantitative antecedent is Bisias, Lo and Watkins (2012). They use historical NIH appropriations and subsequent reductions in years of life lost to proxy return on investment and trace mean–variance-efficient allocations. Their approach assumes a linear relationship between aggregate funding and population health improvements. Instead, I provide a micro-founded structure that maps funding into scientific output and allows me to evaluate counterfactual funding allocations.

The paper is structured as follows. Section 2 formally defines the allocation problem. Section 3 presents a model of scientist reallocation, grant evaluation, and research production. Section 4 derives the marginal return to funding in each field. Section 5 describes the data and the estimation approach. Section 6 reports model fit, parameter estimates, and marginal returns. Section 7 evaluates the efficiency of the current allocation against a range of counterfactual allocations. Section 8 concludes.

2. The Allocation Problem

2.1. NIH Institutes as Research Fields

The allocation problem is how to distribute a fixed NIH budget across institutes, each of which represents a research field. NIH administers funding through institutes and centres (ICs), each focused on a body system, disease class, or broad research mission (Table A.1). This paper covers 19 core institutes and excludes smaller NIH research centres that do not have institute status. Congress sets both the aggregate NIH budget and institute-level allocations, informed by NIH priorities and recommendations. As such, institutes are the policy-relevant funding units as they constitute the choice set of the planner (Congress). This makes them, for this purpose, the most useful proxy for scientific fields. The planner's problem comes down to one question: how much welfare does an extra dollar in one institute generate, once its effects on every field are counted? This section formalises the allocation problem across institutes and shows why the key object to estimate is the marginal return to funding across institutes.

2.2. The Social Planner's Problem

In period t, the planner allocates a finite research budget $\bar{G}_t > 0$ across NIH institutes, indexed by $f \in \{1, \ldots, C\}$:

$$\mathbf{G}_t \equiv (G_{1,t}, \ldots, G_{C,t}) \in \mathbb{R}_+^C \quad \text{and} \quad \sum_{f=1}^{C} G_{f,t} = \bar{G}_t.$$

I denote this budget set by $\Delta(\bar{G}_t)$.

Output depends on current funding and on past output. Let $\mathbf{K}_t$ denote the vector of depreciated knowledge stocks. It summarises past scientific output through the depreciation operator

$$\mathbf{K}_t = \mathcal{K}(\mathbf{K}_{t-1}, \mathbf{Y}_t).$$

Thus, $\mathbf{K}_t$ incorporates current output and becomes available for production in period $t+1$:

$$\mathbf{Y}_{t+1} = \mathbf{Y}(\mathbf{G}_{t+1}, \mathbf{K}_t).$$

Within a period, funding and inherited knowledge determine output in equilibrium:

$$\mathbf{Y}_t = \mathbf{Y}(\mathbf{G}_t, \mathbf{K}_{t-1}), \quad \text{where} \quad \mathbf{Y}(\cdot, \mathbf{K}_{t-1}) : \left\{ \mathbf{G} \in \mathbb{R}_+^C : \sum_{f=1}^{C} G_f = \bar{G}_t \right\} \to \mathbb{R}_+^C.$$

Here $Y_o(\mathbf{G}_t, \mathbf{K}_{t-1})$ is the mean output of field o under funding allocation $\mathbf{G}_t$ and knowledge state $\mathbf{K}_{t-1}$. The planner is risk-neutral and evaluates output through a continuously differentiable welfare function:

$$U : \mathbb{R}_+^C \to \mathbb{R}, \qquad U(\mathbf{Y}_t) \equiv U(Y_{1,t}, \ldots, Y_{C,t}).$$

For a given knowledge state $\mathbf{K}_{t-1}$, the planner chooses $\mathbf{G}_t$ to maximise welfare:[4]

$$\max_{\mathbf{G}_t \in \Delta(\bar{G}_t)} \quad U(\mathbf{Y}(\mathbf{G}_t, \mathbf{K}_{t-1}))\,. \tag{1}$$

At an optimum $(\mathbf{G}_t^\star, \nu^\star)$, the marginal welfare effect of funding is equalised across all institutes.[5]

$$\underbrace{\frac{d}{dG_{d,t}} U(\mathbf{Y}(\mathbf{G}_t, \mathbf{K}_{t-1}))}_{\text{marginal welfare effect for funded field } d} = \nu, \qquad \forall d. \tag{2}$$

Combined with the budget constraint $\sum_{f=1}^{C} G_{f,t} = \bar{G}_t$, this system determines the optimal allocation.

What does the planner need to know to solve this system? Expanding the total derivative gives the answer:[6]

$$\frac{d}{dG_{d,t}} U(\mathbf{Y}(\mathbf{G}_t, \mathbf{K}_{t-1})) = \sum_{o=1}^{C} w_o \cdot \frac{dY_o(\mathbf{G}_t, \mathbf{K}_{t-1})}{dG_{d,t}}, \tag{3}$$

where $w_o \equiv \frac{\partial U(\mathbf{Y})}{\partial Y_o}$ is the social welfare weight placed on field o. These weights can reflect any welfare objective, such as prioritising particular diseases or populations.

The weights w_o encode a normative choice. The central objective of this paper is to estimate $dY_o(\mathbf{G}_t, \mathbf{K}_{t-1})/dG_{d,t}$ for every pair (d, o), where d is the funded field and o the output field. I call this object the *pairwise output effect*. Summing pairwise output effects across output fields o gives funded field d's *marginal return to funding*. Applying the social welfare weights gives its *marginal welfare effect*.

[4] When I estimate marginal returns, I also include discounted future output from knowledge produced in the funded field. For ease of exposition, here we focus on the current-period problem.

[5] If the objective is continuous and the budget set $\Delta(\bar{G}_t)$ is compact, a global maximum $\mathbf{G}_t^\star$ exists. I restrict attention to interior solutions, where $G_{d,t} > 0$ for every d. A corner could in principle dominate an interior solution, but it would close an NIH institute and is therefore unlikely to be politically feasible.

[6] A total derivative is required because a change in $G_{d,t}$ affects equilibrium objects, including application success rates and applicant sorting, and therefore changes every component of $\mathbf{Y}_t$, not only $Y_{d,t}$.

2.3. Why Funding One Field Affects Other Fields

The equilibrium mapping $\mathbf{Y}(\mathbf{G}_t, \mathbf{K}_{t-1})$ encodes the output effects of a budget allocation without specifying the underlying economic mechanisms. Funding in one field affects output in others through two channels. Scientists may move across fields, and knowledge produced in one field may raise productivity in another. The evidence below motivates both channels.

When $G_{d,t}$ rises, funded field d supports additional projects. The pool of biomedical scientists is not perfectly elastic in the short run, so this expansion draws talent away from other fields. Scientists at the margin (those whose comparative advantage does not strongly favour one field) move their research to the better-funded field.

This reallocation raises output in funded field d because more scientists work there. It may reduce output in fields $o \neq d$ because they lose scientists who would otherwise have contributed there. The size of this cross-field effect depends on how substitutable scientific labour is across fields.

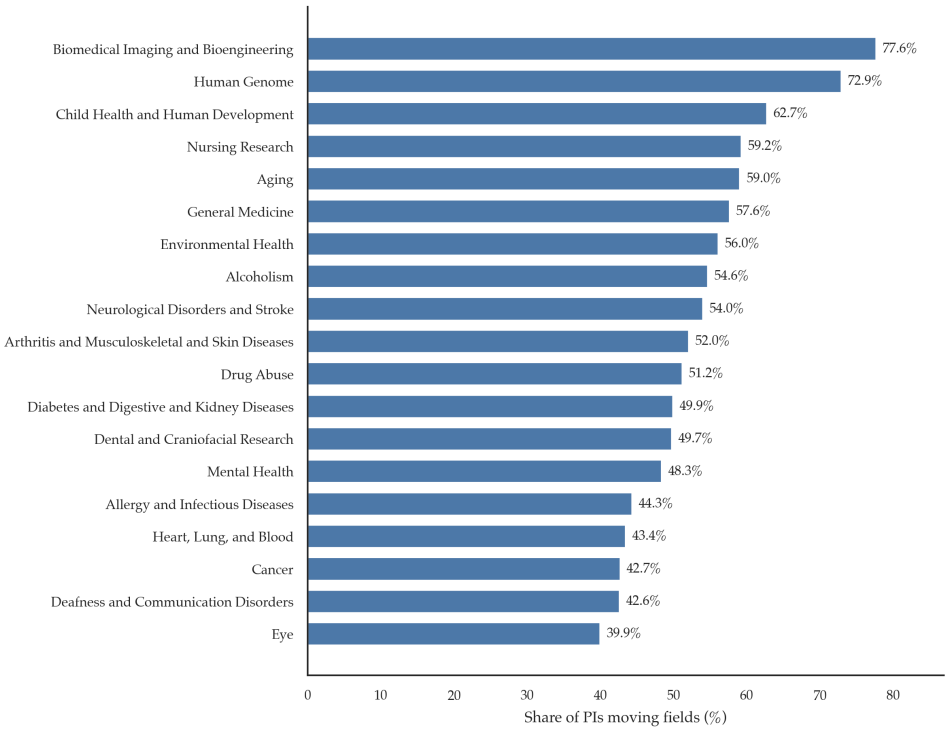
Investigator histories show that such reallocation is frequent but varies across fields. Figure 1 reports the mover share: the share of PIs whose current funding institute differs from their previous funder. Broad institutes have higher mover shares than institutes tied to specific body systems. Biomedical Imaging and Bioengineering and Human Genome have mover shares of 78 and 73 per cent, whereas Eye and Deafness and Communication Disorders have shares of 40 and 43 per cent. Scientists therefore often pursue funding opportunities outside their previous field. The model in Section 3 treats each change of funder as such a choice.[7]

Knowledge also links fields over time through the stock $\mathbf{K}_{t-1}$. Scientific output is not only a final product; it is also an input into future research. For example, a molecular pathway identified by diabetes researchers may spur discoveries in cancer research if the pathway is implicated in oncogenesis. Formally, an increase in output $Y_{d,t}$ from field d raises its knowledge stock and thereby increases future output in fields o.

Citation patterns suggest that knowledge spillovers occur both within and across fields, and that some fields supply far more knowledge than others. Figure 2 reports the share of references made by publications in each citing field (row) to publications in each cited field (column). Three patterns stand out. First, fields draw most on their own research: diagonal entries exceed off-diagonal entries, so that cancer research, for example, is most frequently cited by other cancer research. Second, some fields supply knowledge to many others. General Medicine and Cancer have large entries in the bottom row, which averages each column's off-diagonal entries. Conversely, Nursing Research and Alcoholism are cited less frequently by other fields. Third, cross-field

[7] A change of funder need not be a permanent change of research field. Section 6.3 shows that scientists who switch institutes already work closer to their destination before the switch.

Figure 1. Share of PIs who moved between funding institutes

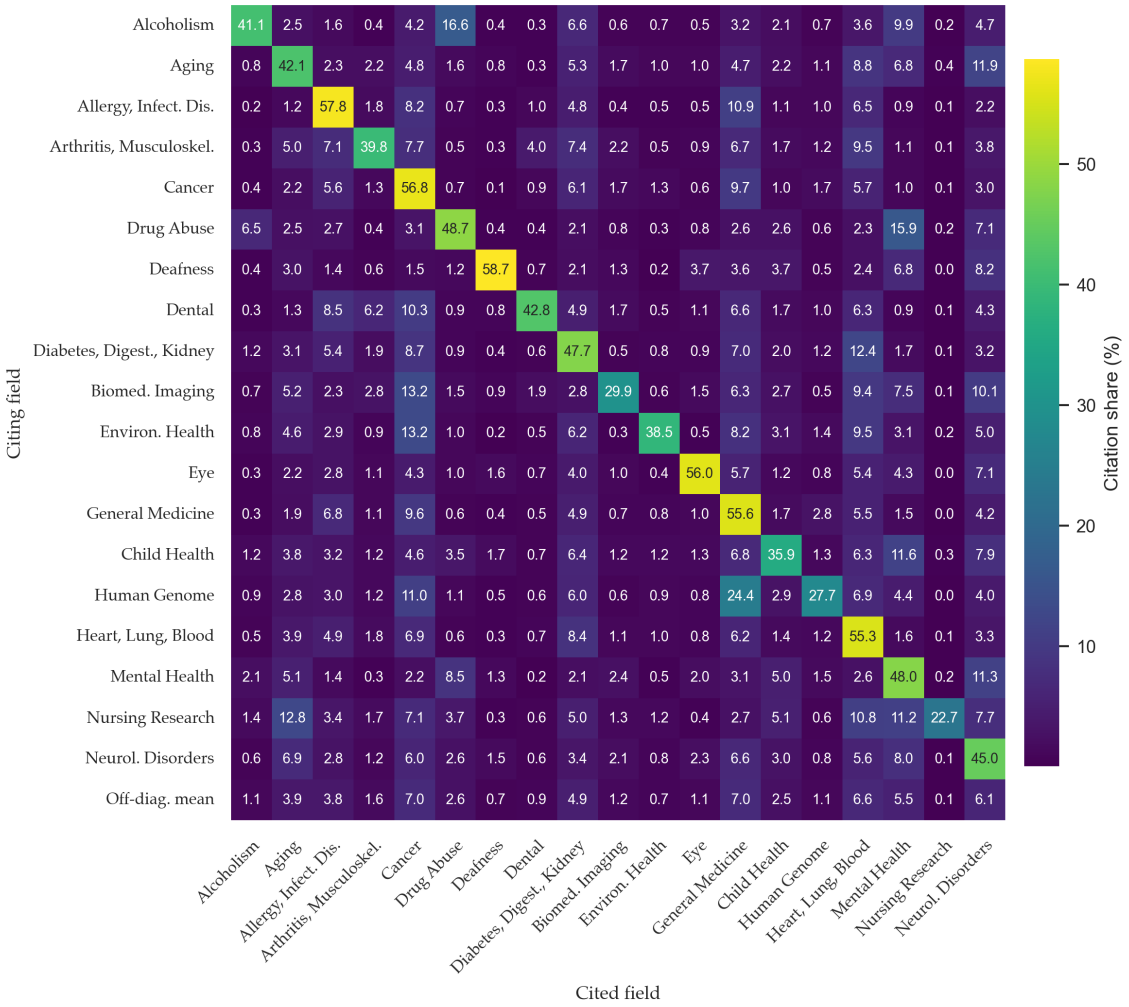


Notes: Each bar reports the percentage of PIs whose current funding institute differs from their previous funder, by destination institute. Institutes are ordered by this mover share. A higher value indicates that the institute draws a larger proportion of its funded investigators from scientists previously funded elsewhere.

flows are concentrated and asymmetric. Alcoholism draws heavily on Drug Abuse, and one field may cite another far more frequently than it is cited in return. These patterns suggest that some fields are more central to the biomedical research ecosystem and generate more knowledge spillovers than others.

Together, these mechanisms imply that estimating the pairwise output effect $dY_o/dG_{d,t}$ requires modelling both how scientists reallocate across fields in response to funding and how knowledge produced in one field raises the productivity of others. The structural model in Section 3 captures both cross-field dependencies.

Figure 2. Citation shares across NIH fields



Notes: Rows report citing fields and columns report cited fields. Each cell is the percentage of references made by publications in the row field that point to publications attributed to the column field. Diagonal cells measure within-field knowledge use. Off-diagonal cells measure cross-field knowledge exposure. The final row averages each cited field's off-diagonal shares across the other 18 citing fields, on the same percentage scale as the matrix.

3. A Model of Grant Applications and Scientific Output

The model captures how funding translates into scientific output. Scientists draw ideas and choose where to apply. Reviewers score the applications, and institute paylines adjust until every available grant is awarded. Funded scientists then combine their ideas, grant funding, and inherited knowledge to produce scientific output. Figure 3 summarises these mechanisms and the equilibrium feedback between applications and paylines; Table 1 defines the model's variables and parameters.

Unless dynamics make it necessary, the theory suppresses the year subscript t. Field subscripts o and d can denote different things. In the scientist choice model, o denotes the origin field of a scientist and d its destination. In the knowledge-spillover analysis, o denotes the citing field and d the cited field. When I analyse pairwise output

effects, o denotes the output field, whose output is affected by d, the funded field. I also use indices f and f' to denote generic fields, when necessary. Within sums, o or d may also be reused as a summation index. This avoids introducing further indices to denote scientific fields (I would need at least five more otherwise). Empirically, the unit of observation is a unique grant–investigator pair, denoted by i. Because observations indexed by i are the most disaggregated in this study, specifying their origin field o, destination field d, and year t is redundant. For ease of exposition, I refer to the corresponding investigator as scientist i and the corresponding grant as grant i, although the model is estimated at the grant–investigator level.

3.1. Funding Rules and the Scientist's Problem

Consider an economy where a fixed number of scientists N work on research projects in C different fields and can redirect their work from an origin field o to a destination field d. There are N_o scientists from each origin field $o \in \{1, \ldots, C\}$.

Each period, scientist i from origin field o draws an idea of quality a_i for each field d, including her origin field $o = d$. Scientist i applies to at most one field d to pursue the corresponding idea and may instead not apply. For example, a scientist whose previous grant came from the National Cancer Institute has origin o = Cancer. She may draw ideas relevant to Cancer and to other fields. If her most promising idea concerns Aging, she may apply to the National Institute on Aging, which becomes her destination field d.

Institute d has a budget G_d exogenously set by the planner, or by Congress in practice. It awards $G_d/\bar{g}_d$ grants, where $\bar{g}_d$ is the average grant size (typically around two million dollars for R01 grants in 2017). The grant size g_i required to pursue idea a_i is exogenous to both the application and funding decisions.[8]

Idea quality a_i is private to the scientist. The evaluation process is noisy, and reviewers assess the grant application imperfectly. They assign the idea a score x_i:

$$x_i = a_i + \eta\, \nu_i, \tag{4}$$

where $\nu_i \sim \mathcal{N}(0, 1)$ and $\eta > 0$ measures noise in the evaluation process.

An application is funded if and only if its score exceeds the institute-specific threshold $\bar{x}_d$, called the *payline*. An application of quality a_i is therefore funded with probability

$$\Pr(x_i > \bar{x}_d) = \Phi\left(\frac{a_i - \bar{x}_d}{\eta}\right). \tag{5}$$

[8] R01 applicants must justify the funds they need in detail. Reviewers are supposed to assess the scientific merit of grants without taking into account their grant size. Modelling grant size as a choice variable would be unrealistic.

Following a standard discrete-choice model (McFadden, 1974), idea quality has an origin–destination mean μ_{od} and an idiosyncratic component ε_i:

$$a_i = \mu_{od} + \varepsilon_i, \tag{6}$$

where the ε_i are independently drawn across destination alternatives from a Type-I Extreme Value (Gumbel) distribution with scale $\sigma > 0$.

Scientists observe their idea qualities and destination paylines before applying, but not the scores they will receive. Scientist i applies to the field d that maximises her funding probability:

$$d^* = \arg \max_{d \in \{0,1,\ldots,C\}} \Pr(\text{funded} \mid a_i, d), \tag{7}$$

where the outside option $d = 0$ yields payoff c, expressed in the same quality-gap units as the research alternatives; higher review scores indicate stronger applications. Because η is common across all origin–destination pairs in the theoretical model,[9] the funding probability $\Phi((a_i - \bar{x}_d)/\eta)$ is strictly increasing in the quality gap $a_i - \bar{x}_d$. Maximising the probability of funding is therefore equivalent to maximising this gap.

The Gumbel errors imply that the probability Π_{od} that any scientist from origin field o applies to field d has a multinomial logit form:

$$\Pi_{od}(\bar{x}) = \frac{\exp\left(\frac{\mu_{od} - \bar{x}_d}{\sigma}\right)}{\exp\left(\frac{c}{\sigma}\right) + \sum_{f=1}^{C} \exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)}. \tag{8}$$

The mass of applications received by institute d is $A_d = \sum_{o=1}^{C} N_o \Pi_{od}(\bar{x})$, where $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_C)$ is the vector of paylines.

3.2. Equilibrium

Conditional on applying to institute d, a scientist's funding probability depends on other applications only through the equilibrium payline $\bar{x}_d$. Let $Z_i \equiv a_i - \bar{x}_d$ denote the quality gap that i will achieve, should she apply to d. Let $f_o(z \mid \text{apply to } d)$ denote the density function of this quality gap, conditional on i applying to d. Appendix C derives this density from the Gumbel structure. Integrating the funding probability over this quality-gap distribution gives the success rate for applicants from origin o to institute d:

$$S_{od}(\bar{x}) = \int_{-\infty}^{\infty} \Phi\left(\frac{z}{\eta}\right) f_o(z \mid \text{apply to } d)\, dz. \tag{9}$$

[9] In estimation, I allow evaluation noise to increase with the distance between the origin field and destination institute. See Section 5.

In equilibrium, the payline vector $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_C)$ is such that every institute disburses all its available grants:

$$\sum_{o=1}^{C} \underbrace{N_o \, \Pi_{od}(\bar{x})}_{\substack{\text{Applications} \\ \text{from } o \text{ to } d}} \underbrace{S_{od}(\bar{x})}_{\substack{\text{Success rate} \\ \text{from } o \text{ to } d}} = \underbrace{\frac{G_d}{\bar{g}_d}}_{\text{Available grants}} \qquad \text{for all } d = 1, \ldots, C. \tag{10}$$

Equation (10) is a market-clearing condition: each payline adjusts until the number of successful applications from all origin fields is exactly equal to the number of grants field d can afford to fund. If institute d receives more funding, it must lower its payline to fund additional grants. If institute d receives less funding, it must increase its payline to lower the application success rate and fund fewer grants. A lower payline makes d more attractive and draws applicants from other institutes; a higher payline reduces its appeal. Changes in applications elsewhere then alter those institutes' clearing paylines. Appendix D proves existence and uniqueness of the equilibrium payline vector. Here paylines behave like prices in a standard model: they adjust until supply equals demand.

3.3. Production Function and Aggregate Output

Upon receiving funding, scientist i combines her idea quality a_i, grant funding g_i, and knowledge stock k_i to produce scientific output y_i:

$$y_i = a_i \, k_i^{\alpha_d} g_i^{\beta_d} + u_i, \tag{11}$$

Here k_i is a knowledge input variable that is specific to grant i. The parameters α_d and β_d are the destination-specific elasticities of knowledge and funding. The mean-zero shock u_i captures the fundamental uncertainty of scientific research.[10] Funding g_i is observed, specific to the grant and exogenous. It may differ from the average grant size in field d, $\bar{g}_d$. In fact, $\bar{g}_d$ is obtained from averaging g_i across all grants funded by d. I assume throughout that the Cobb–Douglas input term $k_i^{\alpha_d} g_i^{\beta_d}$ is mean-independent of idea quality a_i. This assumption can be justified by the R01 evaluation process, in which reviewers assess scientific merit independently of grant size. Integrating over

[10] Because the Gumbel distribution has infinite support, the additive specification can in principle generate negative output y_i. In the empirically relevant parameter range, however, the model's integrals yield strictly positive mean output.

idea quality gives the total output generated by scientists from origin o funded by d:

$$\begin{aligned} Y_{od} \;=\; & \underbrace{N_o\,\Pi_{od}(\bar{x})}_{\substack{\text{Application volume}\\ \text{from } o \text{ to } d}} \quad \mathbb{E}[k_i^{\alpha_d} g_i^{\beta_d}] \\ & \times \int_{-\infty}^{\infty} \underbrace{(z+\bar{x}_d)}_{a_i} \underbrace{\Phi\left(\frac{z}{\eta}\right) f(z \mid \text{apply to } d)\, dz}_{\text{Funding probability}}. \end{aligned} \tag{12}$$

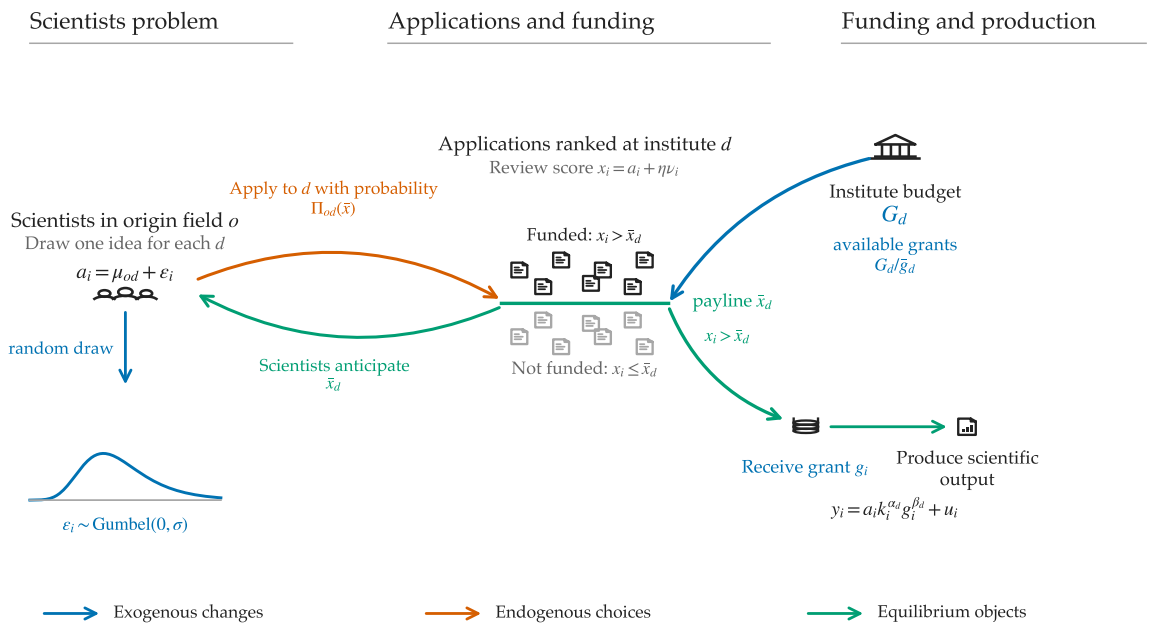
Let $Q_{od}(\bar{x})$ denote mean idea quality conditional on scientist i applying to field d from origin o and receiving funding:

$$\begin{aligned} Q_{od}(\bar{x}) \;&\equiv\; \mathbb{E}[a_i \mid \text{applied to } d \text{ and funded}] \\ &=\; \frac{1}{S_{od}(\bar{x})} \int_{-\infty}^{\infty} (z+\bar{x}_d)\Phi\left(\frac{z}{\eta}\right) f(z \mid \text{apply to } d)\, dz. \end{aligned} \tag{13}$$

Using this expression, we can more cleanly express Y_d, total scientific output produced in field d. For each origin field o, it depends on the number of scientists in that field N_o, the probability that they apply to d $\Pi_{od}(\bar{x})$, the probability that they are funded $S_{od}(\bar{x})$ upon applying, the mean idea quality of those funded $Q_{od}(\bar{x})$, and the mean input term of their production function $\mathbb{E}[k_i^{\alpha_d} g_i^{\beta_d}]$:

$$Y_d(\bar{x}) \;=\; \sum_{o=1}^{C} N_o\,\Pi_{od}(\bar{x})\,S_{od}(\bar{x})\,Q_{od}(\bar{x})\,\mathbb{E}[k_i^{\alpha_d} g_i^{\beta_d}]. \tag{14}$$

Figure 3. Model overview: from applications to scientific output



Notes: The diagram shows scientists' idea draws a_i, application choices d, review scores x_i, paylines $\bar{x}_d$, and funding probabilities S_{od}. Funded scientists combine grant funding g_i and knowledge k_i to produce output y_i, which contributes to future knowledge stocks and spillovers.

Table 1. Model variables and parameters

Symbol	Definition
i	Scientist and corresponding grant–investigator observation.
o, d	Origin and destination fields in the application model; citing and cited fields in spillovers; output and funded fields in pairwise output effects.
f, f'	Generic field indices used within sums.
t	Year, suppressed unless dynamics require it.
C	Number of scientific fields.
N, N_o	Total number of scientists and number from origin field o, respectively.
G_d	Exogenous budget of institute d.
g_i, $\bar{g}_d$	Exogenous funding required by grant i and average grant size in field d, respectively.
$G_d/\bar{g}_d$	Number of available grants at institute d.
a_i	Scientist i's idea quality for a destination field.
μ_{od}	Mean idea-quality parameter for origin–destination pair (o, d).
ε_i	Idiosyncratic idea-quality component, independently drawn across destinations from a Gumbel distribution.
σ	Idea-quality scale of the Gumbel distribution.
x_i	Review score assigned to grant application i.
ν_i	Standard normal evaluation-noise shock.
η	Evaluation-noise standard deviation.
$\bar{x}_d$, $\bar{x}$	Institute d's payline and the vector of all institute paylines, respectively.
d^*	Scientist i's chosen destination; $d^* = 0$ denotes not applying.
c	Outside option, expressed in quality-gap units.
$\Phi(\cdot)$	Standard normal cumulative distribution function.
$\Pi_{od}(\bar{x})$	Probability that a scientist from origin o applies to destination d.
A_d	Number of applications received by institute d.
Z_i, z	Quality gap $a_i - \bar{x}_d$ and its realised value, respectively.
$f_o(\cdot)$, $f(\cdot)$	Density of the quality gap conditional on applying to destination d; f_o indicates origin o.
$S_{od}(\bar{x})$	Funding success rate conditional on applying from origin o to destination d.
k_i	Knowledge input specific to grant i.
α_d, β_d	Destination-specific output elasticities of knowledge and funding, respectively.
u_i	Mean-zero scientific-output shock.
y_i	Scientific output of scientist i's funded grant.
$\mathbb{E}[k_i^{\alpha_d} g_i^{\beta_d}]$	Mean knowledge–funding input term in the production function.
$Q_{od}(\bar{x})$	Mean idea quality conditional on applying from o to d and receiving funding.
Y_{od}, $Y_d(\bar{x})$	Total scientific output from origin-o scientists funded by d and total output produced in field d, respectively.

Notes: The model is defined at the grant–investigator level. Field subscripts have the roles specified above; they may also serve as summation indices. Application probabilities, success rates and funded idea quality depend on the equilibrium payline vector $\bar{x}$.

4. The Marginal Return to Research Funding

This section characterises the paper's main object: the marginal return to research funding. For ease of exposition, d always denotes the *funded field*, which receives the budget increment, and o always denotes the *output field*, whose output response is measured. These roles do not depend on whether either field is an origin or destination in the application model, or a citing or cited field in a knowledge flow.

Consider an increase in funding G_d. Institute d lowers its payline $\bar{x}_d$ to allocate the additional funding. This raises applicants' funding probabilities, makes field d more attractive, and changes the number and quality of projects it funds, by changing the pool of applications it receives. Other fields $o \neq d$ are affected indirectly. As applicants move towards field d, applications to field o fall. Field o must then lower its payline $\bar{x}_o$ to disburse its unchanged budget. The pairwise output effect is therefore the effect of additional funding in d on output in o, at equilibrium. Figure 4 summarises this response and separates its current talent-flow and future knowledge-spillover channels.

$$\frac{dY_o}{dG_d} = \sum_{f=1}^{C} \underbrace{\frac{\partial Y_o}{\partial \bar{x}_f}}_{\substack{\text{Partial effect} \\ \text{of payline } f}} \underbrace{\frac{d\bar{x}_f}{dG_d}}_{\substack{\text{GE payline} \\ \text{adjustment}}}, \qquad \forall d, o. \tag{15}$$

This object is the key focus of the paper.[11]

4.1. From Budgets to Paylines

Equation (15) first requires the response of every payline to the budget of funded field d. The budget increment enters only field d's market-clearing condition. It nevertheless moves every payline, because the number of grants each institute funds depends on all paylines through scientists' application choices. Let

$$E_f(\bar{x}) = \sum_{o=1}^{C} N_o \Pi_{of}(\bar{x}) S_{of}(M_o) \tag{16}$$

denote the expected number of funded grants in field f, where the sum runs over applicants' origin fields. Differentiating the market-clearing condition in Equation (10)

[11] Output Y_o is the reduced-form object $Y_o(\bar{x}(G))$. Funding enters output through the equilibrium payline vector. A partial derivative such as $\partial Y_o / \partial G_d$ would hold $\bar{x}$ fixed and shut down the market-clearing response that defines the general-equilibrium effect.

with respect to G_d gives, for every field $f \in \{1, \ldots, C\}$,

$$\sum_{f'=1}^{C} \underbrace{\frac{\partial E_f}{\partial \bar{x}_{f'}}}_{\substack{\text{Demand shift} \\ \text{in } f \text{ from } \bar{x}_{f'}}} \frac{d\bar{x}_{f'}}{dG_d} = \begin{cases} \frac{1}{\bar{g}_d} & \text{if } f = d, \\ 0 & \text{if } f \neq d, \end{cases} \tag{17}$$

Only field d's equation has a non-zero right-hand side: field d must fund $1/\bar{g}_d$ additional grants per dollar. The payline adjustments $d\bar{x}_f/dG_d$ follow from the entire equilibrium system, not from field d in isolation. Appendix E derives the solution.

4.2. From Paylines to Output

Equation (15) also requires the response of output to each payline. A payline shift changes output in field f through two margins: the volume of applications it receives and the expected quality of the projects it funds. The gradient $\nabla_{\bar{x}} Y_f$ separates the two:

$$\nabla_{\bar{x}} Y_f = \mathbb{E}[f(g_i, k_i)] \sum_{o=1}^{C} N_o \left[\underbrace{\nabla_{\bar{x}} \Pi_{of}}_{\text{Volume shift}} \underbrace{\widetilde{Q}_{of}}_{\text{Base quality}} + \underbrace{\Pi_{of}}_{\text{Base volume}} \underbrace{\nabla_{\bar{x}} \widetilde{Q}_{of}}_{\text{Quality shift}} \right], \tag{18}$$

where the sum runs over applicants' origin fields o and $\widetilde{Q}_{of} = S_{of} Q_{of}$ denotes the quality of applications from origin o to f weighted by their success probability:

$$\widetilde{Q}_{of}(M_o, \bar{x}_f) = \int_{-\infty}^{\infty} (z + \bar{x}_f)\, \Phi\left(\frac{z}{\eta_{of}}\right) f_Z(z; M_o)\, dz. \tag{19}$$

This term integrates over the quality-gap distribution $f_Z(z; M_o)$ and weights each application by its probability $\Phi(z/\eta_{of})$ of securing funding. Appendix E derives the gradient.

In the funded field d, the two effects can pull in opposite directions. The additional budget funds additional projects, which raises output (volume effect). But the lower payline means that the newly funded projects may be of lower quality than the baseline projects (quality effect). The quality effect may offset part of the volume effect; the sign of the total response depends on the fitted production and selection terms.

In other fields $o \neq d$, the response depends on whether their paylines adjust. If they stay fixed, fields $o \neq d$ lose applicants to d, fund fewer projects, and leave part of their budgets unspent. Once all paylines adjust, market clearing restores the number of grants each field $o \neq d$ awards. Output in o can then change through funded-project quality and the mix of origins among its funded scientists. The sign of this general-equilibrium

talent-flow externality is therefore ambiguous. Section 6.3 describes this mechanism in more detail.

4.3. Knowledge Spillovers

I extend the model to allow for knowledge spillovers within and across institutes. Knowledge production is cumulative: new discoveries build upon the existing stock of ideas, which depreciate over time (Griliches, 1979). An extra publication in d raises output in later periods, both in d and in the fields that draw on its knowledge.

Knowledge accumulates as the depreciated sum of past output. The knowledge stock of field f at time t is

$$K_{f,t} \;=\; \sum_{s=0}^{\infty} (1-\delta_f)^s Y_{f,t-s}. \tag{20}$$

A grant producing in period t uses the stocks from period $t-1$, which enter the production function through the grant-specific knowledge index k_i. Let $\boldsymbol{\chi}_i$ be the vector of citation shares for grant i. The knowledge index is the Cobb–Douglas aggregator

$$k_i \;=\; \prod_{f=1}^{C} K_{f,t-1}^{\chi_{if}}, \tag{21}$$

where $\chi_{if} \geq 0$ is the share of the grant's references attributed to field f. Following Acemoglu, Akcigit and Kerr (2016) and Liu and Ma (2021), I use citation shares as proxies for the elasticities of grant knowledge input with respect to the knowledge stock in a particular field.

Aggregating over grants gives each field's exposure to the knowledge of every other field. Here o denotes the citing field and d the cited field. Let $\mathcal{G}_{o,t}$ be the set of grants funded by o in year t, and let $Y_{o,t} = \sum_{i \in \mathcal{G}_{o,t}} y_i$ be the aggregate output in o. The exposure of citing field o to knowledge from cited field d is the output-weighted average citation share

$$\Gamma_{do,t} \;=\; \sum_{i \in \mathcal{G}_{o,t}} \omega_i \, \chi_{id}, \qquad \omega_i \equiv \frac{y_i}{Y_{o,t}}. \tag{22}$$

If the production function in citing field o has knowledge elasticity α_o, the Cobb–Douglas index implies that a 1 per cent increase in d's knowledge stock raises o's output in the next period by $\alpha_o \Gamma_{do,t+1}$ per cent:

$$\frac{\partial \ln Y_{o,t+1}}{\partial \ln K_{d,t}} \;=\; \sum_{i \in \mathcal{G}_{o,t+1}} \omega_i \, \alpha_o \chi_{id} \;=\; \alpha_o \Gamma_{do,t+1}. \tag{23}$$

This effect persists beyond one period. An additional unit of output in field d first raises output in period $t+1$, while its contribution to the stock used in period $t+s$ is depreciated by $(1-\delta_d)^{s-1}$. Let $q=(1+r)^{-1}$ denote the annual discount factor. Assuming that $\Gamma_{do,t+s}$ and $\mathbb{E}[Y_{o,t+s}]/\mathbb{E}[K_{d,t+s-1}]$ are constant over time, the discounted series converges to $q[1-q(1-\delta_d)]^{-1}$. The total knowledge spillover generated by d in field o is

$$\frac{d\sum_{s=1}^{\infty} q^s \mathbb{E}[Y_{o,t+s}]}{dY_{d,t}} = \frac{q\alpha_o \Gamma_{do,t+1}}{1-q(1-\delta_d)} \frac{\mathbb{E}[Y_{o,t+1}]}{K_{d,t}}, \qquad q=\frac{1}{1.05}. \tag{24}$$

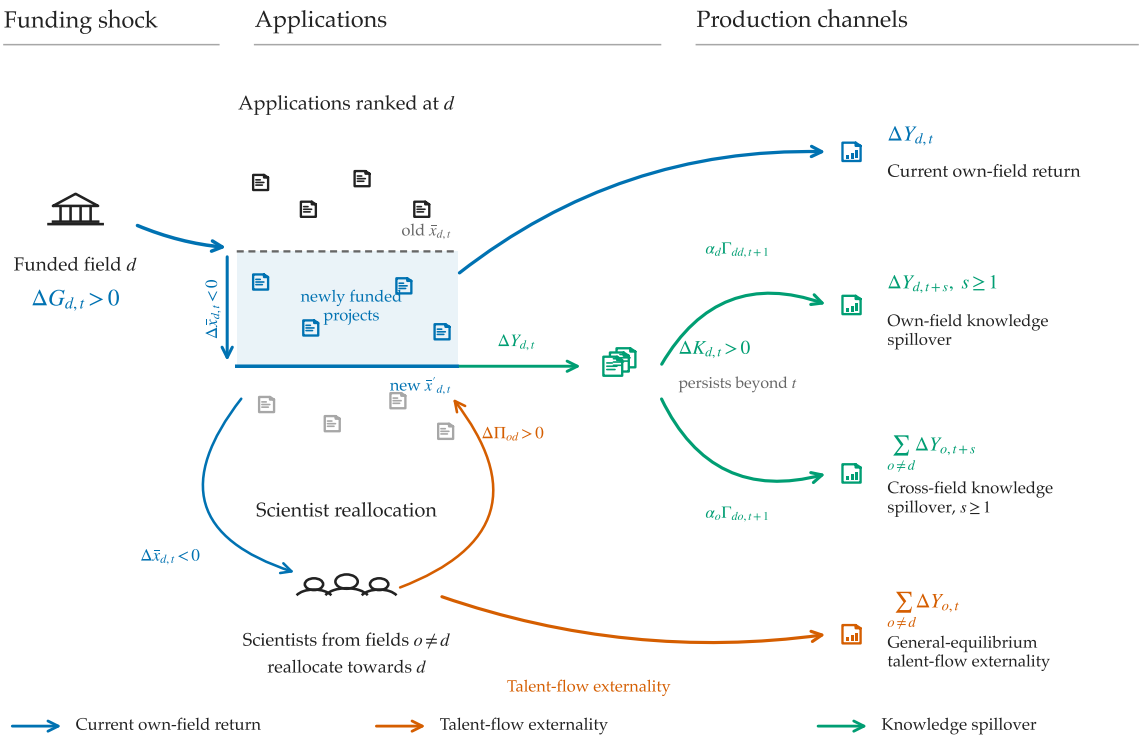
This expression is a first-order approximation. It captures how knowledge produced today raises output in every future period. It omits the further rounds in which this additional future output enlarges the knowledge stock again.

The marginal return combines the current-period pairwise output effects from Section 2 with discounted future effects of the resulting output change in funded field d. Under this focal-field approximation, it decomposes as

$$\begin{aligned} MR_{d,t} = & \underbrace{\frac{dY_{d,t}}{dG_{d,t}}}_{\text{Current own-field return}} \\ & + \underbrace{\sum_{o\neq d} \frac{dY_{o,t}}{dG_{d,t}}}_{\text{General-equilibrium talent-flow externality}} \\ & + \underbrace{\frac{dY_{d,t}}{dG_{d,t}} \frac{q\alpha_d \Gamma_{dd,t+1}}{1-q(1-\delta_d)} \frac{\mathbb{E}[Y_{d,t+1}]}{K_{d,t}}}_{\text{Own-field knowledge spillover}} \\ & + \underbrace{\frac{dY_{d,t}}{dG_{d,t}} \sum_{o\neq d} \frac{q\alpha_o \Gamma_{do,t+1}}{1-q(1-\delta_d)} \frac{\mathbb{E}[Y_{o,t+1}]}{K_{d,t}}}_{\text{Cross-field knowledge spillover}}. \end{aligned} \tag{25}$$

The current own-field return is the output gained in field d from the additional grants it funds. The general-equilibrium talent-flow externality is the output change in fields $o \neq d$ as scientists change their application choices. The own-field and cross-field knowledge spillovers are the discounted output gains from period $t+1$ onwards, as additional output in d enlarges the knowledge stock available to d and other citing fields in the future. Current output changes in other fields are not propagated through their knowledge stocks. Appendix H describes the numerical calculation.

Figure 4. Effects of a budget increase at institute d



Notes: The diagram traces a local budget increase in funded field d through its payline and researchers' application choices to output in field o. It distinguishes current talent-flow effects from discounted future knowledge spillovers.

5. Empirical Strategy

This section describes how I take the model to the data. I construct a grant–investigator-level panel, map each observation to an origin field and destination institute, and use the resulting application counts and scientist-flow shares to estimate the structural parameters listed in Table 3. Appendix Table A.3 summarises the identifying variation and moments for each parameter block.

5.1. Sample Construction

Table 2 summarises the variables and their data sources.

The sample links public NIH grant records to publication metadata from OpenAlex and investigator histories from Dimensions.ai. The NIH Exporter provides per-grant metadata (activity codes, study-section identifiers, project identifiers) and maps NIH project numbers to PubMed IDs, thereby linking grants to publications. Publication records are enriched with OpenAlex data (canonical IDs, journal names and ISSNs, citation counts, parsed authorship) and journal-level metrics, including two-year mean citedness and h-index. Dimensions supplies investigator metadata (identifiers, affiliation, years of activity) and records of all grants received by each investigator,

Table 2. Data variables and their sources

Variable	Description	Source
$A_{d,t}$	Total number of applications received by institute d in year t	NIH Data Book
$\pi_{d,t}$	Aggregate application success rate (funded ÷ applications) for institute d in year t	NIH Data Book
$G_{d,t}$	Total budget disbursed by institute d in year t	NIH R01 grant data
$\bar{g}_{d,t}$	Average grant size for institute d in year t	NIH Reporter
$F_{od,t}$	Number of funded grants from origin o to destination d in year t	Dimensions.ai investigator history
y_{iod}	Publication output of funded grants from o to d in year t	PubMed publications
$N_{o,t}$	Size of the potential applicant pool from origin o in year t (unique first authors of publications in year t funded by origin o grants)	Constructed from publication-grant linkages
D_{od}	Scientific distance between origin o and destination d (one minus cosine similarity)	Constructed from study-section funding

Notes: The table defines the variables used to estimate the application, funding, and production systems. Origin field o and destination institute d range over the $C = 19$ core NIH institutes. Year is indexed by t. Monetary values are converted to constant 2017 US dollars before estimation.

including non-NIH grants. When Dimensions lacks investigator metadata, I use the OpenAlex last-author identifier. The grant–publication mappings also yield the publication stocks used for the knowledge input. These stocks count only publications linked to R01 (R01-equivalent) awards and start in 1980, so that the stock entering the sample window in 2004 reflects prior knowledge production.

Several restrictions define the sample. I use the initial NIH sample, downloaded via the NIH Exporter API in December 2024, and restrict it to individual R01 (and R01-equivalent) awards. I exclude grants not active as of December 2024 or with a start year after 2022. I keep the core NIH institutes listed in Table A.1 and exclude smaller NIH centres (e.g., the Fogarty International Center) to improve comparability. I also exclude the National Institute on Minority Health and Health Disparities (NIMHD) because it was only established in 2010. For simplicity, I refer to each institute by a short research-field label rather than its official NIH name; Table A.1 provides the mapping.

The sample begins in 2001 to ensure the best institute coverage, as the National Institute of Biomedical Imaging and Bioengineering (NIBIB) was only established in 2000. The estimation sample spans 2005–2018, although the underlying grant data cover 2001–2022. This window excludes the pre-2005 NIH budget-doubling period and the post-2018 disruption attributable to the COVID-19 pandemic.

Each observation in the final dataset is a unique investigator–grant pair. For grants funded by more than one IC, I sum per-IC contributions into a grant-level total and

assign the grant to the institute contributing the largest funding share. When a grant has multiple investigators, I split the award equally across them. I deflate all monetary values to 2017 constant dollars using the FRED GDP deflator. After cleaning, the sample contains 106,664 investigator–grant pairs, 53,175 unique investigators, and 85,290 unique grants.

The applicant pool $N_{o,t}$ is the number of unique first authors of publications published in year t that are linked to grants whose origin is o. Counting first authors avoids double-counting PIs who hold multiple grants. It also anchors the pool to scientists active at the margin of applying rather than to incumbent grant holders, and therefore allows many scientists never to secure a grant.[12]

Institute-level application success rates come from NIH Data Book tables, merged by institute and reference year (National Institutes of Health, n.d.[b]). The NIH does not publish these rates for R01 grants specifically, so I proxy them with rates for all investigator-initiated mechanisms. This category includes R01s as well as smaller mechanisms (R03, R21) and mechanisms targeting under-served institutions (R15). Non-R01 mechanisms typically have higher success rates than pure R01s, so the proxy likely overstates R01 funding probabilities. What matters here, however, is the relative success rate across institutes, not its level. The bias is therefore not problematic so long as it affects every field in the same way.

5.2. Adapting the Model for Estimation

Three changes help the model of Section 3 fit the data. The first two improve the realism of the model while preserving its core structure and tractability. The third imposes the normalisations required for identification.

Time-varying outside option. I allow the outside option to vary over time and estimate it as $c_t = c_0 + c_1 t$. The time trend c_1 lets the model accommodate secular changes in the overall propensity to apply, arising, for example, from changes in career incentives or non-NIH funding opportunities.

Distance-dependent evaluation noise. Grant reviewers in a given field may find it harder to evaluate proposals from scientifically distant fields. I allow evaluation noise to depend on the scientific distance D_{od} between the origin field and destination field:

$$\eta_{od} = \eta_0 \exp(\gamma D_{od}). \tag{26}$$

[12] In the small number of years where the implied aggregate pool $\sum_o N_{o,t}$ does not strictly exceed total applications A_t, I scale all origin counts $N_{o,t}$ up by a common yearly factor so that $\sum_o N_{o,t} > A_t$, preserving relative origin shares. The mass of non-applicants from origin o in year t is $N_{o,t} - \sum_d F_{od,t}/\pi_{d,t}$, where $F_{od,t}/\pi_{d,t}$ approximates the total number of applications from o to d.

With $\gamma \geq 0$, noise rises with distance. This is economically motivated: grant reviewers in a given field may face greater difficulty evaluating proposals from scientifically distant fields. The parametrisation does not force field-specific noise. If $\hat{\gamma} \approx 0$, the model reduces to common evaluation noise; under the normalisation below, $\eta_{od} = 1$ for all origin–destination pairs.

Normalisation. The application and funding probabilities are invariant to destination-specific shifts in μ_{od} and to a common rescaling of all idea quality distributions. I therefore impose

$$\sum_{o=1}^{C} \mu_{od} = 0 \quad \text{for every } d, \qquad \eta_0 = 1. \tag{27}$$

The first restriction centres mean idea quality within each destination. The second measures idea quality in units of baseline evaluation noise.

Scientific distance. How far apart are two institutes scientifically? I measure distance by emulating the technology-proximity measure of firm distance in patent space (Bloom, Schankerman and Van Reenen, 2013). Instead of the overlap in patent portfolios between firms, I use the overlap in study-section funding between ICs. Let S_c be the vector of shares of institute c's funding allocated to each NIH study section in a given year. The proximity between institutes o and d is the cosine similarity of their share vectors:

$$\rho_{od} = \frac{S_o \cdot S_d}{\|S_o\| \|S_d\|}. \tag{28}$$

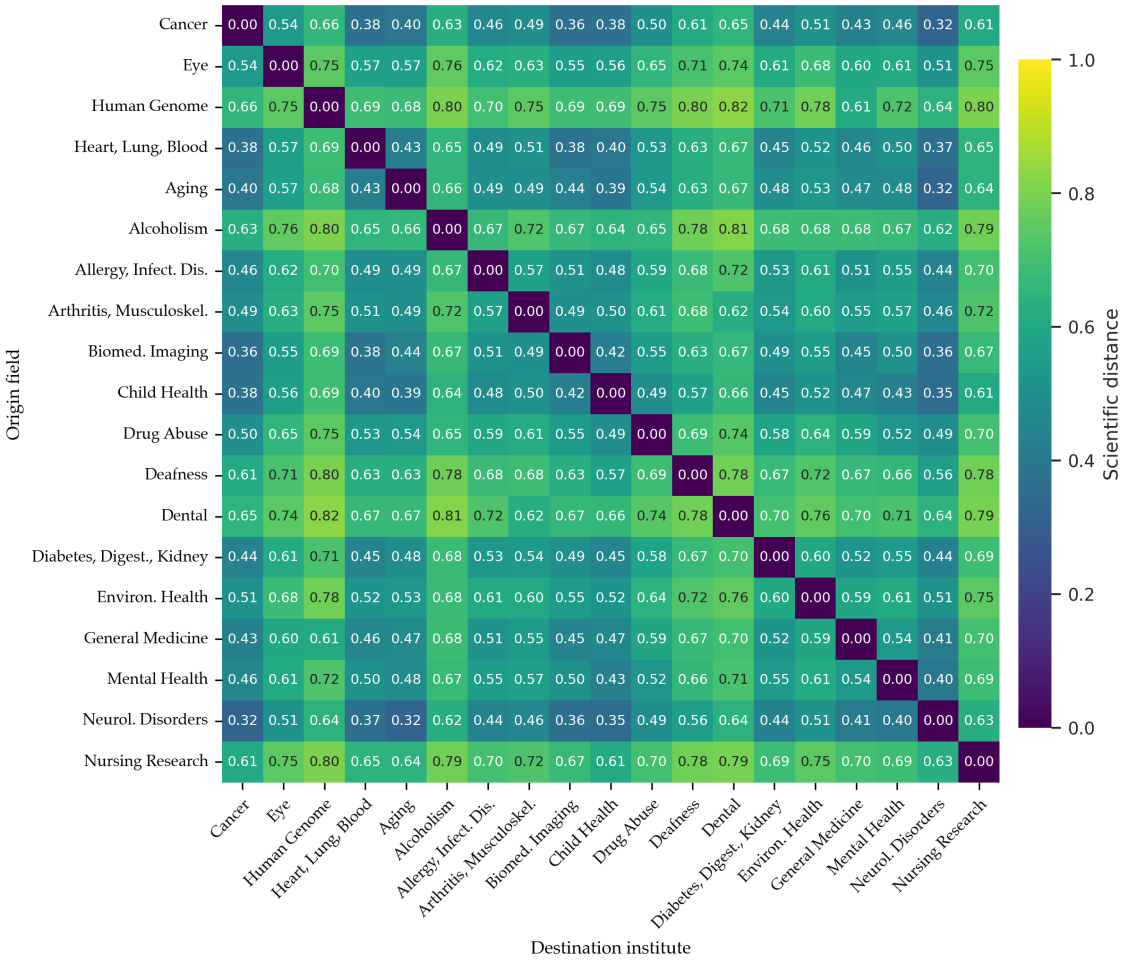
Scientific distance is one minus cosine similarity:

$$D_{od} = 1 - \rho_{od}. \tag{29}$$

Thus $D_{od} = 0$ denotes identical study-section funding shares and $D_{od} = 1$ denotes disjoint shares.

Figure 5 displays the resulting average distance matrix over the 2005–2018 estimation window.

Figure 5. Scientific distance between institutes, 2005–2018



Notes: Each cell reports scientific distance D_{od} between origin field o (row) and destination institute d (column), averaged over 2005–2018. Distance equals one minus the cosine similarity of the two institutes' study-section funding-share vectors. Values near zero indicate similar funding portfolios; larger values indicate greater scientific distance.

Table 3. Structural parameters

Parameter	Description
μ_{od}	Mean idea quality; $C \times C$ matrix with $\sum_o \mu_{od} = 0$ for each d
σ	Idea-quality scale of the Gumbel distribution; $\sigma > 0$
c_0	Intercept of the outside option
c_1	Time slope of the outside option; $c_t = c_0 + c_1 \cdot t$
η_0	Baseline evaluation noise standard deviation, fixed to one
γ	Distance gradient of evaluation noise; $\eta_{od} = \eta_0 \exp(\gamma D_{od})$, constrained $\gamma \geq 0$

Notes: The $C \times C$ matrix μ_{od} governs mean idea quality for each origin–destination pair and contains $C(C-1)$ free contrasts after destination-column centring. The estimator fits σ, c_0, c_1, and γ. The restriction $\eta_0 = 1$ fixes the latent-quality unit and is not an estimated coefficient.

5.3. Moment Conditions

I match two broad blocks of model-implied moments to their observed analogues. The flow-share block contains three disjoint moment sets. All moment conditions are expressed as normalised (relative) deviations, placing observations with different scales on a common metric. Throughout, estimated model-implied quantities are written with hats and observed quantities without hats.

Aggregate applications. The model predicts the total number of applications received by institute d in year t as:

$$\hat{A}_{d,t}(\theta) \;=\; \sum_{o=1}^{C} N_{o,t}\, \Pi_{od,t}(\bar{x}_t(\theta); \theta). \tag{30}$$

I match this to the observed application count $A_{d,t}$ using a relative moment condition for each valid (d, t) cell:

$$m^{A}_{d,t}(\theta) \;=\; \frac{\hat{A}_{d,t}(\theta) - A_{d,t}}{A_{d,t}} \;=\; 0. \tag{31}$$

Expressing the gap as a fraction of observed applications protects against over-weighting large institutes.

Bilateral scientist-flow shares. The model predicts funded grants from origin o to destination d in year t as:

$$\hat{F}_{od,t}(\theta) \;=\; N_{o,t}\, \Pi_{od,t}(\bar{x}_t(\theta); \theta)\, S_{od}(M_{o,t}(\theta);\, \eta_{od}). \tag{32}$$

Because bilateral scientist flows are sparse at the annual frequency, I pool them within three disjoint sets of years: τ_1 = 2005–2009, τ_2 = 2010–2014, and τ_3 = 2015–2018. For each destination d, the observed share of origin o in epoch e is:

$$s_{od,e} \;=\; \frac{\sum_{t \in \tau_e} F_{od,t}}{\sum_{t \in \tau_e} \sum_{f} F_{fd,t}}, \tag{33}$$

and the corresponding moment condition is a relative (proportional) deviation:

$$m^{F}_{od,e}(\theta) \;=\; \frac{\hat{s}_{od,e}(\theta) - s_{od,e}}{s_{od,e}} \;=\; 0, \tag{34}$$

where $\hat{s}_{od,e}(\theta)$ is the analogous model-implied share using model-predicted flows pooled within epoch e. In each epoch, I retain an origin–destination pair only if

its observed share exceeds 0.1% and its scientist-flow count is at least five. These rules yield three flow-moment sets with 248, 262, and 261 moments, respectively. These moments are the primary source of identification for mean idea quality μ_{od}: conditional on total application volumes to each destination institute (disciplined by the application moments) and evaluation noise, the distribution of funded grants across origins identifies relative mean idea quality for any pair of origin fields (o, f).

5.4. GMM Estimation

The estimator matches 1,037 moments to 346 free parameters: 266 institute–year application moments, 771 bilateral flow-share moments across the three epochs, 342 mean-idea-quality parameters, and four scalars $(\sigma, c_0, c_1, \gamma)$. For each candidate θ, the equilibrium payline system is solved year by year before evaluating the moments. I estimate θ by two-stage block-weighted GMM.

Stage 1. For the first stage of the GMM, I use an equal-block weighting matrix that assigns total weight one to each of the two moment blocks and distributes that weight uniformly within block:

$$W_{rr}^{(1)} = \frac{1}{n_b} \quad \text{for moment } r \text{ in block } b, \tag{35}$$

where $b \in \{\text{applications, flows}\}$ and n_b is the number of moments in block b. This weighting allows each block to contribute equally to the first-stage estimates.

Stage 2. In the second stage, I reweight each block by the inverse of its first-stage residual variance:

$$\hat{v}_b = \mathrm{Var}_{r \in b}\left(m_r(\hat{\theta}_1)\right), \qquad W_{rr}^{(2)} = \frac{1}{\hat{v}_b} \quad \text{for moment } r \in \text{block } b. \tag{36}$$

The second-stage estimator minimises

$$J_2(\theta) = m(\theta)^\top W^{(2)} m(\theta), \tag{37}$$

starting from $\hat{\theta}_1$ without ridge regularisation. Appendix G provides more detail about the estimation procedure.

5.5. Estimating the Production Function

Dependent variable construction. Y_i denotes the dependent variable, with i indexing a unique grant–investigator observation and therefore fixing its origin, destination, and year. The observation is not a unique grant because grants can have multiple

investigators, although the vast majority of sampled grants have only one investigator (see Figure A.3).

Y_i counts the publications funded by the grant in observation i. For multiple-investigator grants, the publication count is divided by the number of investigators to avoid double-counting. When multiple grants fund one publication, it is split equally across the contributing grants, so each receives $1/n$ of a publication. Each publication is then weighted according to the output measure. Hence Y_i is closer to a continuous variable than a count variable. Appendix I gives the exact outcome construction.

Independent variable and functional form. I estimate the production function by negative binomial maximum likelihood to account for overdispersion that would invalidate a standard Poisson approach (see Figure A.3(c)). Non-negativity constraints are imposed on all destination-specific knowledge and funding elasticities $\alpha_d \geq 0$ and $\beta_d \geq 0$ to avoid economically absurd results. In the raw-publication specification, the funding-elasticity bound binds only for Dental and Craniofacial Research. Appendix J details the estimator and how standard errors are computed at the boundary. The estimating equation is:

$$\mathbb{E}[Y_i] = \exp\big(\phi_d + \tau_t + \alpha_d \ln k_i + \beta_d \ln g_i + \lambda \ln Q_{od,t}\big), \tag{38}$$

where k_i is the knowledge stock available to observation i, g_i is its realised grant funding, ϕ_d and τ_t are destination and year fixed effects, and α_d, β_d are destination-specific knowledge and funding elasticities.

The term $\ln Q_{od,t}$ is a structural control for applicant selection. The selection object is the conditional expected idea quality among funded scientists, shifted to ensure non-negativity:

$$\begin{aligned} Q_{od,t} &\equiv \mathbb{E}\big[a_i - \underline{a} \mid \text{applied to } d, \text{ funded, } t\big] \\ &= \frac{1}{S_{od,t}} \int (z + \bar{x}_{d,t} - \underline{a})\, \Phi\!\left(\frac{z}{\eta_{od}}\right) f_Z(z;\, M_{o,t})\, dz, \end{aligned} \tag{39}$$

where $\underline{a} = \min_{o,d,z}(\mu_{od} + z)$ is a global floor computed over the quadrature support, ensuring $Q_{od,t} > 0$ for all pairs. The shift is a normalisation of the quality scale and remains implicit in the notation $Q_{od,t}$. The object is computed from the estimated structural parameters. Including $\ln Q_{od,t}$ absorbs cross-cell variation in funded-project quality. The coefficients α_d and β_d are conditional elasticities of output with respect to knowledge and funding in this fitted specification. This is more parsimonious than a full set of origin-destination fixed effects whilst retaining structural discipline on the selection correction. The funding elasticity β_d and the knowledge elasticity α_d are both destination-specific.

I construct the grant-level knowledge stock k_i using field-specific depreciation rates and grant-level citation shares. Current research benefits from the accumulated output of related fields, depreciated over time. Rather than imposing a common depreciation rate across all fields, I estimate rates δ_f from the age distribution of references to cited field f (details in Appendix F). The stock used in year t is measured before that year's output is produced.

Following Acemoglu, Akcigit and Kerr, 2016, I define the knowledge input to grant i as a citation-weighted aggregate of field-level stocks. Let χ_i be its vector of cited-field shares, normalised to sum to one. Each share is computed by tracing every cited publication back to the institute that funded it. This construction follows the logic of shift-share instruments, where each grant is differentially exposed to a common shock (Borusyak, Hull and Jaravel, 2024), and exploits within-field heterogeneity in citation profiles across grants. The knowledge stock is a Cobb–Douglas aggregate of field-level knowledge stocks:

$$k_i = \prod_{f=1}^{C} \left(1 + K_{f,t-1}\right)^{\chi_{if}}, \tag{40}$$

where $K_{f,t-1}$ is the depreciated publication stock of cited field f, computed using δ_f. Output produced in year t enters $K_{f,t}$. The log knowledge stock entering the production function is therefore $\ln k_i = \sum_{f=1}^{C} \chi_{if} \ln(1 + K_{f,t-1})$.

To address potential endogeneity, I construct citation shares using only publications that were cited and published at least one year before the grant's start date. This ensures that χ_i is predetermined and not a function of the grant's output, avoiding the reverse causality whereby higher-output grants mechanically cite more publications.

6. Results

This section presents the model's estimates and fit. It reports and decomposes estimated marginal returns by institute, then examines why changes in scientist flows contribute little to those returns.

6.1. Model Estimates and Fit

I start with the parameters of the application and review model. Table 4 reports the four estimated scalar parameters and their standard errors. The estimated idea-quality scale is $\hat{\sigma} = 0.3635$. The outside option starts at $\hat{c}_0 = -2.0024$ and rises by $\hat{c}_1 = 0.0150$ per year. The estimated distance gradient of evaluation noise is $\hat{\gamma} = 1.5777$, suggesting that evaluation noise is larger for applications to more distant fields. Appendix Figure A.1 reports the full 19×19 matrix of mean idea quality $\hat{\mu}_{od}$.

Mean idea quality depends on the direction of the move. The off-diagonal entries

of $\hat{\mu}_{od}$ range from −1.532 to 1.830. The largest positive mean idea quality runs from Human Genome to Aging; the smallest runs from Nursing Research to Eye. These values compare origin fields within a destination. The destination-column parameter restriction rules out comparisons of mean idea-quality levels across destinations.

Table 4. GMM scalar parameter estimates

Parameter	Estimate	SE
σ (idea-quality scale)	0.3635	0.1053
c_0 (outside option intercept)	-2.0024	0.3423
c_1 (outside option time slope)	0.0150	0.0070
η_0 (baseline evaluation noise; fixed)	1.0000	fixed
γ (distance slope in η_{od})	1.5777	0.4116

Notes: The table reports four estimated scalars from the two-stage GMM estimator over 2005–2018, their standard errors, and the fixed normalisation $\eta_0 = 1$. The scale σ governs idiosyncratic idea-quality shocks; c_0 and c_1 determine the outside option; and γ determines the distance gradient of evaluation noise. Appendix G.2 specifies the covariance estimator.

Within every destination, incumbents have better ideas than the average entrant. Every diagonal entry $\hat{\mu}_{dd}$ is positive (Table 5). The destination-column restriction $\sum_o \hat{\mu}_{od} = 0$ forces each column to average to zero. A positive $\hat{\mu}_{dd}$ therefore implies that incumbent scientists (those whose origin field matches destination d) have ideas of higher quality than the average entrant into d. The magnitude of $\hat{\mu}_{dd}$ measures the incumbent's quality gap relative to the mean across all origins. This value is largest in Human Genome ($\hat{\mu}_{dd} = 2.622$), Deafness and Communication Disorders (2.120), and Nursing Research (1.633), and smallest in Aging (0.731) and Cancer (0.876). More niche and specific fields seem to have larger gaps. This may reflect the larger advantage of incumbents embedded in more specialised research areas.

Figure 6 compares predicted and observed values for the two targeted blocks. Across 266 institute–year cells, the application fit has an RMSE of 3,041.9 applications, a MAPE of 12.616 per cent, and an R^2 of 0.887. Across 771 scientist-flow shares pooled separately within the three epochs, the fit has an RMSE of 0.0230 in share units (2.30 percentage points), a MAPE of 15.940 per cent, and an R^2 of 0.974. The plot shows underprediction in the largest application cells. These cells all belong to Cancer after 2012, suggesting that they are a set of outliers.

The model also predicts a moment it does not target: the aggregate application success rate. For institute d in year t, this is the ratio of total funded grants to total applications:

$$\hat{\pi}_{d,t}(\theta) \;=\; \frac{\sum_o \hat{F}_{od,t}(\theta)}{\hat{A}_{d,t}(\theta)}. \tag{41}$$

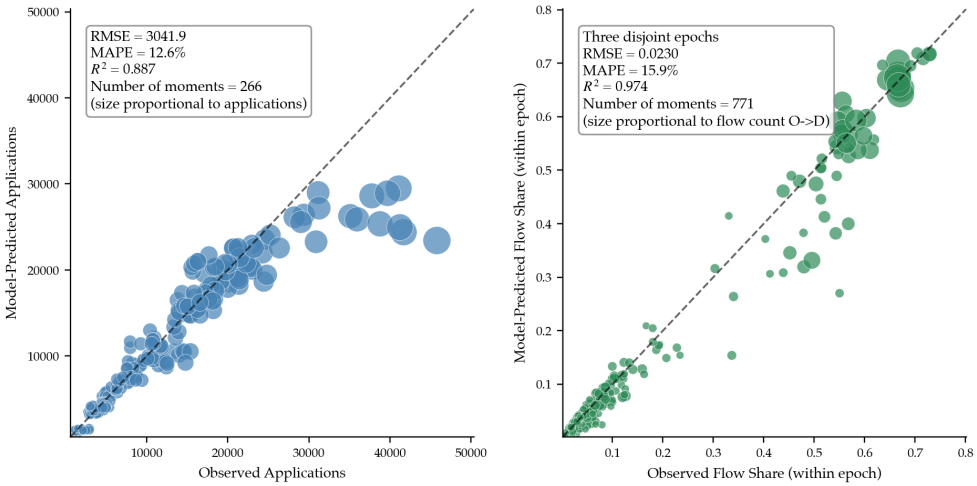
Panel (b) of Figure 7 compares predicted and observed success rates. The RMSE is 0.0297, the MAPE is 14.557 per cent, and the R^2 is 0.488. Here some of the discrepancy seems to arise from the static nature of the model. The range of predicted success rates

Table 5. Mean idea quality of incumbents ($\hat{\mu}_{dd}$)

Institute	Mean idea quality ($\hat{\mu}_{dd}$)
Human Genome	2.622
Deafness and Communication Disorders	2.120
Nursing Research	1.633
Eye	1.606
Alcoholism	1.595
Dental and Craniofacial Research	1.465
Environmental Health	1.326
Drug Abuse	1.261
Allergy and Infectious Diseases	1.224
Biomedical Imaging and Bioengineering	1.049
Child Health and Human Development	1.020
Mental Health	0.988
Neurological Disorders and Stroke	0.953
General Medicine	0.944
Diabetes and Digestive and Kidney Diseases	0.940
Heart, Lung, and Blood	0.932
Arthritis and Musculoskeletal and Skin Diseases	0.931
Cancer	0.876
Aging	0.731

Notes: The table reports the diagonal entries $\hat{\mu}_{dd}$ of the estimated mean-idea-quality matrix, ordered from largest to smallest. Each entry measures the incumbent-origin mean relative to the equal-weight mean across all origins in destination d; that column mean is normalised to zero.

Figure 6. Model fit: targeted moments



Notes: The left panel compares model-predicted and observed institute–year application totals. The right panel compares model-predicted and observed bilateral origin–destination flow shares pooled separately over 2005–2009, 2010–2014, and 2015–2018. These are the three disjoint moment sets used by the GMM. Both panels show targeted moments. The dashed 45-degree line denotes perfect fit. Marker area is proportional to the underlying application or scientist-flow count.

(13–25 per cent) is much narrower than the observed range (5–30 per cent). This likely reflects the static nature of the model that returns to equilibrium instantaneously after a budget shock. In practice, success rates are not arbitraged away as quickly and therefore can fluctuate more widely than in the model.

To further validate the model, I hold the estimated parameters fixed and recompute applications and scientist-flow shares for 2002–2004 and 2019–2022. Application MAPE is 12.261 per cent in the years before the estimation window and 17.183 per cent in the years after it. The pooled untargeted application moments have a MAPE of 15.073 per cent and an R^2 of 0.830. Pooled untargeted flow-share moments have an RMSE of 0.0243 and an R^2 of 0.964, but a MAPE of 101.016 per cent.

Why is the flow-share MAPE so high when the R^2 is close to one? The MAPE divides each error by the observed share, without weights. Rare positive flow shares near the 0.1 per cent threshold therefore generate large percentage errors from small absolute deviations. A high MAPE can reflect large relative errors for a handful of rare origin–destination combinations rather than an overall failure of the model to capture the general pattern of scientist flows.

Finally, the model recovers the paylines that clear each grant market. Appendix Figure A.2 plots the model-implied equilibrium paylines $\bar{x}_{dt}$ over the sample period. They generally tighten over 2008–2013, reflecting an increasingly competitive NIH funding environment, and ease over 2014–2018. This pattern is broadly consistent with observed trends in overall application success rates at the NIH.

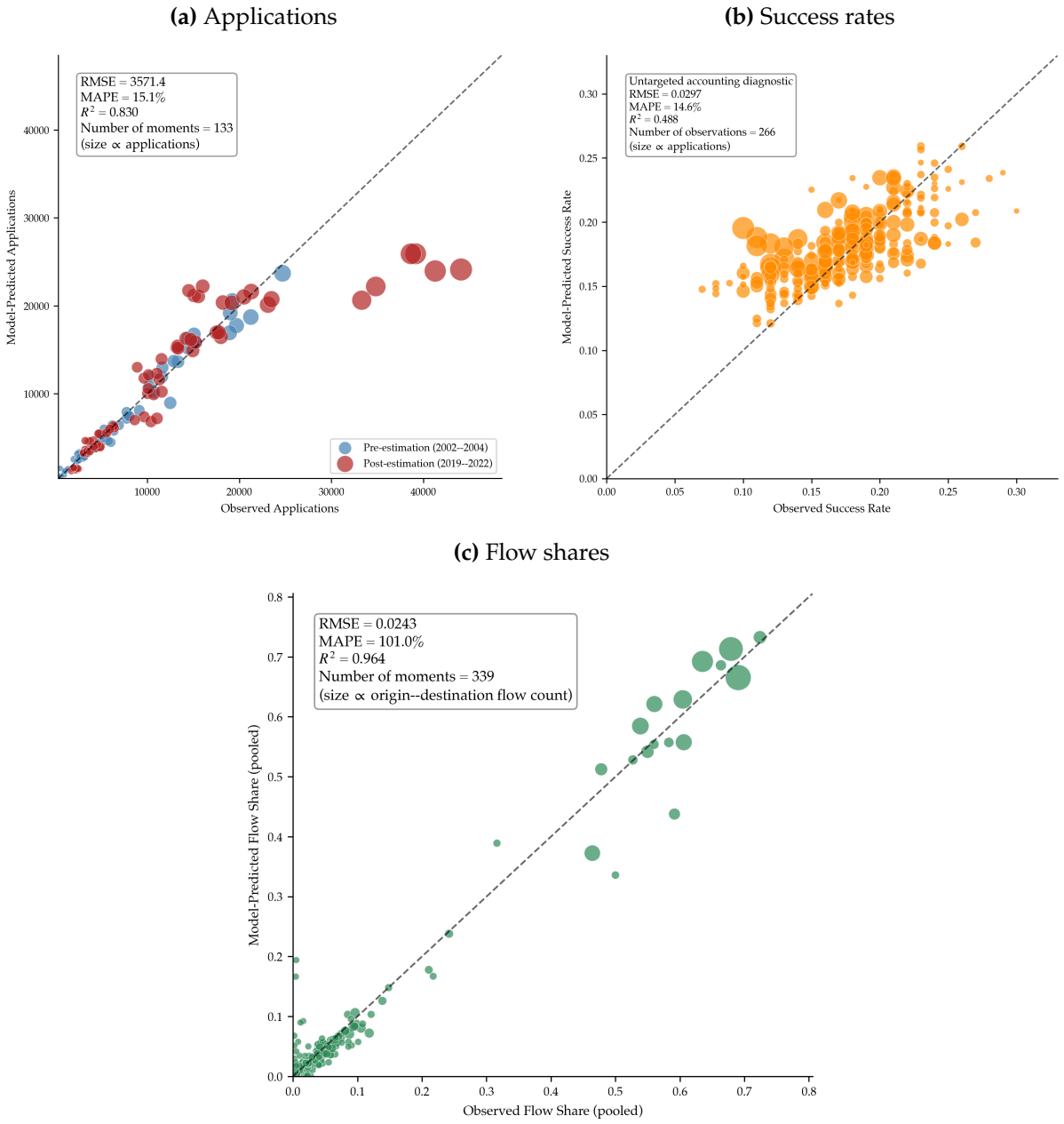
6.2. Marginal Returns Estimates

I estimate marginal returns in raw publications. The marginal return combines the current output gain in funded field d, the talent-flow effect in other fields, and discounted own-field and cross-field knowledge spillovers. Only the current output change in d is propagated through the knowledge channel. Appendix H describes the numerical calculation, and Appendix H.1 reports the return intervals.

Figure 8 reports these components at the 2018 R01 allocation. Point estimates and pointwise 95 per cent confidence intervals sit at the tip of each bar. Knowledge spillovers include future depreciated output, discounted at 5 per cent per year. The field-specific lifetime multipliers range from 4.78 to 5.71.

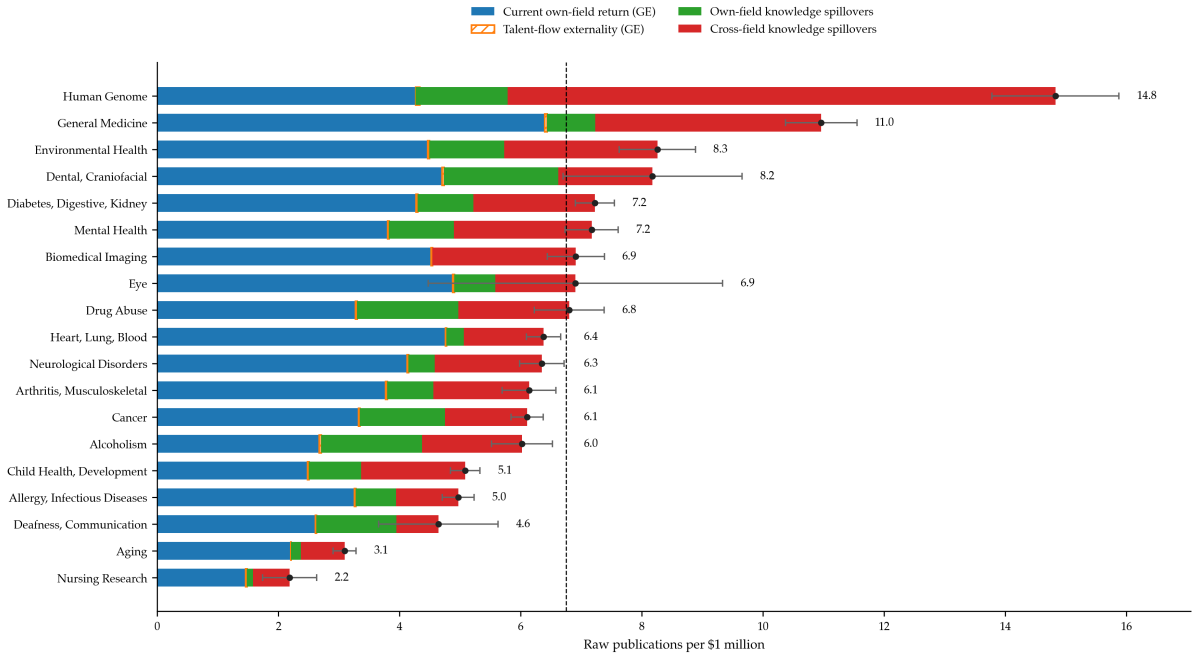
Knowledge spillovers account for 71.2 per cent of Human Genome's raw-publication return, including 61 per cent generated in other fields (Table 6). Cross-field spillovers exceed own-field spillovers for 15 of the 19 institutes.

Figure 7. Model fit: untargeted moments



Notes: Structural parameters are estimated on 2005–2018 and then held fixed. The vertical axis reports model predictions and the horizontal axis reports observations; the dashed 45-degree line denotes perfect fit. Panels (a) and (c) report untargeted application and bilateral flow-share moments in 2002–2004 and 2019–2022. Panel (b) reports the in-sample success-rate accounting diagnostic, which is not a targeted GMM moment. Blue and red markers distinguish the pre-estimation years 2002–2004 from the post-estimation years 2019–2022 in panels (a) and (c). Marker area is proportional to applications in panel (a), applications in panel (b), and raw scientist-flow counts in panel (c).

Figure 8. Marginal returns estimates, 2018



Notes: The bars decompose the additional raw publications from a $1 million increment at the indicated institute into the current own-field return, general-equilibrium talent-flow externality, own-field knowledge spillover and cross-field knowledge spillover. The dashed line marks the cross-institute mean; dots and whiskers report point estimates and pointwise 95 per cent confidence intervals.

Table 6. Share of the raw-publication marginal return from knowledge spillovers, by institute.

Institute	Own-field knowledge spillovers	Cross-field knowledge spillovers	Total knowledge spillovers
Human Genome	10.2%	61.0%	71.2%
Alcoholism	27.6%	27.4%	55.1%
Drug Abuse	24.6%	26.9%	51.5%
Child Health and Human Development	17.0%	33.8%	50.9%
Mental Health	15.0%	31.7%	46.7%
Environmental Health	15.0%	30.6%	45.6%
Cancer	23.1%	22.2%	45.3%
Deafness and Communication Disorders	28.4%	15.0%	43.4%
Dental and Craniofacial Research	23.0%	19.1%	42.1%
General Medicine	7.3%	34.0%	41.3%
Diabetes and Digestive and Kidney Diseases	12.7%	27.8%	40.6%
Arthritis and Musculoskeletal and Skin Diseases	12.5%	25.7%	38.2%
Neurological Disorders and Stroke	6.9%	27.8%	34.8%
Biomedical Imaging and Bioengineering	0.0%	34.3%	34.3%
Allergy and Infectious Diseases	13.4%	20.7%	34.1%
Nursing Research	4.7%	27.6%	32.3%
Eye	9.8%	19.2%	29.0%
Aging	5.3%	23.4%	28.7%
Heart, Lung, and Blood	4.5%	20.6%	25.1%

Notes: Each row is a funded institute at the 2018 R01 allocation. The first two numeric columns report own-field and cross-field knowledge spillovers as percentages of its total marginal return to a $1 million increment. Their sum is the total knowledge-spillover share.

6.3. Why the Talent-Flow Externality Is Small

Scientists follow funding, yet their moves barely change output in the fields they leave. PIs frequently change funder (Figure 1). In the model, a budget increase at d raises the share of its grants awarded to scientists from other fields in 17 of the 19 institutes, even after all paylines clear (Table 7). Other fields therefore lose scientists to d. Yet the general-equilibrium (GE) talent-flow externality is close to zero for every institute (Figure 8). Three features explain this result: payline adjustment, filtering at application and review, and the identity of the scientists who leave.

Table 7. Composition of funded grants and its budget elasticity

Institute	Baseline off-field share	PE Elasticity	GE Elasticity	GE largest off-field origin
Allergy and Infectious Diseases	32.7%	0.321	0.287	General Medicine
Deafness and Communication Disorders	26.3%	0.313	0.308	Neurological Disorders and Stroke
Eye	27.2%	0.291	0.283	Neurological Disorders and Stroke
Cancer	35.8%	0.194	0.162	General Medicine
Heart, Lung, and Blood	31.5%	0.190	0.160	Diabetes, Digestive, and Kidney Diseases
General Medicine	40.2%	0.177	0.146	Cancer
Neurological Disorders and Stroke	45.2%	0.149	0.131	Mental Health
Diabetes, Digestive, and Kidney Diseases	41.5%	0.129	0.108	Heart, Lung, and Blood
Drug Abuse	44.0%	0.109	0.095	Mental Health
Alcoholism	48.3%	0.098	0.090	Drug Abuse
Dental and Craniofacial Research	45.3%	0.098	0.093	Cancer
Child Health and Human Development	51.9%	0.091	0.080	Mental Health
Environmental Health	47.8%	0.080	0.074	Cancer
Mental Health	42.0%	0.074	0.056	Drug Abuse
Arthritis, Musculoskeletal, and Skin Diseases	58.5%	0.039	0.029	Eye
Nursing Research	59.5%	0.032	0.030	General Medicine
Biomedical Imaging and Bioengineering	66.3%	0.011	0.007	Cancer
Aging	65.2%	0.006	-0.008	Human Genome
Human Genome	58.9%	-0.010	-0.007	General Medicine

Notes: Each row is a funded institute d. The off-field share is the percentage of funded grants at d awarded to scientists whose origin field differs from d. The first numeric column reports its model-implied value in the 2018 baseline equilibrium, before the budget shock. This baseline can differ from the observed 2018 share as the model's fit is not perfect. The elasticities report the percentage change in this baseline share after a 1 per cent increase in G_d. Partial-equilibrium (PE) elasticities hold all other paylines fixed; general-equilibrium (GE) elasticities allow all paylines to clear. The final column identifies the non-d origin field contributing the largest GE increase in funded scientists at d.

Payline adjustment absorbs almost all of the loss. If paylines in fields $o \neq d$ stayed fixed, a \$1 million increment at d would cost these fields between 0.988 and 4.215

publications (Table 8). This is the partial-equilibrium (PE) externality. Once all paylines clear, each field $o \neq d$ again awards the same number of grants, and the GE externality lies between −0.065 and 0.039 publications. The scientists who replace the movers are therefore, on average, as productive as the movers. Two questions follow: why are the replacements as productive, and why is the GE externality slightly positive for most institutes?

Table 8. Decomposition of the talent-flow externality

Institute	Talent-flow externality (PE)	Talent-flow externality (GE)	GE: lower paylines	GE: composition	Returnee share	Returnee quality gap
Human Genome	-4.040	-0.065	-0.015	-0.050	71.5%	0.727
General Medicine	-4.215	0.039	-0.016	0.055	80.5%	-0.767
Environmental Health	-2.829	0.031	-0.011	0.042	70.3%	-0.890
Dental and Craniofacial Research	-2.922	0.038	-0.012	0.050	77.8%	-0.684
Diabetes and Digestive and Kidney Diseases	-2.264	0.028	-0.009	0.037	85.2%	-0.815
Mental Health	-2.221	0.026	-0.009	0.035	71.2%	-0.647
Biomedical Imaging and Bioengineering	-3.537	0.019	-0.014	0.033	61.4%	-0.536
Eye	-2.717	0.030	-0.011	0.040	78.6%	-0.617
Drug Abuse	-2.094	0.029	-0.008	0.038	65.9%	-0.666
Heart, Lung, and Blood	-1.469	0.019	-0.005	0.024	86.5%	-0.732
Neurological Disorders and Stroke	-2.285	0.017	-0.009	0.026	69.1%	-0.586
Arthritis and Musculoskeletal and Skin Diseases	-2.786	0.029	-0.011	0.040	65.5%	-0.944
Cancer	-1.866	0.019	-0.007	0.025	80.5%	-0.699
Alcoholism	-2.157	0.035	-0.009	0.043	67.0%	-0.737
Child Health and Human Development	-2.198	0.018	-0.008	0.027	72.1%	-0.635
Allergy and Infectious Diseases	-1.788	0.018	-0.007	0.024	82.6%	-0.647
Deafness and Communication Disorders	-2.214	0.020	-0.009	0.029	79.8%	-0.439
Aging	-0.988	0.006	-0.004	0.010	49.0%	-0.845
Nursing Research	-1.640	0.023	-0.007	0.030	74.9%	-0.763

Notes: Each row is a funded institute at the 2018 R01 allocation. Talent-flow externalities and their two GE components are changes in raw publications per $1 million; PE holds other paylines fixed and GE allows them to clear. Returnee share is the percentage of lost grants elsewhere held by scientists originating in the funded institute. Returnee quality gap compares their funded-project quality with the average in the fields they leave.

Filtering helps answer the first question. Quality differences between origin fields may shrink between application and funding. Scientists do not apply unless their idea is good enough, and reviewers reject weak applications. Either filter would make funded scientists from different origins more alike than their ideas were before application.

Table 9 tests this prediction. The two stages measure dispersion on different scales, so I divide each cross-origin standard deviation by the total standard deviation across

individual scientists. Before application, this relative standard deviation compares $\hat{\mu}_{od}$ across origin fields with the dispersion of individual idea quality $\mu_{od} + \varepsilon_i$. After funding, it compares mean publications per PI across the same origins with their dispersion across funded grants.

The gap between the two stages is large. The median relative standard deviation falls from 0.687 before application to 0.147 after funding. Origin fields thus account for 47 per cent of the variance in idea quality but for 2 per cent of the variance in realised output. The gap is positive in all 19 institutes. It is largest in Human Genome (0.816), Deafness and Communication Disorders (0.755) and Nursing Research (0.731), and smallest in Aging (0.338) and Diabetes and Digestive and Kidney Diseases (0.404). Large differences in expected idea quality across origin fields *ex ante* translate into small differences in productivity *ex post*.

Table 9. Cross-origin dispersion in idea quality and funded output

Institute	Relative cross-origin SD		Gap
	Before application: idea quality	After funding: publications per PI	
Human Genome	0.924	0.108	0.816
Deafness and Communication Disorders	0.845	0.090	0.755
Nursing Research	0.811	0.080	0.731
Drug Abuse	0.773	0.135	0.638
Dental and Craniofacial Research	0.755	0.161	0.594
General Medicine	0.677	0.089	0.588
Alcoholism	0.792	0.205	0.586
Environmental Health	0.689	0.143	0.546
Eye	0.774	0.230	0.545
Biomedical Imaging and Bioengineering	0.687	0.156	0.532
Allergy and Infectious Diseases	0.665	0.153	0.512
Cancer	0.618	0.133	0.485
Mental Health	0.709	0.244	0.465
Heart, Lung, and Blood	0.608	0.167	0.441
Arthritis and Musculoskeletal and Skin Diseases	0.582	0.145	0.437
Child Health and Human Development	0.583	0.147	0.436
Neurological Disorders and Stroke	0.588	0.181	0.407
Diabetes and Digestive and Kidney Diseases	0.550	0.146	0.404
Aging	0.505	0.167	0.338
Median	0.687	0.147	0.532

Notes: Each row is a destination institute d. The sample contains the origin fields o with at least 20 funded grant–PI observations at d over 2005–2018, which gives 231 origin–destination cells and 54,732 observations. Each relative standard deviation equals $\sqrt{V_O/(V_O + V_W)}$, where V_O is the variance of origin means across origins and V_W is the mean variance within origins. Before application, the origin mean is the estimated mean idea quality $\hat{\mu}_{od}$ and $V_W = \pi^2\hat{\sigma}^2/6$ is the variance of the Gumbel idea-quality shock. After funding, both variances are computed from publications per PI of funded grants, where each grant's output is divided by the mean output of its destination and fiscal year. The square of each relative standard deviation is the share of individual-level variance accounted for by origin fields. The gap is the before-application value minus the after-funding value. Institutes are ordered by this gap.

Filtering does not answer the second question. The answer lies in who leaves the

other fields. Section 2.3 argued that the scientists who move are those at the margin between fields. If so, a change of funder need not be a permanent change of field. A scientist whose funding moves from, say, Aging to Cancer may simply work at the intersection of the two fields. Her research should then already resemble Cancer research before she switches.

Grant abstracts allow a model-free test of this prediction. I compare the grants of scientists who leave o for d (future switchers) with those of scientists who stay in o (matched stayers). For each grant, I measure the distance between its title and abstract and the centroid $c_{d,t}$ of all grants funded by d in year t, in a word-embedding space.[13]

Future switchers already work closer to their destination. Figure 9 shows that grants preceding a switch are on average 10 per cent closer to their eventual destination than the grants of those who stay.[14] The gap holds in every institute, though its size varies. Before the switch, future switchers' research is therefore more aligned with their destination than that of their peers. This is consistent with future switchers being more likely to operate at the margin between fields. Appendix B details the sample construction, embedding procedure and distance measures.

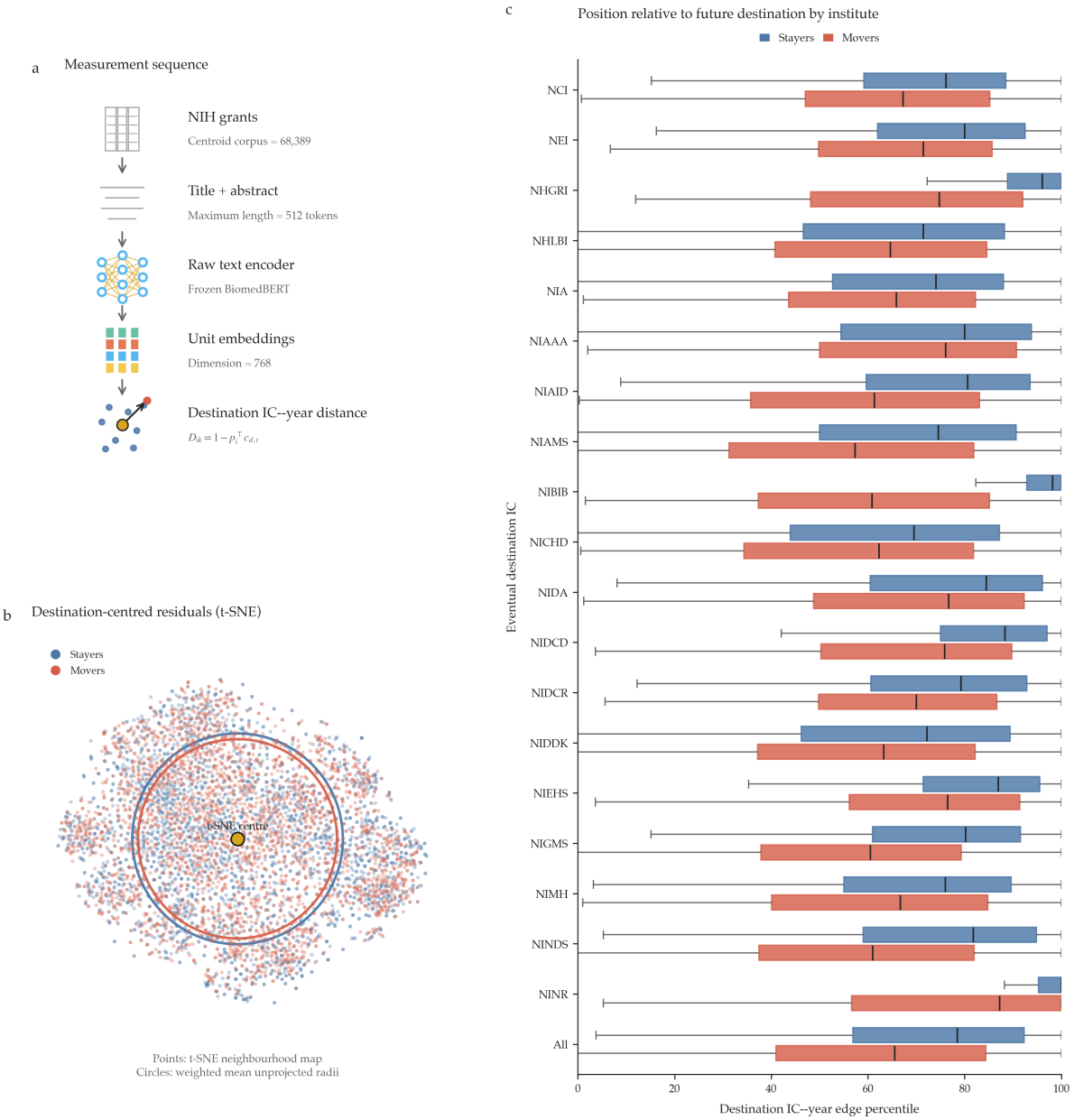
The model also ties leavers to their destination. Between 49 and 86.5 per cent of the grants that fields $o \neq d$ lose belong to scientists whose origin field is d (Table 8). These scientists had moved from d to o, and the larger budget at d draws them back; I call them returnees. In the fields they leave, their funded-project quality lies 0.44 to 0.94 units below the field average. Back in d, they join the incumbents, whose estimated mean idea quality exceeds the destination average in every institute (Table 5).

Returnees determine the sign of the GE externality. Table 8 splits it into two effects. Lower paylines in fields $o \neq d$ admit marginal projects and reduce output by 0.004 to 0.016 publications. The replacement of below-average returnees raises average funded-project quality, and this composition effect adds 0.010 to 0.055 publications. The GE externality is therefore small and positive for most institutes.

[13] I embed each grant's title and abstract with the BiomedBERT encoder, a version of BERT trained by Microsoft on biomedical vocabulary. The model is open source and available from the Hugging Face model page.

[14] The weighted mean distance is 0.004502 for future switchers and 0.005035 for matched stayers.

Figure 9. Semantic alignment with the future destination



Notes: The matched sample contains 8,989 future switchers across 3,110 origin–destination–year cells and 15,327 unique stayer grants from the corresponding origin–year cells. Panel (a) summarises the construction of 768-dimensional BiomedBERT embeddings and destination–year centroid distances. Panel (b) applies t-distributed stochastic neighbour embedding (t-SNE) to the distances between the centroid and the grants in the embedding space, for a deterministic balanced sample of 3,000 matched stayers and 3,000 future switchers. It illustrates that future switchers already operate close to their future fields before the switch. Panel (c) reports the distribution of each grant's distance percentile based on its distance from the destination institute's centroid for that year. A lower percentile indicates greater semantic similarity to the destination's typical research, whereas a higher percentile indicates greater distance.

7. Policy Implications

In this section I explore the implications of the marginal-return estimates for welfare and the allocative efficiency of NIH funding. I construct three welfare-relevant measures of scientific output and re-estimate the production function and marginal returns under each. Using these measures, I compare and evaluate different rules to allocate a $1 million NIH budget increment. Finally, I recover the weights that rationalise the existing NIH allocation.

7.1. From Publications to Welfare

Raw publication counts do not map one-for-one into welfare. Publications differ along many dimensions, such as scientific influence, commercial application, or relevance to population health. I therefore construct three measures of scientific output: academic impact, commercial potential, and disease burden. Throughout this section, $m \in \{\text{IF}, \text{CP}, \text{DB}\}$ indexes these measures. Each measure weights publications differently: the academic-impact measure uses journal impact factor; the commercial-potential measure uses the probability of receiving a patent citation; and the disease-burden measure uses the burden of the disease targeted by the underlying grant. This weighting scheme is analogous to the welfare weights described in Section 2.

The academic-impact measure weights each publication by its journal impact factor, treating publication in a more influential journal as having greater scientific value. I use the two-year impact factor from 2014 as computed in the OpenAlex dataset. Using journal impact factor rather than publication-level measures such as citations smooths differences in citation practices across fields and avoids overweighting older publications that have had more time to accumulate citations.

The commercial-potential measure weights each publication by the *ex ante* probability that the publication is cited by at least one renewed patent, estimated by Masclans, Hasan and Cohen (2023) using word embeddings of its title and abstract. Eighty-nine per cent of the publications in my sample are covered by Masclans, Hasan and Cohen (2023). Observations without a commercial-potential score are assigned a weight of zero. Their work provides a measure of commercial potential *ex ante*, which is preferable to *ex post* measures such as patent citations for this purpose. The large time lag between academic publication and patent citation would have overweighted older publications and biased the measure against more recent research.

The disease-burden measure differs from the other two. Throughout the paper, *disease burden* denotes disability-adjusted life years (DALYs) for the US population unless stated otherwise. Here, *average disease burden* denotes the mean disease burden of an institute's grant portfolio, whereas *marginal disease burden* denotes its marginal return

measured in disease-burden-weighted scientific output. I do not observe the disease burden of each publication. Instead, I link each grant to the burden of the disease it targets and apply the same disease-burden weight to all publications funded by that grant. This weighting places greater value on research directed towards conditions associated with greater loss of healthy life and more disability.

To map grants to diseases, I use the RCDC disease tags that the NIH attaches to each grant. I match these tags to the Global Burden of Disease taxonomy. I do so using the crosswalk manually curated in National Institutes of Health (n.d.[c]). For grants without a matched tag, I impute the mean matched burden among grants funded by the same institute. I use the overall matched mean if the institute has no matched grant. The same grant-level data serve a second purpose: aggregated to the institute level, they yield the average disease-burden profiles used to benchmark the funding allocation below. Appendix I gives the full outcome construction, including the RCDC–DALY mapping and imputation.

The disease-burden measure is constrained by incomplete grant–DALY linkage. Only 45 per cent of grants in the analysis sample have a direct RCDC–DALY match; 33.4 per cent have no RCDC tag and 21.6 per cent have tags without a match in the burden workbook. Some institutes have a match rate lower than 10 per cent. I impute each unmatched grant with the mean burden among directly matched grants funded by the same institute or centre, using the overall matched mean only when no institute mean exists. Appendix K describes the missing-link problem, the text classifier, and its limitations; Appendix I gives the exact original construction.

7.2. Marginal Returns by Output Measure

I estimate the production function separately for each of these three measures. Knowledge elasticities $\hat{\alpha}_d^m$ are positive for all 19 institutes and all three measures. Table 10 reports the mean and maximum destination-specific knowledge and funding elasticities. The knowledge elasticity ranges from 0.098 to 0.553 for academic impact, from 0.107 to 0.650 for commercial potential, and from 0.055 to 0.425 for disease burden. Funding elasticities $\hat{\beta}_d^m$ are smaller on average, and all estimates lie weakly below 0.3, indicating decreasing returns for both funding and knowledge. The non-negativity constraints bind only for selected funding elasticities: Dental and Craniofacial Research in all three specifications, and Alcoholism and Drug Abuse in the commercial-potential specification. Appendix J provides more details on the estimation and reports the full set of estimates.

I next estimate marginal returns for all three output measures. Figure 10 shows that their distribution across fields is broadly similar to the distribution for raw publications. Return rankings are remarkably stable: Human Genome ranks first and General

Table 10. Production-function elasticities by output measure

	Knowledge elasticity ($\hat{\alpha}_d$)		Funding elasticity ($\hat{\beta}_d$)	
Output measure	Mean	Max	Mean	Max
Academic impact	0.307	0.553	0.116	0.220
Commercial potential	0.375	0.650	0.078	0.234
Disease burden	0.212	0.425	0.104	0.260

Notes: Each row summarises the 19 destination-specific production-function estimates for one measure of scientific output. The table reports the cross-institute mean and maximum of the knowledge elasticity $\hat{\alpha}_d$ and funding elasticity $\hat{\beta}_d$. Appendix Table A.7 reports all institute-specific estimates.

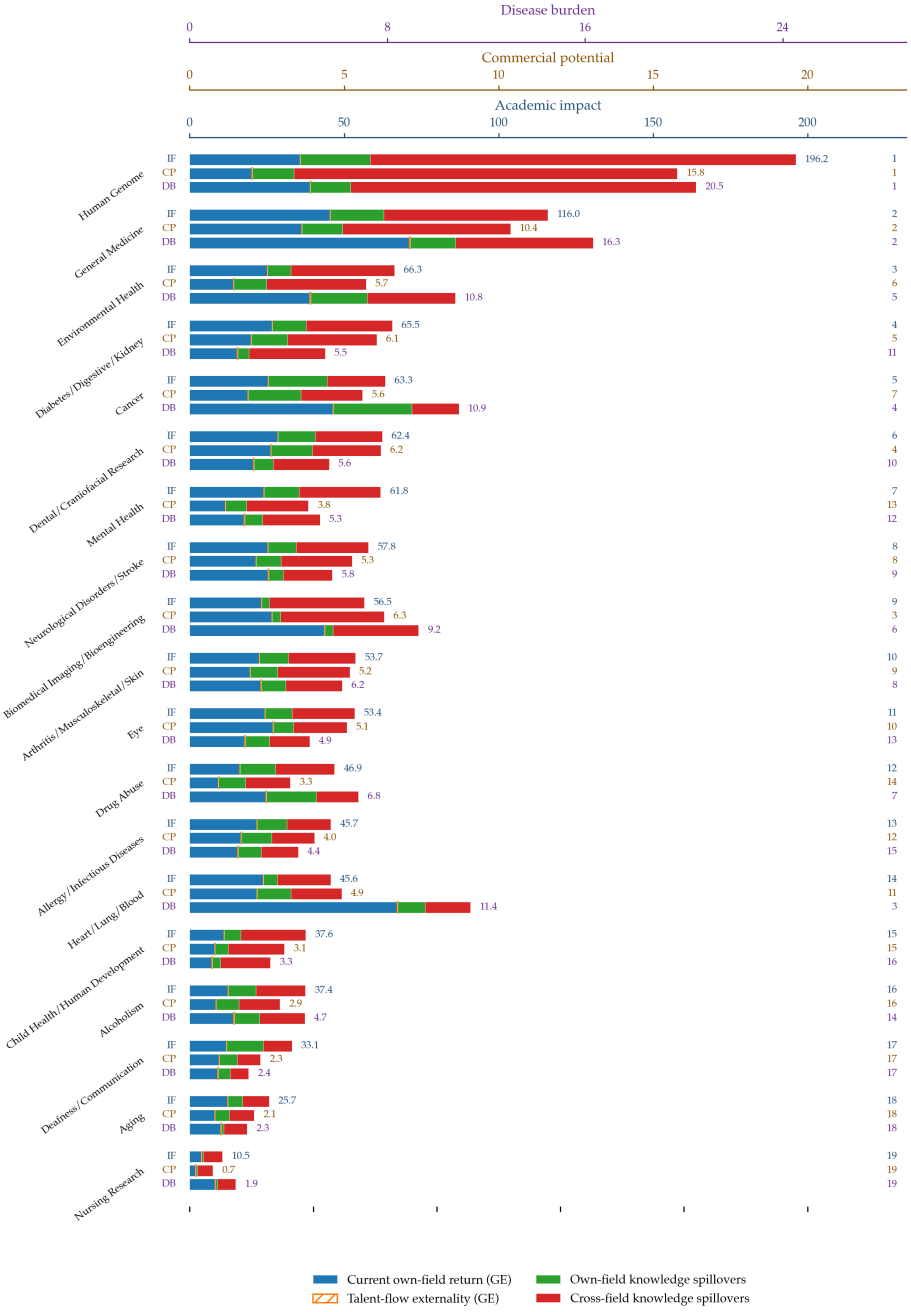
Medicine second under all three measures, while Nursing Research ranks nineteenth and Aging eighteenth. Rankings also vary little in the middle of the distribution.

This stability reflects that each measure retains publication counts as its quantity component and changes only the weights attached to publications. It is reinforced by the correlation between weighted publication outputs, especially for academic impact and commercial potential. Figure 11 reports the grant-level correlations among the measures: academic-impact and commercial-potential output correlate at 0.81, compared with 0.04 between disease burden and academic impact and 0.11 between disease burden and commercial potential. All correlations are positive. The close academic–commercial relationship is consistent with evidence that scientifically influential research is also linked to more valuable inventions (Marx and Fuegi, 2022; Poege et al., 2019) and is more likely to be used by inventors (Bikard, 2018; Bikard and Marx, 2020).

While most fields have similar returns across output measures, there are some notable exceptions. Heart, Lung, and Blood ranks third on disease burden, compared with fourteenth on academic impact and eleventh on commercial potential. Its relative performance reflects the very high burden of cardiovascular disease in the United States (GBD 2017 DALYs and HALE Collaborators, 2018). Dental and Craniofacial Research ranks fourth on commercial potential. Biomedical Imaging and Bioengineering ranks third on commercial potential, compared with ninth on academic impact and sixth on disease burden. Research in bioengineering and biomedical imaging lends itself to commercial application more than most other scientific fields.

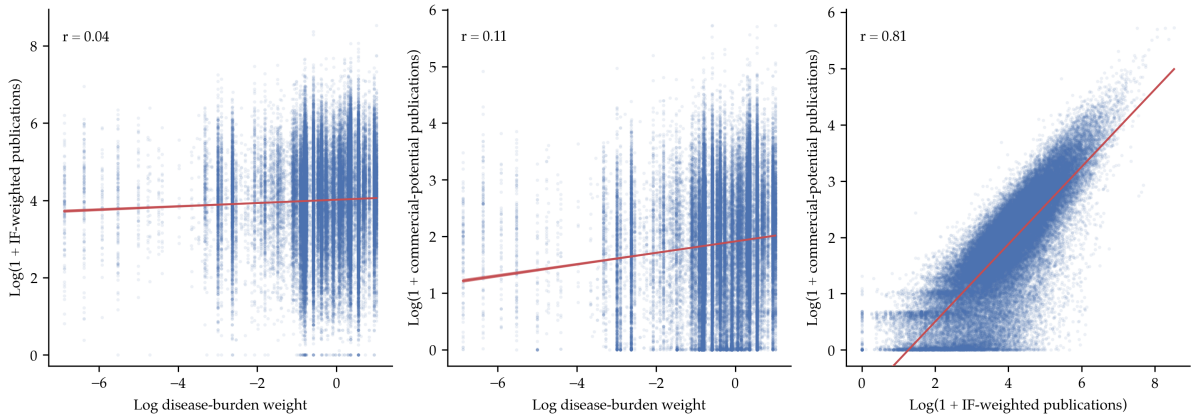
The institute-mean imputation could affect measured marginal returns, particularly for institutes with limited direct linkage. I therefore construct a second version of the disease-burden measure that assigns predicted disease categories to projects without an observed match. It preserves every direct RCDC–DALY match. For the remaining projects, I use a BiomedBERT-based classifier to read the title and abstract of each unmatched grant and predict which of the 74 disease categories apply to that project. These predicted DALY values replace the institute-mean imputation. I train the classifier on projects with observed RCDC–DALY matches and then re-estimate the disease-burden production function and marginal returns. On a held-out test sample, it attains

Figure 10. Marginal returns by output measure



Notes: Each institute has one stacked horizontal bar for academic impact (IF), commercial potential (CP), and disease burden (DB), evaluated for a $1 million increment at the 2018 R01 allocation. Blue denotes the current own-field return, orange the general-equilibrium talent-flow externality, green the own-field knowledge spillover, and red the cross-field knowledge spillover. Knowledge effects propagate only the funded institute's current output change and include discounted future output at an annual rate of 5 per cent. Institutes are ordered by the academic-impact return. The three upper axes use measure-specific scales, so bar lengths are comparable within but not across output measures. Labels beside bars report marginal returns; labels on the right report within-measure ranks.

Figure 11. Grant-level correlations among output measures



Notes: Each point is a unique grant in the 2002–2018 analysis sample. The left and centre panels relate the logarithm of the normalised grant-level disease-burden weight to $\log(1+y)$ for impact-factor-weighted and commercial-potential-weighted publications. The right panel relates the two publication measures. Red lines are fitted linear trends. The reported r values are Pearson correlations in the plotted logarithmic variables.

a micro F1 score of 0.847 and a Spearman correlation of 0.868 between predicted and observed project DALYs. Human Genome remains first and Nursing Research remains last. The broad ranking is robust to missing links in the data. Table 11 reports the institute-specific comparison. Appendix K describes the classifier, its performance, and its limitations; Appendix K.6 reports the full analysis using model-assigned categories.

Table 11. Disease-burden marginal returns: institute-mean imputation versus classifier-based imputation

Institute	Institute-mean imputation	Classifier-based imputation	Δ	Δ (%)	Institute-mean rank	Classifier-based rank
Human Genome	20.484	25.631	+5.146	+25.1	1	1
General Medicine	16.319	17.528	+1.209	+7.4	2	2
Heart, Lung, and Blood	11.354	11.153	-0.201	-1.8	3	4
Cancer	10.890	10.415	-0.475	-4.4	4	5
Environmental Health	10.752	11.237	+0.485	+4.5	5	3
Biomedical Imaging and Bioengineering	9.249	9.492	+0.243	+2.6	6	6
Drug Abuse	6.820	6.433	-0.387	-5.7	7	8
Arthritis and Musculoskeletal and Skin Diseases	6.162	6.778	+0.616	+10.0	8	7
Neurological Disorders and Stroke	5.771	5.833	+0.062	+1.1	9	12
Dental and Craniofacial Research	5.639	6.179	+0.540	+9.6	10	10
Diabetes and Digestive and Kidney Diseases	5.474	5.881	+0.407	+7.4	11	11
Mental Health	5.268	4.994	-0.274	-5.2	12	13
Eye	4.857	6.294	+1.437	+29.6	13	9
Alcoholism	4.657	4.367	-0.290	-6.2	14	15
Allergy and Infectious Diseases	4.386	4.855	+0.469	+10.7	15	14
Child Health and Human Development	3.266	3.615	+0.349	+10.7	16	16
Deafness and Communication Disorders	2.387	2.887	+0.501	+21.0	17	17
Aging	2.318	2.449	+0.131	+5.7	18	18
Nursing Research	1.862	2.012	+0.150	+8.0	19	19

Notes: Each row reports the total increase in US-DALY-weighted publications from a $1 million increment to the indicated institute at the 2018 R01 allocation. "Institute-mean imputation" retains observed RCDC–DALY matches and uses the institute-mean imputation for unmatched grants. "Classifier-based imputation" preserves the observed matches and uses classifier-based imputation for unmatched projects. Δ is Classifier-based imputation minus Institute-mean imputation. Both grant-level burden weights are normalised to mean one across unique projects.

7.3. Designing Counterfactual Allocations

Equipped with marginal return estimates for all fields and output measures, I can now compare the performance of different allocation rules. I study how best to allocate a $1 million increment to the 2018 NIH R01 budget across the 19 institutes. I compare different ways to split the increment across institutes. Two design choices need justification: allocating a local increment rather than reallocating the whole budget, and splitting the increment proportionally rather than giving it all to one field.

I study a local increment because a wholesale reallocation is politically unrealistic. In practice, NIH budgets usually adjust through differential growth across institutes, not through a complete redesign. Institute funding is shaped by path dependence and lobbying pressures (Sampat, 2012). The distribution of funding across institutes is mostly stable over time, with occasional disproportionate increases. Over the past twenty years, aggregate NIH funding has risen in most years. Nominal cuts are rare, and prolonged nominal declines do not occur (Figures A.4 and A.5). Budget adjustment operates mainly through relative growth: some institutes stagnate in real or relative terms, while others receive discrete increases. But a complete redesign never happens.

A local increment also requires weaker modelling assumptions. The model is identified from marginal variation around the observed allocation. It therefore speaks to local budget increments, not to the output of a budget rebuilt from scratch. Evaluating a wholesale redesign would require the estimated parameter values to hold globally.

The second choice is to split the increment proportionally because a corner allocation is likely to be politically infeasible. The optimal allocation would place the entire $1 million increment in the field with the largest marginal return, namely Human Genome. Figure 10 reports the return from spending $1 million on any single institute. But spending an entire budget increase in one field would likely face strong political resistance. A proportional split is more palatable: every institute receives additional funding, but more productive fields receive more.

I compare four allocation rules with a benchmark that preserves current budget shares. The benchmark allocates the local $1 million budget increment in proportion to current institute budget shares. Let s_d^0 denote funded field d's observed 2018 R01 budget share. The benchmark gives field d an additional s_d^0 dollars for each dollar in the increment. This is the local analogue of maintaining the current allocation.

Three marginal-return rules split the increment according to each field's marginal return. Let $a \in \{\text{IF}, \text{CP}, \text{DB}\}$ index the output measure used as the allocation criterion for a marginal-return rule. Let MR_d^a denote funded field d's marginal return to a $1 million budget increment under criterion a, evaluated at the 2018 comparative-statics

baseline and summed over output fields as in Section 6.2:

$$MR_d^a = \sum_o \left. \frac{dY_o^a}{dG_d} \right|_{t^\star=2018}. \tag{42}$$

The marginal-return rule based on criterion a splits the increment in proportion to these returns:

$$s_d^{\mathrm{MR},a} = \frac{MR_d^a}{\sum_f MR_f^a}, \qquad \sum_d s_d^{\mathrm{MR},a} = 1. \tag{43}$$

Thus, the academic-impact, commercial-potential, and disease-burden marginal-return rules use the corresponding output measures as their allocation criteria.

A fourth rule allocates the budget increment in proportion to each institute's average disease-burden profile, echoing medical studies that benchmark NIH funding against the burden of the diseases it targets (Gross, Anderson and Powe, 1999). I construct $\bar{B}_d$ as the average disease burden of the diseases addressed by field d's grant portfolio, using the data introduced in Section 7.1. Unlike the marginal-return rules, this rule uses the average disease-burden profile directly as its allocation criterion and does not account for production efficiency. It assigns the increment according to

$$s_d^B = \frac{\bar{B}_d}{\sum_f \bar{B}_f}. \tag{44}$$

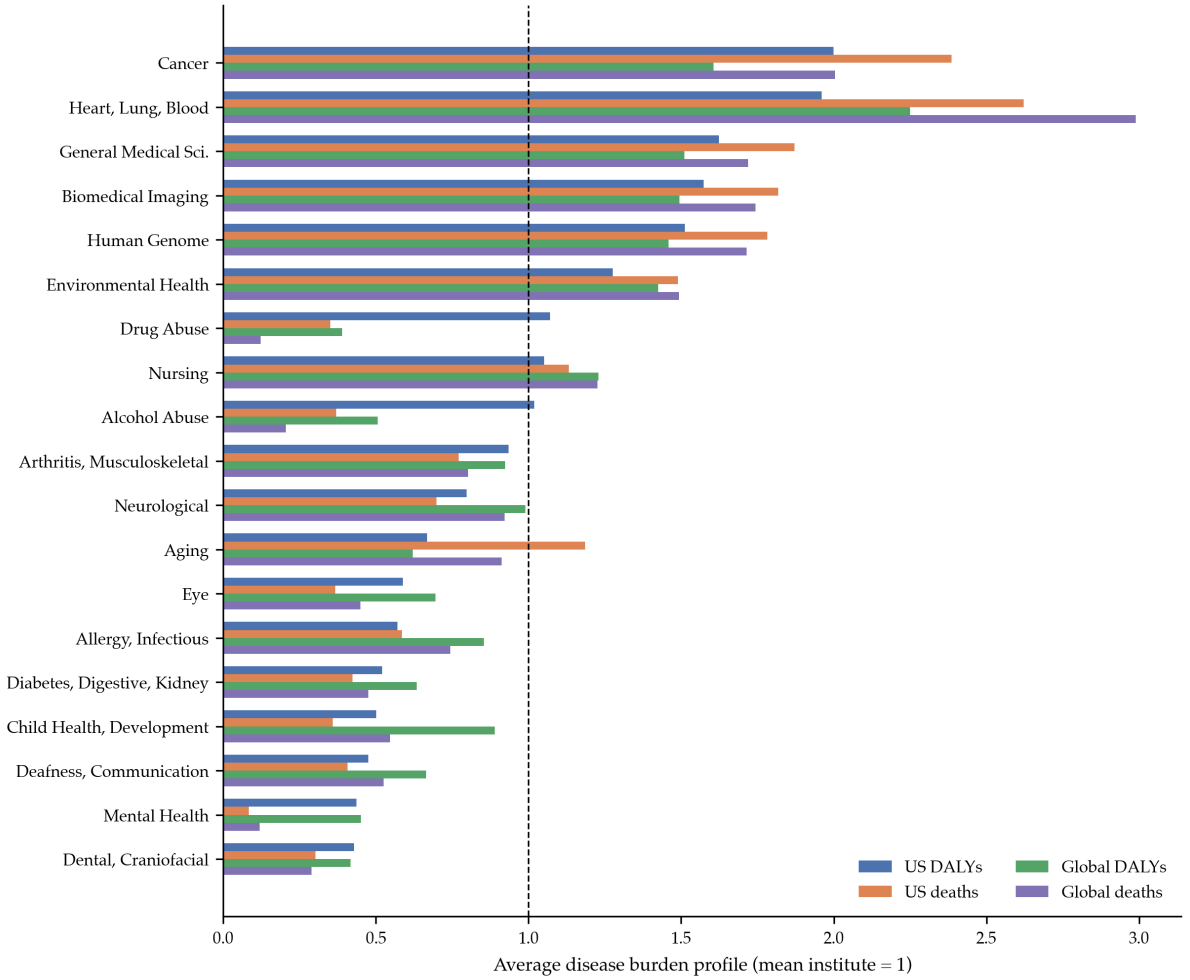
The highest average disease-burden profiles belong to Cancer; Heart, Lung, and Blood; General Medicine; Biomedical Imaging and Bioengineering; and Human Genome. Appendix I provides the construction details, while Figure 12 reports the full profiles by institute.

7.4. Evaluating Counterfactual Allocations

I next evaluate every allocation rule under each output measure. Let r denote an allocation rule and let $m \in \{\mathrm{IF}, \mathrm{CP}, \mathrm{DB}\}$ index the evaluation measure. Then $\Delta Y^{r,m}$ is the total output gain from applying rule r to the local \$1 million increment, evaluated under measure m. Table 12 reports this five-by-three comparison. Rows identify allocation rules; columns identify evaluation measures. Figure 13 shows the allocations implied by the rules, and Figure 14 compares their resulting output gains with the current-allocation benchmark.

Three results stand out. First, the average disease-burden rule improves on the current allocation. It raises academic-impact output by 22.3 per cent, commercial-potential output by 21.9 per cent, and disease-burden-weighted publication output by 25.7 per cent. Second, the corresponding marginal-return rules still dominate it. Relative to the average disease-burden rule, the academic-impact marginal-return rule

Figure 12. Average disease-burden profiles by institute

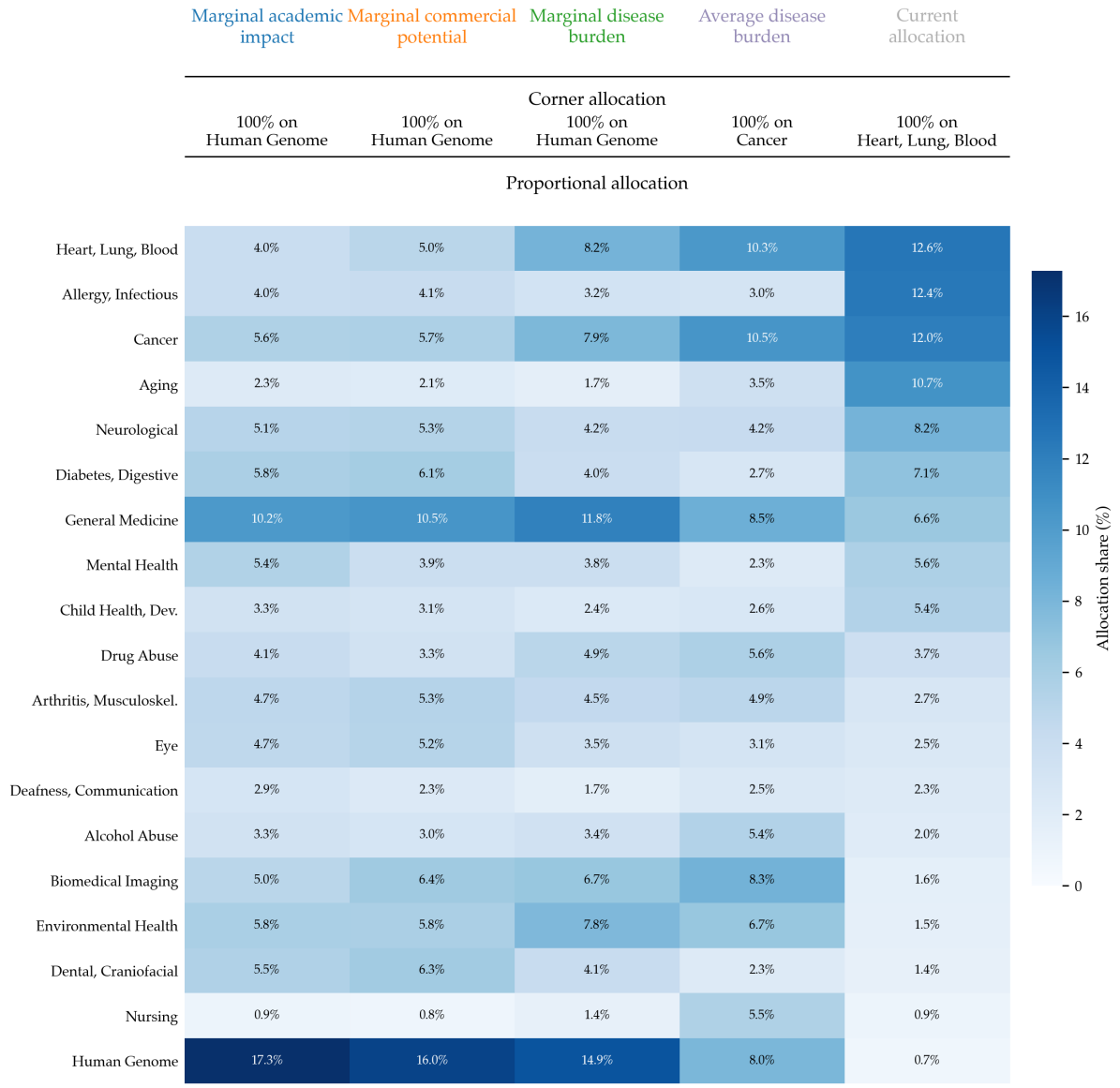


Notes: Each institute's profile averages burden across its grant portfolio in 2002–2018 using matched RCDC tags, with grants split equally across multiple matched categories; unmatched grants do not enter these profiles. Blue and green report US and global DALYs lost, while orange and purple report US and global deaths. Each series is normalised to a mean of one across institutes, marked by the dashed line. The US DALY profile determines $\bar{B}_d$ in Equation 44.

raises academic-impact output by 26.4 per cent, the commercial-potential marginal-return rule raises commercial-potential output by 22.2 per cent, and the disease-burden marginal-return rule raises disease-burden-weighted publication output by 15.1 per cent. Third, the same logic applies when corner allocations are allowed, but the efficiency gaps between the allocations widen, as reported in the upper panel of Figure 14. The average disease-burden rule selects Cancer and delivers 1.52 times the current-allocation benchmark in disease-burden-weighted publication output. The disease-burden marginal-return rule selects Human Genome and delivers 2.85 times the benchmark, 88 per cent more than the average disease-burden choice. This exercise highlights the importance of accounting for production efficiency, even under a disease-burden welfare objective: all three marginal-return allocation rules dominate the average disease-burden rule, even when evaluated using disease-burden-weighted publication output. Human Genome ranks first under all three marginal-return criteria and so

receives the largest share under every marginal-return rule; Appendix L.3 repeats this counterfactual exercise without Human Genome and shows that each marginal-return rule still dominates the average disease-burden rule under its own evaluation measure.

Figure 13. Institute shares of a $1 million increment under each rule



Notes: Cells report the percentage of the local $1 million increment assigned to each institute under the proportional allocation rule named by the column. Rows are ordered by the current-allocation share. The annotation above each column reports the institute selected when the corresponding rule assigns the full increment to one institute; the resulting disease-burden-weighted output relative to the current-allocation benchmark appears in the upper panel of Figure 14. Darker cells denote larger proportional allocation shares.

7.5. What Weights Rationalise the Current Allocation?

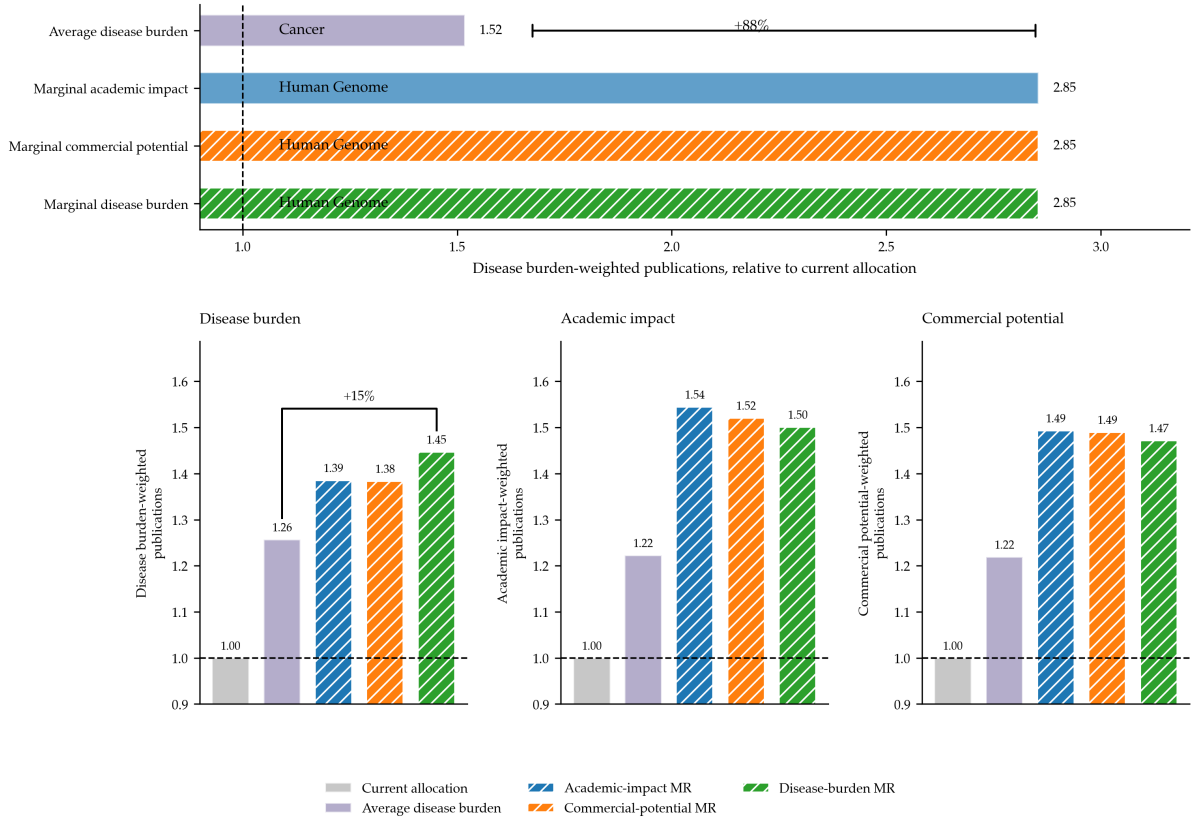
For the three welfare objectives considered here, the current allocation seems inefficient. But it may embody other priorities outside academic impact, commercial

Table 12. Output gains by allocation rule and evaluation measure

Allocation rule	Academic impact	Commercial potential	Disease burden
Current allocation	54.53 (1.000)	4.83 (1.000)	7.18 (1.000)
Average disease burden	66.67 (1.223)	5.88 (1.219)	9.02 (1.257)
Academic-impact MR	84.25 (1.545)	7.21 (1.493)	9.94 (1.386)
Commercial-potential MR	82.95 (1.521)	7.19 (1.490)	9.93 (1.383)
Disease-burden MR	81.85 (1.501)	7.10 (1.471)	10.38 (1.447)

Notes: Each row identifies the allocation rule used to divide a local $1 million increment in the 2018 R01 budget across institutes. The current-allocation rule uses observed institute budget shares. The average disease-burden rule uses institute-level disease-burden profiles. The three marginal-return (MR) rules use the output measure named by the rule as their allocation criterion. Columns identify evaluation measures. Cell entries report output gains $\Delta Y^{r,m}$ under the column measure; parentheses report the ratio to the current-allocation benchmark under that measure.

Figure 14. Output gains relative to the current allocation



Notes: The upper panel assigns the full $1 million increment to the institute selected by each allocation rule and evaluates the resulting output using the disease-burden measure. The three lower panels implement proportional allocation rules and evaluate each rule under disease burden, academic impact, and commercial potential output measures. They report $\Delta Y^{r,m}$ from Table 12 relative to the current-allocation benchmark. Numbers beside or above bars report these ratios. Dashed lines mark parity with the current allocation. Hatched bars denote marginal-return rules.

potential, and disease burden. I explore this possibility by using the marginal return estimates in raw publication to back out the set of welfare weight needed to rationalise the current allocation. At an interior optimum, the planner equalises the weighted marginal return across institutes. As shown in Section 2, the first-order condition is

$$\omega_d \cdot MR_d^P = \nu \quad \forall\, d,$$

where MR_d^P is funded field d's marginal return in raw publications and ω_d is its rationalising weight. I evaluate MR_d^P at the 2018 allocation using Figure 8 and invert this condition. Normalising the weights to average one across the 19 institutes yields

$$\omega_d = 19 \cdot \frac{1/MR_d^P}{\sum_f 1/MR_f^P}. \tag{45}$$

Assuming that the current allocation chosen by Congress is optimal and that the marginal return estimates in this paper are accurate, these weights may provide some indication of congressional preferences. They absorb any omitted objectives required to reconcile the observed allocation with the model. Table 13 reports the weights alongside average academic impact, commercial potential, and disease-burden value per publication. Rationalising weights are normalised to average one throughout.

Table 13. Rationalising weights and publication-value profiles

Institute	Raw MR (publications/$1M)	Rationalising weight ω_d	Academic impact	Commercial potential	Disease burden
Nursing Research	2.2	2.62	0.61	0.40	1.03
Aging	3.1	1.85	0.97	0.83	0.65
Deafness, Communication	4.6	1.23	0.82	0.88	0.47
Allergy, Infectious Diseases	5.0	1.15	1.17	1.33	0.66
Child Health, Development	5.1	1.13	0.79	0.62	0.49
Alcoholism	6.0	0.95	0.74	0.60	1.01
Cancer	6.1	0.94	1.15	1.29	2.01
Arthritis, Musculoskeletal	6.1	0.93	0.95	1.21	0.87
Neurological Disorders	6.3	0.90	1.07	1.19	0.79
Heart, Lung, Blood	6.4	0.90	0.89	1.09	2.04
Drug Abuse	6.8	0.84	0.83	0.58	1.07
Eye	6.9	0.83	0.94	1.16	0.55
Biomedical Imaging	6.9	0.83	0.97	1.38	1.55
Mental Health	7.2	0.80	1.01	0.63	0.48
Diabetes, Digestive, Kidney	7.2	0.79	1.05	1.14	0.52
Dental, Craniofacial	8.2	0.70	0.99	1.23	0.44
Environmental Health	8.3	0.69	1.01	0.83	1.23
General Medicine	11.0	0.52	1.32	1.37	1.68
Human Genome	14.8	0.39	1.72	1.24	1.46
Mean	6.7	1.00	1.00	1.00	1.00

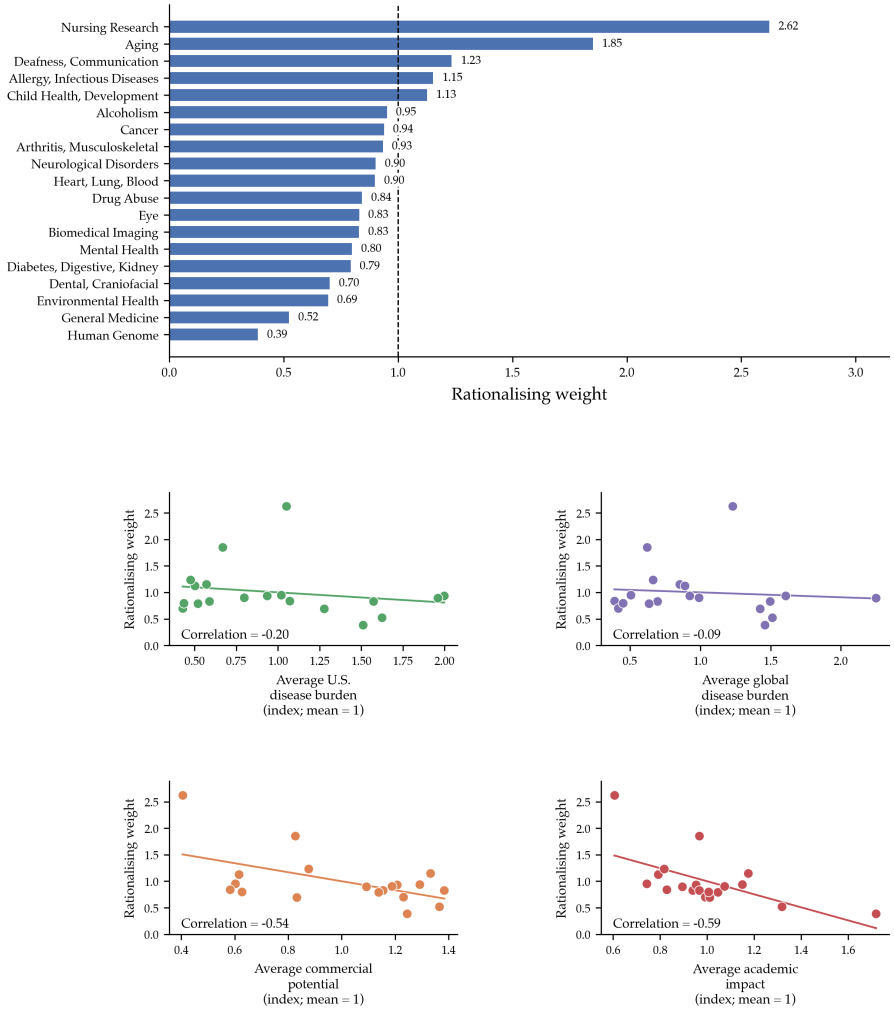
Notes: Each row reports an institute's raw-publication marginal return and the rationalising weight ω_d that rationalises its 2018 R01 allocation under equalised weighted marginal returns, as defined in Equation 45. The final three columns report average academic impact, commercial potential, and disease burden per publication. The weight and each publication-value profile are normalised to a cross-institute mean of one. Institutes are ordered by ω_d.

Figure 15 relates the rationalising weights to average field characteristics. The

correlations are negative for academic impact (−0.59) and commercial potential (−0.54), but weak for average disease burden (−0.20) and average global disease burden (−0.09). The burden profiles use the mix of RCDC disease categories in each institute's grant portfolio and are distinct from the disease-burden measure.

The upper tail is suggestive of omitted priorities. The weights on Nursing Research, Aging, Child Health, and Alcoholism, for example, would need relatively high rationalising weights attached to them to justify their budgets, given their relatively low returns. These relatively high weights may reflect priorities attached to nursing, healthier ageing, child health, or reduced alcohol harm that the three output measures used here do not capture. This interpretation is suggestive, not conclusive: the weights rationalise the allocation under the model but do not identify the objectives that produced it.

Figure 15. Rationalising weights versus field characteristics



Notes: The upper panel orders institutes by the rationalising weight ω_d for the 2018 R01 allocation. The four lower panels plot ω_d against normalised average US disease burden, global disease burden, commercial potential, and academic impact; solid lines are fitted linear relationships and labels report Pearson correlations. Disease-burden profiles use each institute's RCDC grant portfolio, splitting grants equally across multiple matched categories. Academic impact and commercial potential are average values per publication. Every horizontal profile is normalised to a cross-institute mean of one.

8. Conclusion

How should the NIH divide its budget across institutes? This paper examines the question by estimating the marginal return to funding in each field. The estimates come from a micro-founded model of scientists' application choices, grant evaluation, and knowledge spillovers, estimated using NIH R01 grant data and linked publication records.

Measured in publications, marginal returns differ substantially across fields. An extra $1 million produces 14.8 publications in Human Genome but 2.2 in Nursing Research. Knowledge spillovers, rather than scientist reallocation, drive this gap. Scientists move towards better-funded fields, but their replacements have similar productivity, leaving a small net effect. Cross-field knowledge spillovers exceed own-field spillovers in 15 of the 19 institutes. Upstream and general-purpose fields, including Human Genome, Environmental Health, and Biomedical Imaging and Bioengineering, generate particularly large cross-field spillovers.

Measuring a field's scientific output in terms of publication count does not allow a welfare interpretation. To evaluate allocations in terms of social value, I weight each field's output by the academic impact and commercial potential of the publications it produces, and by the US disease burden that its research addresses. I then evaluate alternative allocation schemes under academic-impact, commercial-potential, and disease-burden output measures. This exercise highlights the importance of allocating funding based on marginal rather than average returns: each marginal-return allocation outperforms both the current allocation and an allocation based on the average disease burden of each field, *regardless of the way scientific output is measured*. Even when I evaluate output in terms of disease-burden-weighted publication, the marginal-return rule produces 15.1 per cent more output than the average disease-burden rule. The marginal-return rules outperform because they account for differences in production efficiency and knowledge spillovers across fields. When the social planner is allowed to pour all additional funds into one institute, the institute with the highest marginal disease burden (Human Genome) delivers 88 per cent more disease-burden-weighted publication than the institute with the highest average disease burden (Cancer).

These numbers depend on the planner's objective function. The current allocation may simply reflect an objective function different from those considered in this paper. Indeed, the current allocation can be rationalised by assigning high weights to Nursing Research, Aging, Child Health and Human Development, and Alcoholism. This may reflect concerns among members of Congress and the public that extend beyond academic impact, commercial potential or disease burden. This does not mean, however, that these wider concerns should motivate more research funding in these fields. Other policy instruments may be better suited to promoting healthy ageing or reducing

alcohol-related harm, for example.

Three extensions would broaden the framework. First, incorporating non-NIH research funders into my system would capture the output forgone when researchers move to the NIH from the National Science Foundation, private foundations, or industry. Doing so would provide a fuller picture of the opportunity cost generated by the NIH when it expands funding. Second, adding a dynamic dimension to the model would highlight the trade-off between funding more productive established investigators and training younger scientists to improve their future productivity (Qiu, 2023). Third, considering the risk profile of different research fields would add a further dimension to the analysis and may significantly affect the optimal allocation, depending on the planner's risk preferences.

The broader lesson of this paper is that research fields are not independent production units. They draw on a shared scientific workforce, build on a common stock of knowledge, and respond jointly to changes in funding conditions. Evaluating an institute only by the research produced within its own portfolio therefore misses a central part of the allocation problem. A complete assessment must trace the full matrix of responses: the research created in the funded field, the research displaced elsewhere, and the knowledge transmitted across fields. Ignoring these margins may lead to a suboptimal allocation of resources at a time when science funding is particularly scarce.

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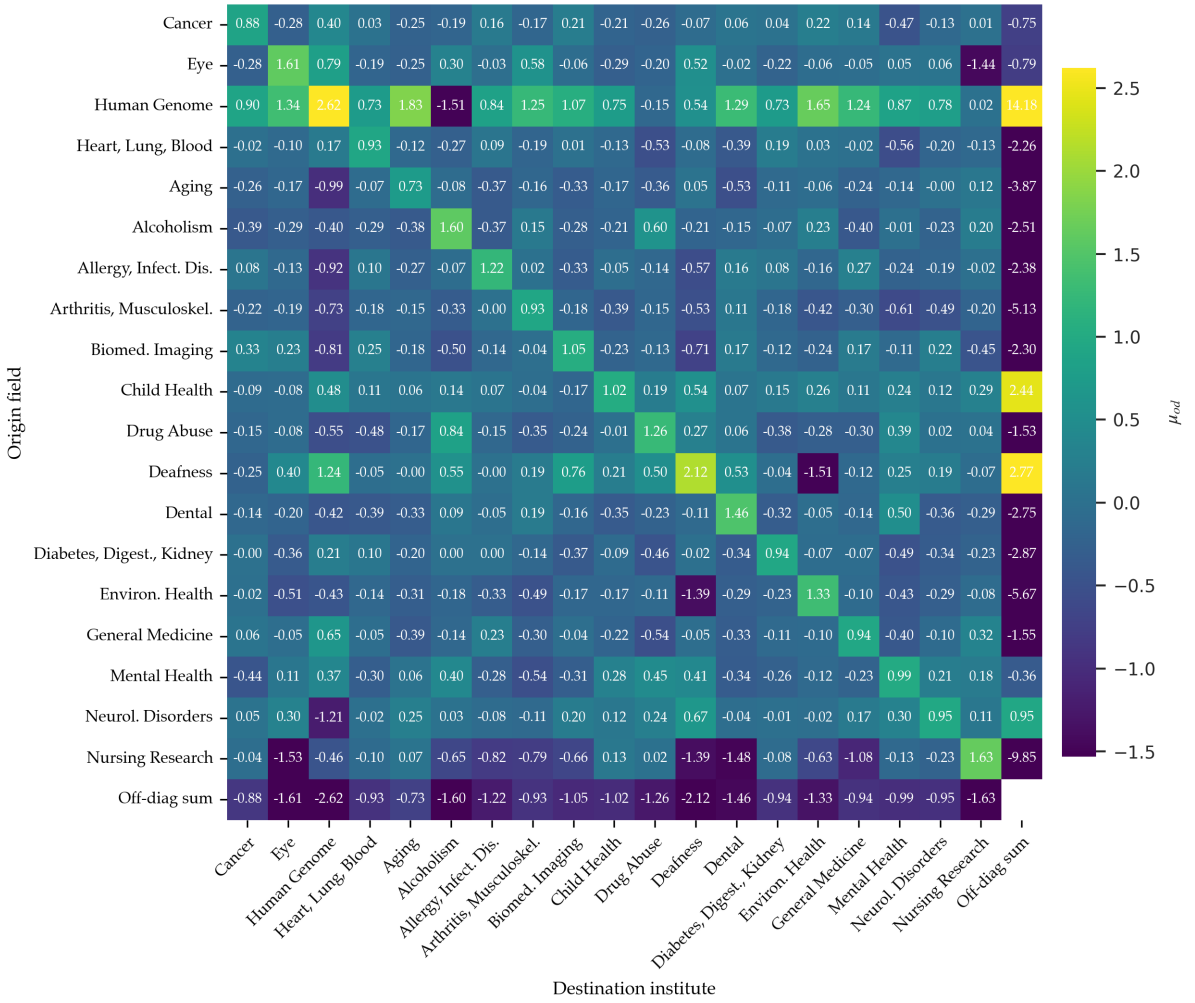
A. Supplementary Figures and Tables

Table A.1. NIH institutes and centres

Acronym	Official Name	Name in this paper
FIC	Fogarty International Center	NA
NCCIH	National Center for Complementary and Integrative Health	NA
NCI	National Cancer Institute	Cancer
NCRR	National Center for Research Resources	NA
NIDCD	National Institute on Deafness and Other Communication Disorders	Deafness and Communication Disorders
NEI	National Eye Institute	Eye
NHGRI	National Human Genome Research Institute	Human Genome
NHLBI	National Heart, Lung, and Blood Institute	Heart, Lung, and Blood
NIA	National Institute on Aging	Aging
NIAAA	National Institute on Alcohol Abuse and Alcoholism	Alcoholism
NIAID	National Institute of Allergy and Infectious Diseases	Allergy and Infectious Diseases
NIAMS	National Institute of Arthritis and Musculoskeletal and Skin Diseases	Arthritis and Musculoskeletal and Skin Diseases
NIBIB	National Institute of Biomedical Imaging and Bioengineering	Biomedical Imaging and Bioengineering
NICHD	National Institute of Child Health and Human Development	Child Health and Human Development
NIDA	National Institute on Drug Abuse	Drug Abuse
NIDCR	National Institute of Dental and Craniofacial Research	Dental and Craniofacial Research
NIDDK	National Institute of Diabetes and Digestive and Kidney Diseases	Diabetes and Digestive and Kidney Diseases
NIEHS	National Institute of Environmental Health Sciences	Environmental Health
NIGMS	National Institute of General Medical Sciences	General Medicine
NIMH	National Institute of Mental Health	Mental Health
NINDS	National Institute of Neurological Disorders and Stroke	Neurological Disorders and Stroke
NINR	National Institute of Nursing Research	Nursing Research

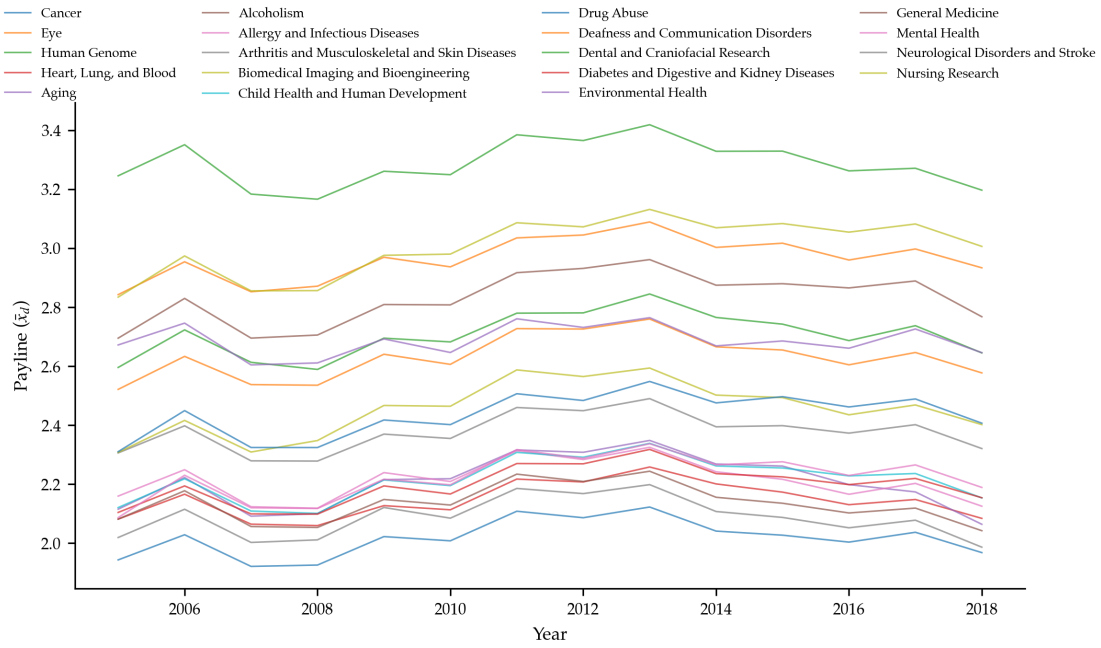
Notes: The table maps NIH institute and centre acronyms to official names and to the shorter field labels used throughout the paper (National Institutes of Health, n.d.[a]). "NA" identifies organisations excluded from the 19-field estimation sample.

Figure A.1. Estimated mean idea quality (μ_{od})



Notes: The heatmap reports the structurally estimated mean-idea-quality matrix $\hat{\mu}_{od}$ over 2005–2018. Rows index scientists' origin fields o and columns index destination institutes d. Each destination column sums to zero. The bottom row sums off-diagonal origin means within each destination, so it equals the negative diagonal; the right column sums off-diagonal entries across destinations, whose levels have separate normalisations. Values are measured in units of baseline evaluation noise because $\eta_0 = 1$. Cell labels are point estimates rounded to two decimals. Table 5 reports the diagonal point estimates.

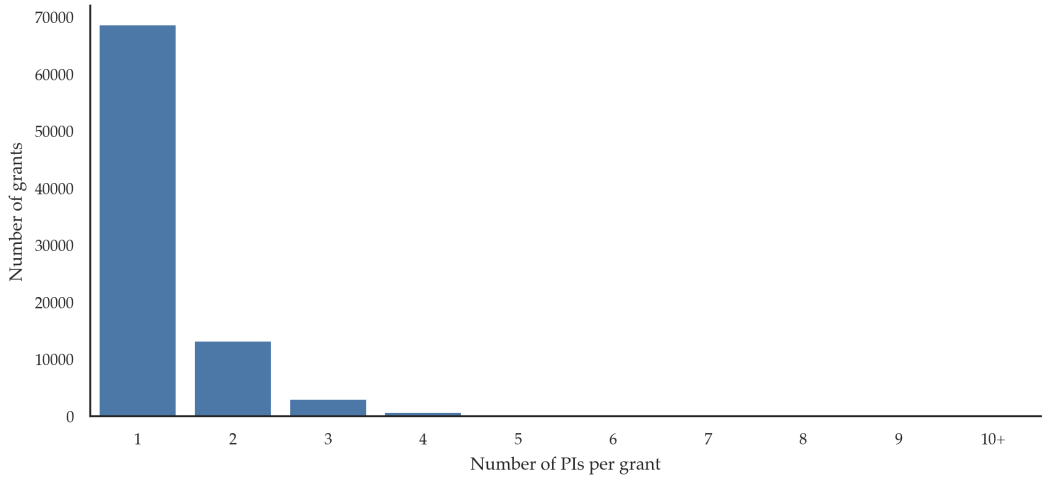
Figure A.2. Model-implied equilibrium paylines



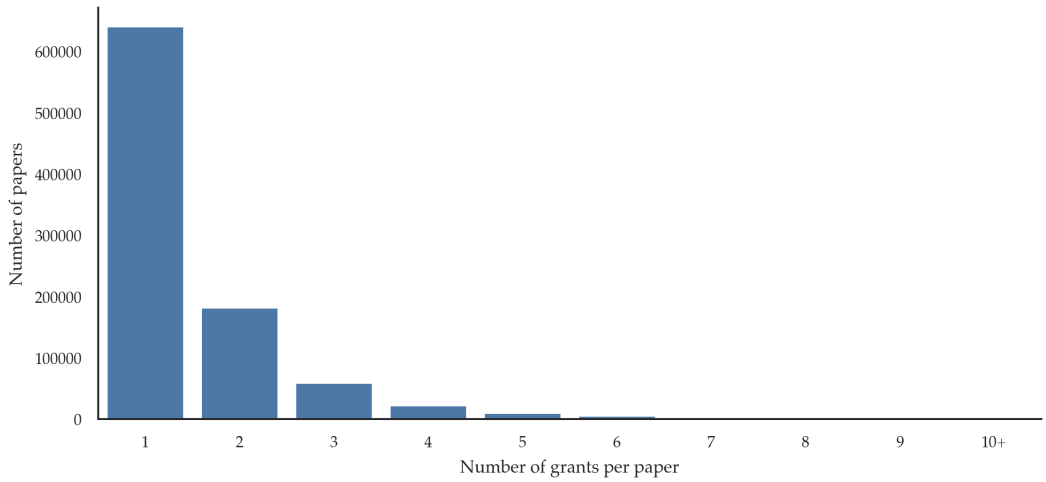
Notes: The figure plots the model-implied equilibrium payline $\bar{x}_{d,t}$ for each NIH institute over the 2005–2018 estimation period. A higher value denotes a more stringent funding threshold.

Figure A.3. Many-to-many links between PIs, grants, and publications

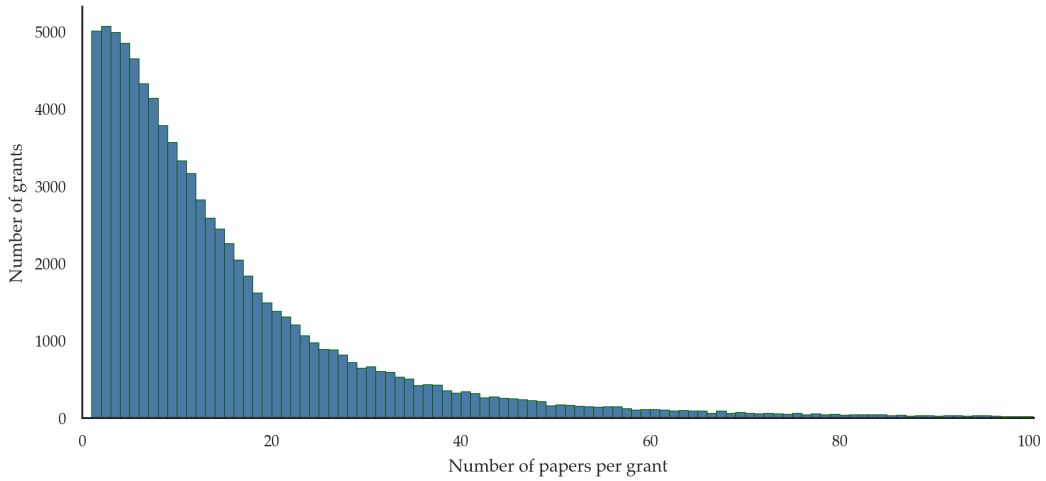
(a) PIs per grant



(b) Grants per publication

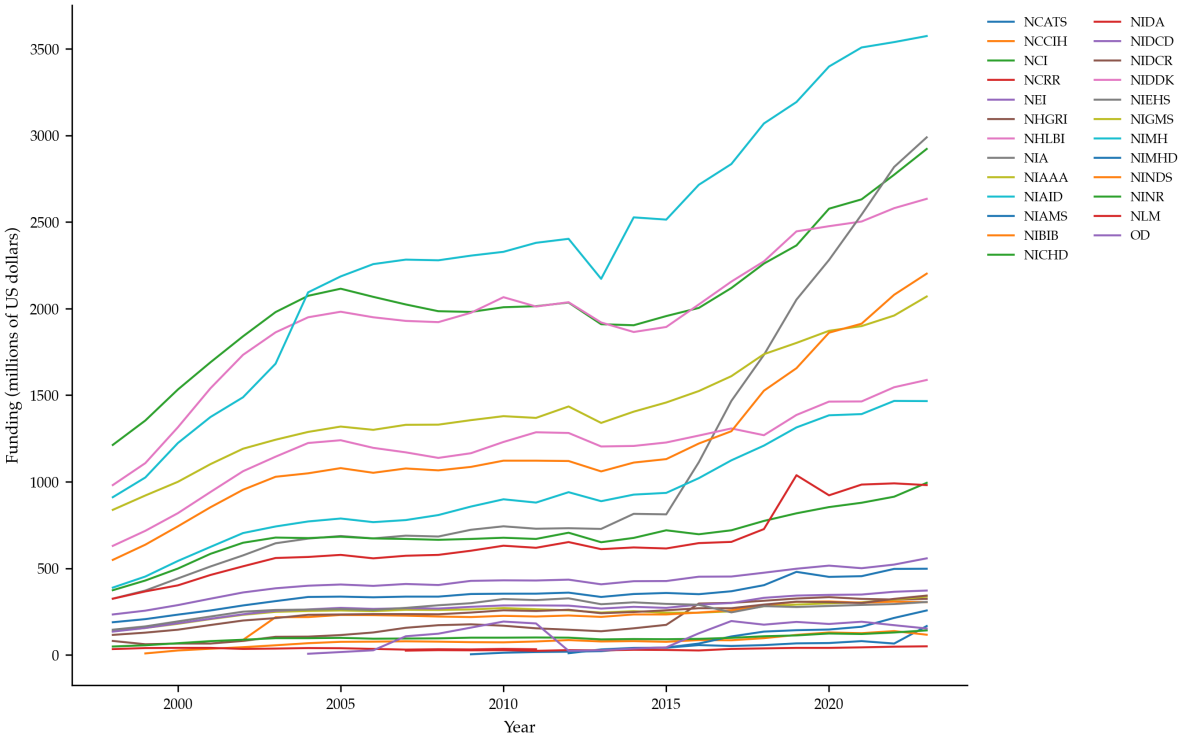


(c) Publications per grant



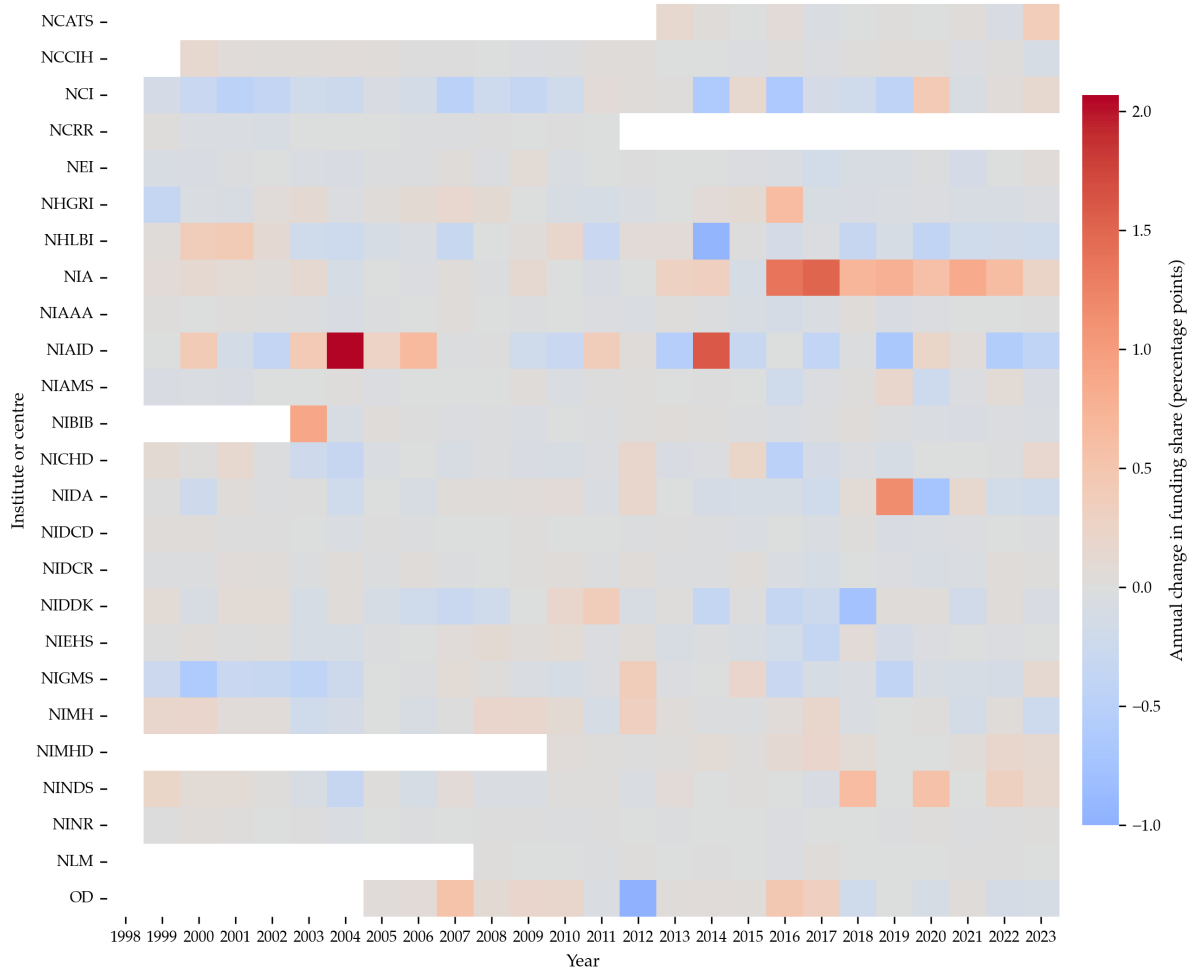
Notes: Panel (a) reports the distribution of the number of principal investigators attached to each grant. Panel (b) reports the number of grants acknowledged by each linked publication. Panel (c) reports the number of linked publications per grant and truncates the horizontal axis at 100 for readability. Bar heights are observation counts in the linked NIH grant–publication data.

Figure A.4. NIH funding trends



Notes: Each line reports annual nominal Research Project Grant funding for one NIH institute or centre from 1998 to 2023, measured in millions of US dollars. The series are not adjusted for inflation. Source: NIH Data Book, Report 209 (National Institutes of Health, n.d.[d]).

Figure A.5. Yearly changes in NIH funding shares



Notes: Rows identify NIH institutes or centres and columns identify fiscal years from 1998 to 2023. The 1998 column is the initial share baseline; subsequent cells report annual changes in each organisation's share of total NIH Research Project Grant funding, in percentage points. Red cells indicate an increase in budget share and blue cells a decrease; colour intensity records magnitude. Source: NIH Data Book, Report 209 (National Institutes of Health, n.d.[d]).

B. Semantic Distance Before an Institute Switch

This section documents the analysis reported in Figure 9. Its purpose is to test whether a grant preceding a change in funding institute is more aligned with the eventual destination than grants that remain in the same origin institute and year.

B.1. Sample and Future-Switcher Definition

Within each investigator, I order grants by start year, parsed project start date, and project number, then attach the next observed grant. Both the focal and next grants must have one PI. Let f_i be focal grant i's current institute and let f_i^+ be the funding institute of the next eligible observed grant. The future-switcher indicator is

$$S_i = \mathbf{1}\{f_i^+ \neq f_i\}. \tag{46}$$

Thus, $S_i = 1$ records a change between two consecutive observed grants. It does not require the PI to remain in the new field. Investigators without a later observed grant are excluded because S_i is undefined. Grants without a full abstract are excluded from both the comparison and embedding samples.

The cleaned panel contains 106,664 grant–PI observations and 85,290 projects. Restricting focal grants to one PI leaves 68,545 observations. Of these, 31,201 have no subsequent observed grant, 12,892 are followed by a multi-PI grant, 11 fail the cleaned-link consistency restriction, and 74 lack a full abstract. The retained candidate sample therefore comprises 9,036 future switchers and 15,331 future stayers over 2001–2022. For the centroids, I use the broader population of 68,389 single-investigator grants with a full abstract, a valid start year, and one of the 19 core institutes.

B.2. Raw BiomedBERT Embeddings

Each grant is represented by a unit-normalised mean of its token embeddings. I pass the project title first and the abstract second, cap the combined input at 512 tokens, and truncate only the abstract. I use the raw BiomedBERT base encoder without task-specific fine-tuning.[15] Let $h_{im} \in \mathbb{R}^{768}$ denote the final hidden state of token m for grant i, and let a_{im} equal one for a non-padding token and zero otherwise. The representation is

$$\widetilde{p}_i = \frac{\sum_m a_{im} h_{im}}{\sum_m a_{im}}, \qquad p_i = \frac{\widetilde{p}_i}{\|\widetilde{p}_i\|_2}. \tag{47}$$

The procedure yields 68,389 vectors with 768 dimensions and a Euclidean norm of 1.

[15] The model identifier is `microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`; the version used is `e1354b7a3a09615f6aba48dfad4b7a613eef7062`.

B.3. Destination–Year Centroids and Distance

To compare each grant with its eventual destination, I construct a separate centroid for each destination–year cell. For destination d and focal year t, let $\mathcal{G}_{dt}$ denote the grants in the centroid population and let $N_{dt} = |\mathcal{G}_{dt}|$. I average their unit embeddings and then normalise the mean:

$$\overline{p}_{dt} = \frac{1}{N_{dt}} \sum_{j \in \mathcal{G}_{dt}} p_j, \qquad c_{dt} = \frac{\overline{p}_{dt}}{\|\overline{p}_{dt}\|_2}. \tag{48}$$

For a transition cell $k = (o, d, t)$, both switchers and matched stayers originate in o but are evaluated against c_{dt}. Their outcome is the cosine distance

$$D_{ik} = 1 - p_i' c_{dt}. \tag{49}$$

Because both p_i and c_{dt} have unit norm, the corresponding Euclidean radius is

$$\left\| p_i - c_{dt} \right\|_2 = \sqrt{2D_{ik}}. \tag{50}$$

All centroids and distances are computed for the exact destination and focal year.

B.4. Comparison Specification

Let $\mathcal{S}_{odt}$ denote focal grants that switch from o to d after year t, and let $\mathcal{C}_{ot}$ denote future stayers in the same origin and focal year. Write $n^S_{odt} = |\mathcal{S}_{odt}|$ and $n^C_{ot} = |\mathcal{C}_{ot}|$. Each switcher enters its observed transition cell once. Every stayer in $\mathcal{C}_{ot}$ enters each supported destination transition from (o, t). The analysis weight is

$$w_{ik} = \begin{cases} 1, & i \in \mathcal{S}_{odt}, \\ n^S_{odt} / n^C_{ot}, & i \in \mathcal{C}_{ot}. \end{cases} \tag{51}$$

Hence, switchers and matched stayers receive equal mass in every transition cell:

$$\sum_{i \in \mathcal{S}_{odt}} w_{ik} = \sum_{i \in \mathcal{C}_{ot}} w_{ik} = n^S_{odt}. \tag{52}$$

Because each transition cell assigns equal total weight to switchers and stayers, the pooled estimand averages observed transitions in proportion to their number of switchers, without giving extra influence to origin–year cells that contain more stayers.

I estimate the matched comparison by weighted least squares with transition-cell fixed effects:

$$D_{ik} = \alpha_k + \beta S_i + \varepsilon_{ik}, \qquad k = (o, d, t). \tag{53}$$

With these weights, the fixed-effects coefficient can be written as

$$\widehat{\beta} = \frac{\sum_k n_k^S \left(\overline{D}_k^S - \overline{D}_k^M \right)}{\sum_k n_k^S}, \tag{54}$$

where $\overline{D}_k^S$ and $\overline{D}_k^M$ are the switcher and matched-stayer means in cell k. A negative coefficient indicates that switchers are closer to their eventual destinations. I cluster standard errors by investigator, so a cluster can contain repeated grants and every destination assignment of the same stayer.

Table A.2 reports the estimates. Weighted mean distance is 0.004502 for future switchers and 0.005035 for matched stayers. The difference is −0.000533, with an investigator-clustered standard error of 0.000020 and a 95 per cent confidence interval from −0.000572 to −0.000493. Future switchers are therefore 10.6 per cent closer to their eventual destination than matched stayers. Their weighted mean destination–year distance percentile is 61.1, compared with 71.8 for matched stayers.

Table A.2. Distance to the destination before an institute switch

	Matched stayers	Future switchers
Unique grants	15,327	8,989
Investigators	10,280	6,172
Stacked observations	172,256	8,989
Effective weight	8,989	8,989
Weighted mean semantic distance	0.005035	0.004502
Mean semantic distance difference	-0.000533***	
Weighted standard deviation	0.001637	0.001569
Weighted median semantic distance	0.004797	0.004295
Transition cells	3,110	3,110

Notes: Distance is one minus cosine similarity between a focal grant's raw-BiomedBERT embedding and the focal-year centroid of the institute to which switchers move. Stayers are drawn from the same origin institute and focal year. Within each origin–destination–year cell, switchers have unit weight and each stayer has weight equal to the number of switchers divided by the number of available stayers. The mean-distance difference is switchers minus stayers from a weighted transition-cell fixed-effects regression. Stars denote significance levels from investigator-clustered inference: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.5. Construction of the Visualisation

Figure 9 is designed to connect the high-dimensional distance measure to its two-dimensional display. I adapt the embedding-space presentation of Hao et al. (2026), which uses scikit-learn's t-distributed stochastic neighbour embedding (t-SNE) to visualise a random sample of 10,000 publications split equally by artificial-intelligence status, while measuring knowledge extent in the unflattened 768-dimensional space. Because that study reports no further t-SNE parameters, I specify the display procedure

here. I use a deterministic balanced sample of 3,000 matched-stayer rows and 3,000 future switchers: switchers are sampled uniformly, whereas matched-stayer rows are sampled without replacement in proportion to their analysis weights. Panel (a) records the measurement sequence. In panel (b), I subtract the exact destination–year centroid from each sampled grant's 768-dimensional embedding. Status labels do not enter the t-SNE fit.

t-SNE is used here to display local neighbourhoods, not to define or measure distance. It converts high-dimensional local similarities into two-dimensional similarities with a heavy-tailed Student-t kernel and minimises the Kullback–Leibler divergence between the resulting similarity distributions. I use scikit-learn for the implementation. I centre the returned coordinates at their pooled mean and apply one scalar so that the balanced sample's mean projected radius equals the full matched sample's weighted mean exact radius $\sqrt{2D_{ik}}$. This calibration makes the display scale transparent but does not make any individual t-SNE radius metric-preserving. The axes, global distances, radial distances, cluster areas, and apparent density therefore have no analytical interpretation.

The circles provide the display's only quantitative summary from the original space. For each status, a circle marks the analysis-weighted arithmetic mean of the observation-level exact radii $\sqrt{2D_{ik}}$ in the full matched sample, rather than in the balanced display sample. The exact mean radii are 0.0994 for matched stayers and 0.0940 for future switchers. Thus, only the circles encode the unprojected radial metric in panel (b); the regression continues to use the exact 768-dimensional cosine distance D_{ik}, and no t-SNE coordinate enters estimation or inference.

Panel (c) complements the map with a destination–year distance percentile. Each matched row is ranked against all centroid grants funded by its assigned destination in the focal year. The panel reports analysis-weighted distributions for matched stayers and future switchers across all 19 eventual destinations and in the pooled sample. Horizontal lines mark weighted medians, boxes span weighted interquartile ranges, and whiskers reach the most extreme observations within 1.5 times the weighted interquartile range; outliers are omitted. All observations remain in the percentile, distance, and regression calculations. The white background and dark axes are stylistically neutral, while colour distinguishes the two statuses.

Figure A.6 repeats the display separately for all 19 eventual destinations. Each panel uses a deterministic balanced sample of at most 1,000 rows per status and a label-free t-SNE fit with the parameters from panel (b), except that perplexity is kept below the displayed sample size. I centre each map and apply a destination-specific scalar so that its mean projected radius equals the full destination sample's weighted mean exact radius. Because separate t-SNE fits have arbitrary orientation, shape, and scale, neither point coordinates nor cloud dimensions are comparable across destinations. I impose a common plotting boundary using the pooled 99.5th percentile of calibrated two-

dimensional point radii, clipping 89 more distant points, or 0.50 per cent of displayed points, at that boundary. Exact circles are neither clipped nor recalculated from the point clouds. The future-switcher exact mean-radius circle lies inside the matched-stayer circle for all 19 destinations. These patterns are descriptive and do not identify an effect of funding opportunities on research direction.

Figure A.6. Distance to the destination before a switch, by institute



Notes: Each panel corresponds to the labelled eventual destination. Panel headings report full matched-row counts, where n_M denotes matched-stayer rows and n_S denotes switchers; a stayer may appear under several destinations. Points show deterministic balanced samples of at most 1,000 rows per status. Matched stayers are sampled in proportion to their analysis weights. Each residual is formed relative to the exact destination–focal-year centroid. Separate label-free t-SNE fits are centred and scalar-calibrated to their destination's full-sample weighted mean exact radius for display only. Point coordinates, orientation, shape, scale, global distances, radial distances, cluster areas, and apparent density are not comparable across destinations and are not used for inference. The common outer boundary is the pooled 99.5th percentile of calibrated point radii; more distant points are retained analytically but clipped at the boundary in the image. Coloured circles mark status-specific weighted arithmetic means of the exact unprojected radii $\sqrt{2D_{ik}}$ using all matched rows. Only these circles, not cloud dimensions, are quantitative in the 768-dimensional radial metric. Future switching denotes a different institute on the next eligible observed grant.

C. Derivation of the Realised Quality Gap Distribution

This section derives the distribution of the realised quality gap z_i for scientist i from origin o, conditional on applying to field d, and defines the inclusive value M_o.

Assuming that, for each field d, scientist i draws ε_i independently from a Type-I Extreme Value (Gumbel) distribution with location 0 and scale $\sigma > 0$, standard multinomial-logit properties imply that the distribution of the realised shock, conditional on applying to field d, is identical to the unconditional distribution of the maximum shock across all fields and the outside option.

Consequently, the realised quality gap $z_i \equiv \mu_{od} - \bar{x}_d + \varepsilon_i$, conditional on scientist i applying to field d, follows a Gumbel distribution with scale σ and location parameter given by the inclusive value M_o:

$$M_o \;=\; \sigma \ln\left(\exp\left(\frac{c}{\sigma}\right) + \sum_{f=1}^{C} \exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)\right), \tag{55}$$

where c represents the outside option. The conditional density of the realised quality gap z is therefore:

$$f_Z(z; M_o) \;=\; \frac{1}{\sigma} \exp\left(-\frac{z - M_o}{\sigma} - \exp\left(-\frac{z - M_o}{\sigma}\right)\right). \tag{56}$$

This density is shifted right when M_o increases (i.e. when scientists from origin o face a more favourable landscape overall, their chosen quality gaps tend to be larger).

D. Equilibrium Existence and Uniqueness

D.1. Existence

Proposition 1 (Existence of an Equilibrium)**.** *Let $s_d = G_d/\bar{g}_d$ denote the number of available grants in destination d. Suppose $s_d > 0$ for all d and the aggregate available-grant vector is feasible, so that $\sum_{d=1}^{C} s_d < \sum_{o=1}^{C} N_o$. Suppose further that there exists a compact rectangle $\mathcal{X} = \prod_{d=1}^{C} [L_d, U_d]$ such that, for every destination d,*

$$E_d(L_d, \bar{x}_{-d}) \geq s_d \quad \textit{and} \quad E_d(U_d, \bar{x}_{-d}) \leq s_d \quad \textit{for all } \bar{x}_{-d} \in \prod_{f \neq d} [L_f, U_f]. \tag{57}$$

Then there exists an equilibrium payline vector $\bar{x}^* \in \mathbb{R}^C$ solving (10).

Proof. Let the excess demand function for destination d be $Z_d(\bar{x}) = E_d(\bar{x}) - s_d$, where $E_d(\bar{x}) = \sum_{o=1}^{C} N_o \Pi_{od}(\bar{x}) S_{od}(M_o(\bar{x}))$. The equilibrium condition is $Z(\bar{x}) = \mathbf{0}$. The aggreg-

ate feasibility condition is necessary because each scientist can apply to at most one destination and $S_{od} \leq 1$, so $\sum_d E_d(\bar{x}) \leq \sum_o N_o$ for any payline vector.

The stated face conditions are the uniform version of the one-dimensional limit argument. On an upper face, the application probability to destination d vanishes as $\bar{x}_d$ becomes large, uniformly over bounded values of $\bar{x}_{-d}$. Since $0 \leq S_{od} \leq 1$, this makes E_d arbitrarily small. The lower-face condition imposes the corresponding capacity requirement: when destination d's payline is sufficiently loose relative to the other coordinates in $\mathcal{X}$, expected funded mass in d weakly exceeds its slot capacity.

The functions $Z_d(\bar{x})$ are continuous because the logit probabilities, the inclusive value, and the success integrals are continuous in $\bar{x}$. By assumption, $Z_d(L_d, \bar{x}_{-d}) \geq 0$ and $Z_d(U_d, \bar{x}_{-d}) \leq 0$ for every d and every $\bar{x}_{-d}$ on the corresponding face of $\mathcal{X}$. The Poincaré-Miranda theorem therefore implies that there is some $\bar{x}^* \in \mathcal{X}$ such that $Z(\bar{x}^*) = \mathbf{0}$. This vector clears all grant markets. □

D.2. Uniqueness

The model primitives imply the substitution inequality required for uniqueness.

Proposition 2 (Global Uniqueness of Equilibrium)**.** *Under the Gumbel application shocks and Gaussian evaluation noise, the equilibrium $\bar{x}^*$ is globally unique.*

Proof. By the Gale–Nikaido univalence theorem, a continuously differentiable mapping defined on a rectangular region is globally injective if its Jacobian matrix is a P-matrix everywhere on that region. We apply this theorem to the mapping $-Z$, not to Z itself. Since the Jacobian of Z is $J(\bar{x})$, the Jacobian of $-Z$ is $-J(\bar{x})$. We show that $-J(\bar{x})$ is an M-matrix, which is a subclass of P-matrices.

Recall the elements of the Jacobian J derived in Section E:

$$J_{dd} = -\sum_{o=1}^{C} N_o \Pi_{od} \left[\frac{1}{\sigma}(1 - \Pi_{od}) S_{od} + \Pi_{od} S'_{od} \right]$$

$$J_{df} = \sum_{o=1}^{C} N_o \Pi_{od} \Pi_{of} \left[\frac{1}{\sigma} S_{od} - S'_{od} \right] \quad \text{for } f \neq d.$$

By Lemma 2, $J_{df} > 0$ for all strictly off-diagonal elements, making $-J_{df} < 0$. Thus, $-J$ has positive diagonal elements and non-positive off-diagonal elements.

To determine if $-J$ is strictly diagonally dominant, we sum the entries of a row of J:

$$\begin{aligned}\sum_{f=1}^{C} J_{df} &= J_{dd} + \sum_{f\neq d} J_{df} \\ &= \sum_{o=1}^{C} N_o \Pi_{od} \left[-\frac{1}{\sigma}(1-\Pi_{od}) S_{od} - \Pi_{od} S'_{od} + \sum_{f\neq d} \Pi_{of} \left(\frac{1}{\sigma} S_{od} - S'_{od} \right) \right] \\ &= \sum_{o=1}^{C} N_o \Pi_{od} \left[\frac{1}{\sigma} S_{od} \left(\sum_{f\neq d} \Pi_{of} - (1-\Pi_{od}) \right) - S'_{od} \left(\Pi_{od} + \sum_{f\neq d} \Pi_{of} \right) \right].\end{aligned}$$

Let Π_{o0} denote the probability of not applying. By the properties of the multinomial logit, $\sum_{f=1}^{C} \Pi_{of} + \Pi_{o0} = 1$. Therefore, $\sum_{f\neq d} \Pi_{of} = 1 - \Pi_{od} - \Pi_{o0}$. Substituting this yields:

$$\sum_{f=1}^{C} J_{df} = \sum_{o=1}^{C} N_o \Pi_{od} \left[-\frac{1}{\sigma} S_{od} \Pi_{o0} - S'_{od}(1-\Pi_{o0}) \right]. \tag{58}$$

Since $\Pi_{o0} > 0$, $S_{od} > 0$, and $S'_{od} > 0$, we have $\sum_{f=1}^{C} J_{df} < 0$ for every row d. This implies that $-J$ is strictly diagonally dominant by rows.

Any strictly diagonally dominant matrix with non-positive off-diagonal elements and positive diagonal elements is a non-singular M-matrix, and consequently a P-matrix. Because $-J(\bar{x})$ is a P-matrix everywhere, Gale–Nikaido implies that $-Z$ is injective on any rectangle in $\mathbb{R}^C$. Since injectivity of $-Z$ is equivalent to injectivity of Z, the system $Z(\bar{x}) = \mathbf{0}$ admits at most one solution. Combined with existence, this establishes that $\bar{x}^*$ is globally unique. □

E. Comparative Statics Derivation

This section derives the elasticity of each pairwise output effect, $\mathrm{CS1}_{do} = \frac{d \ln Y_o}{d \ln G_d}$, step by step from the primitives of the model. In this pairwise object, d is the funded field and o is the output field. Rather than reuse the output index as a destination index, the intermediate application equations use f for a generic destination field. Within those equations, o retains its application-model role as the origin field, and f' denotes another generic field when required.

Step 1: Total derivative via the chain rule. Output Y_o depends on the budget vector G only through the equilibrium paylines $\bar{x}(G)$. By the chain rule:

$$\frac{dY_o}{dG_d} = \sum_{f=1}^{C} \frac{\partial Y_o}{\partial \bar{x}_f} \cdot \frac{d\bar{x}_f}{dG_d}. \tag{59}$$

Converting to an elasticity by multiplying both sides by G_d/Y_o:

$$\frac{d \ln Y_o}{d \ln G_d} = \frac{G_d}{Y_o} \sum_{f=1}^{C} \frac{\partial Y_o}{\partial \bar{x}_f} \cdot \frac{d\bar{x}_f}{dG_d}. \tag{60}$$

Steps 2–5 derive the output gradient for a generic field. Step 8 applies that gradient to output field o. Steps 6–7 derive the GE payline adjustments $\frac{d\bar{x}_f}{dG_d}$.

Step 2: Differentiate total output with respect to paylines. Using the $\widetilde{Q}_{of}$ convention in Section 4, where $\widetilde{Q}_{of} = S_{of} Q_{of}$ includes the funding-probability weighting, total output in a generic field f can be written as:

$$Y_f = \mathbb{E}[f(g_i, k_i)] \sum_{o=1}^{C} N_o \, \Pi_{of}(\bar{x}) \, \widetilde{Q}_{of}(M_o(\bar{x}), \bar{x}_f).$$

The prefactor $\mathbb{E}[f(g_i, k_i)]$ does not depend on $\bar{x}$. Inside the sum, for each origin field o, the term $\Pi_{of} \cdot \widetilde{Q}_{of}$ is a product of two functions that both depend on $\bar{x}_{f'}$:

$$\frac{\partial Y_f}{\partial \bar{x}_{f'}} = \mathbb{E}[f(g_i, k_i)] \sum_{o=1}^{C} N_o \left[\frac{\partial \Pi_{of}}{\partial \bar{x}_{f'}} \cdot \widetilde{Q}_{of} + \Pi_{of} \cdot \frac{\partial \widetilde{Q}_{of}}{\partial \bar{x}_{f'}} \right]. \tag{61}$$

We need to compute $\frac{\partial \Pi_{of}}{\partial \bar{x}_{f'}}$ and $\frac{\partial \widetilde{Q}_{of}}{\partial \bar{x}_{f'}}$ separately.

Step 3: Differentiate the application probabilities. The application probability from origin field o to generic destination field f is given by the multinomial logit formula (8):

$$\Pi_{of}(\bar{x}) = \frac{\exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)}{\exp\left(\frac{c}{\sigma}\right) + \sum_{\ell=1}^{C} \exp\left(\frac{\mu_{o\ell} - \bar{x}_\ell}{\sigma}\right)}.$$

Write the denominator as $D_o = \exp(c/\sigma) + \sum_\ell \exp((\mu_{o\ell} - \bar{x}_\ell)/\sigma)$, so that $\Pi_{of} = \exp((\mu_{of} - \bar{x}_f)/\sigma)/D_o$.

When $f' = f$, both the numerator and denominator depend on $\bar{x}_f$:

$$\begin{aligned}
\frac{\partial \Pi_{of}}{\partial \bar{x}_f} &= \frac{1}{D_o} \frac{\partial}{\partial \bar{x}_f} \exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right) - \frac{\exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)}{D_o^2} \frac{\partial D_o}{\partial \bar{x}_f} \\
&= \frac{1}{D_o} \cdot \left(-\frac{1}{\sigma}\right) \exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right) - \frac{\exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)}{D_o^2} \cdot \left(-\frac{1}{\sigma}\right) \exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right) \\
&= -\frac{1}{\sigma} \Pi_{of} + \frac{1}{\sigma} \Pi_{of}^2 \\
&= -\frac{1}{\sigma} \Pi_{of} (1 - \Pi_{of}).
\end{aligned}$$

When $f' \neq f$, the numerator $\exp((\mu_{of} - \bar{x}_f)/\sigma)$ does not depend on $\bar{x}_{f'}$, so only the denominator changes:

$$\begin{aligned}
\frac{\partial \Pi_{of}}{\partial \bar{x}_{f'}} &= -\frac{\exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)}{D_o^2} \cdot \frac{\partial D_o}{\partial \bar{x}_{f'}} \\
&= -\frac{\exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)}{D_o^2} \cdot \left(-\frac{1}{\sigma}\right) \exp\left(\frac{\mu_{of'} - \bar{x}_{f'}}{\sigma}\right) \\
&= \frac{1}{\sigma} \cdot \frac{\exp\left(\frac{\mu_{of} - \bar{x}_f}{\sigma}\right)}{D_o} \cdot \frac{\exp\left(\frac{\mu_{of'} - \bar{x}_{f'}}{\sigma}\right)}{D_o} \\
&= \frac{1}{\sigma} \Pi_{of} \Pi_{of'}.
\end{aligned}$$

Step 4: Differentiate the inclusive value M_o. Before differentiating $\widetilde{Q}_{of}$, we need $\frac{\partial M_o}{\partial \bar{x}_{f'}}$. Recall from (55):

$$M_o(\bar{x}) = \sigma \ln D_o = \sigma \ln\left(\exp\left(\frac{c}{\sigma}\right) + \sum_{\ell=1}^{C} \exp\left(\frac{\mu_{o\ell} - \bar{x}_\ell}{\sigma}\right)\right).$$

Differentiating with respect to $\bar{x}_{f'}$:

$$\begin{aligned}\frac{\partial M_o}{\partial \bar{x}_{f'}} &= \sigma \cdot \frac{1}{D_o} \cdot \frac{\partial D_o}{\partial \bar{x}_{f'}} \\ &= \sigma \cdot \frac{1}{D_o} \cdot \left(-\frac{1}{\sigma}\right) \exp\left(\frac{\mu_{of'} - \bar{x}_{f'}}{\sigma}\right) \\ &= -\frac{\exp\left(\frac{\mu_{of'} - \bar{x}_{f'}}{\sigma}\right)}{D_o} \\ &= -\Pi_{of'}.\end{aligned}$$

Step 5: Differentiate the funded-quality term $\widetilde{Q}_{of}$. The funded-quality contribution from origin field o in generic destination field f is:

$$\widetilde{Q}_{of}(M_o, \bar{x}_f) = \int_{-\infty}^{\infty} (z + \bar{x}_f)\, \Phi\left(\frac{z}{\eta_{of}}\right) f_Z(z; M_o)\, dz.$$

This depends on $\bar{x}_{f'}$ through two channels: (i) directly via $\bar{x}_f$ in the integrand (only when $f' = f$), and (ii) indirectly via $M_o(\bar{x})$ in the density f_Z. We treat each in turn.

The direct effect of $\bar{x}_f$ follows because it appears linearly inside the integrand:

$$\begin{aligned}\left.\frac{\partial}{\partial \bar{x}_f}\right|_{\text{direct}} \widetilde{Q}_{of} &= \frac{\partial}{\partial \bar{x}_f} \int_{-\infty}^{\infty} (z + \bar{x}_f)\, \Phi\left(\frac{z}{\eta_{of}}\right) f_Z(z; M_o)\, dz \\ &= \int_{-\infty}^{\infty} 1 \cdot \Phi\left(\frac{z}{\eta_{of}}\right) f_Z(z; M_o)\, dz \\ &= S_{of}(M_o).\end{aligned}$$

The indirect effect operates through M_o. Recall that $f_Z(z; M_o) = f_{Z_0}(z - M_o)$ where $Z_0 \sim \text{Gumbel}(0, \sigma)$. Using the change of variables $z_0 = z - M_o$, rewrite $\widetilde{Q}_{of}$ as:

$$\widetilde{Q}_{of} = \int_{-\infty}^{\infty} (z_0 + M_o + \bar{x}_f)\, \Phi\left(\frac{z_0 + M_o}{\eta_{of}}\right) f_{Z_0}(z_0)\, dz_0.$$

Since f_{Z_0} does not depend on M_o, differentiating under the integral sign yields:

$$\begin{aligned}\frac{\partial \widetilde{Q}_{of}}{\partial M_o} &= \int_{-\infty}^{\infty} \Phi\left(\frac{z_0 + M_o}{\eta_{of}}\right) f_{Z_0}(z_0)\, dz_0 + \int_{-\infty}^{\infty} (z_0 + M_o + \bar{x}_f)\, \frac{1}{\eta_{of}}\, \phi\left(\frac{z_0 + M_o}{\eta_{of}}\right) f_{Z_0}(z_0)\, dz_0 \\ &= S_{of}(M_o) + \int_{-\infty}^{\infty} (z + \bar{x}_f)\, \frac{1}{\eta_{of}}\, \phi\left(\frac{z}{\eta_{of}}\right) f_Z(z; M_o)\, dz,\end{aligned}$$

where the last line reverts to the original variable $z = z_0 + M_o$. The first term arises because

a rightward shift in the quality gap distribution increases the mass of funded projects. The second term captures the change in the success-weighted quality contribution.

The chain rule combines the direct and indirect effects:

For $f' = f$ (own payline):

$$\frac{\partial \widetilde{Q}_{of}}{\partial \bar{x}_f} = \underbrace{S_{of}(M_o)}_{\text{direct payline effect}} + \underbrace{\frac{\partial \widetilde{Q}_{of}}{\partial M_o}}_{\text{through } M_o} \cdot \underbrace{\frac{\partial M_o}{\partial \bar{x}_f}}_{=-\Pi_{of} \text{ (inclusive-value derivative)}}$$

$$= S_{of}(M_o) - \Pi_{of} \frac{\partial \widetilde{Q}_{of}}{\partial M_o}.$$

For $f' \neq f$ (cross payline), the direct effect is zero because $\bar{x}_{f'}$ does not appear in the integrand of $\widetilde{Q}_{of}$:

$$\frac{\partial \widetilde{Q}_{of}}{\partial \bar{x}_{f'}} = \frac{\partial \widetilde{Q}_{of}}{\partial M_o} \cdot \frac{\partial M_o}{\partial \bar{x}_{f'}}$$

$$= -\Pi_{of'} \frac{\partial \widetilde{Q}_{of}}{\partial M_o}.$$

Step 6: Differentiate the market-clearing condition. We now derive the GE payline adjustments $\frac{d\bar{x}_f}{dG_d}$. The market-clearing condition for generic destination field f is:

$$E_f(\bar{x}) \equiv \sum_{o=1}^{C} N_o \, \Pi_{of}(\bar{x}) \, S_{of}(M_o(\bar{x})) = \frac{G_f}{\bar{g}_f}.$$

To find how $\bar{x}$ responds to a marginal change in G_d, totally differentiate each equation with respect to G_d:

$$\sum_{f'=1}^{C} \frac{\partial E_f}{\partial \bar{x}_{f'}} \frac{d\bar{x}_{f'}}{dG_d} = \begin{cases} \frac{1}{\bar{g}_d} & \text{if } f = d, \\ 0 & \text{if } f \neq d. \end{cases}$$

We now compute $\frac{\partial E_f}{\partial \bar{x}_{f'}}$. Using the product rule on $\Pi_{of} \cdot S_{of}$:

$$\frac{\partial E_f}{\partial \bar{x}_{f'}} = \sum_{o=1}^{C} N_o \left[\frac{\partial \Pi_{of}}{\partial \bar{x}_{f'}} \cdot S_{of} + \Pi_{of} \cdot \frac{\partial S_{of}}{\partial \bar{x}_{f'}} \right].$$

The first term uses the logit derivatives from Step 3. The second term requires $\frac{\partial S_{of}}{\partial \bar{x}_{f'}}$, which operates through M_o:

$$\frac{\partial S_{of}}{\partial \bar{x}_{f'}} = \frac{\partial S_{of}}{\partial M_o} \cdot \frac{\partial M_o}{\partial \bar{x}_{f'}} = S'_{of}(M_o) \cdot (-\Pi_{of'}) = -\Pi_{of'} \, S'_{of}(M_o),$$

where $S'_{of}(M_o) = \int_{-\infty}^{\infty} \frac{1}{\eta_{of}} \phi\left(\frac{z}{\eta_{of}}\right) f_Z(z; M_o)\, dz > 0$.

When $f' = f$, substitute the own-payline logit derivative and the derivative of S_{of}:

$$\begin{aligned}
\frac{\partial E_f}{\partial \bar{x}_f} &= \sum_{o=1}^{C} N_o \left[-\frac{1}{\sigma} \Pi_{of}(1 - \Pi_{of}) \cdot S_{of} + \Pi_{of} \cdot \left(-\Pi_{of}\, S'_{of} \right) \right] \\
&= \sum_{o=1}^{C} N_o \left[-\frac{1}{\sigma} \Pi_{of} S_{of} + \frac{1}{\sigma} \Pi_{of}^2 S_{of} - \Pi_{of}^2 S'_{of} \right] \\
&= -\sum_{o=1}^{C} N_o \Pi_{of} \left[\frac{1}{\sigma} (1 - \Pi_{of}) S_{of} + \Pi_{of} S'_{of} \right].
\end{aligned}$$

This is strictly negative: raising the payline in f both (i) drives applicants away (Π_{of} falls) and (ii) lowers M_o, reducing their funding probability.

When $f' \neq f$, substitute the cross-payline logit derivative:

$$\begin{aligned}
\frac{\partial E_f}{\partial \bar{x}_{f'}} &= \sum_{o=1}^{C} N_o \left[\frac{1}{\sigma} \Pi_{of} \Pi_{of'} \cdot S_{of} + \Pi_{of} \cdot \left(-\Pi_{of'}\, S'_{of} \right) \right] \\
&= \sum_{o=1}^{C} N_o \Pi_{of} \Pi_{of'} \left[\frac{1}{\sigma} S_{of} - S'_{of} \right].
\end{aligned}$$

The results from Step 6 are collected in the following lemma.

Lemma 1 (Jacobian of the Equilibrium System)**.** *Let J denote the $C \times C$ Jacobian matrix of the equilibrium system (10) with respect to $\bar{x}$, with representative element $J_{ff'} = \frac{\partial E_f}{\partial \bar{x}_{f'}}$. Then:*

$$J_{ff} = -\sum_{o=1}^{C} N_o \Pi_{of} \left[\frac{1}{\sigma} (1 - \Pi_{of}) S_{of}(M_o) + \Pi_{of} S'_{of}(M_o) \right] < 0, \tag{62}$$

$$J_{ff'} = \sum_{o=1}^{C} N_o \Pi_{of} \Pi_{of'} \left[\frac{1}{\sigma} S_{of}(M_o) - S'_{of}(M_o) \right] \quad \text{for } f' \neq f. \tag{63}$$

Lemma 2 (Substitution Dominance)**.** *For every origin o, destination f, and finite M_o,*

$$\frac{1}{\sigma} S_{of}(M_o) - S'_{of}(M_o) > 0. \tag{64}$$

Proof. Let $\xi_{of} \sim \mathcal{N}(0, \eta_{of}^2)$ denote evaluation noise and define $T = \exp((M_o + \xi_{of})/\sigma) > 0$. Conditional on ξ_{of}, the Gumbel distribution implies

$$S_{of}(M_o) = \mathbb{E}[1 - e^{-T}], \qquad S'_{of}(M_o) = \frac{1}{\sigma} \mathbb{E}[T e^{-T}].$$

Hence

$$\frac{1}{\sigma}S_{of}(M_o) - S'_{of}(M_o) = \frac{1}{\sigma}\mathbb{E}[1-(1+T)e^{-T}] > 0,$$

because $e^T > 1+T$ for $T > 0$. □

Thus $J_{ff'} > 0$ for $f' \neq f$ and $\sum_{f'} J_{ff'} < 0$, so $-J$ is a strictly diagonally dominant M-matrix.

Step 7: Solve the linear system. The system from Step 6 is:

$$\sum_{f'=1}^{C} \frac{\partial E_f}{\partial \bar{x}_{f'}} \frac{d\bar{x}_{f'}}{dG_d} = \mathbf{1}_{f=d} \cdot \frac{1}{\bar{g}_d}, \quad f = 1, \ldots, C.$$

To see the logic cleanly, define the equilibrium mapping

$$F_f(\bar{x}, G) \equiv E_f(\bar{x}) - \frac{G_f}{\bar{g}_f}, \qquad f = 1, \ldots, C.$$

An equilibrium is any vector $\bar{x}^*(G)$ such that $F(\bar{x}^*(G), G) = 0$. The paylines are therefore not free arguments once G is given; they are endogenous functions pinned down implicitly by the full system. Differentiating the identity $F(\bar{x}^*(G), G) = 0$ with respect to G_d gives

$$\sum_{f'=1}^{C} \frac{\partial F_f}{\partial \bar{x}_{f'}} \frac{d\bar{x}_{f'}}{dG_d} + \frac{\partial F_f}{\partial G_d} = 0.$$

Since $\partial F_f/\partial \bar{x}_{f'} = \partial E_f/\partial \bar{x}_{f'}$ and $\partial F_f/\partial G_d = -\mathbf{1}_{f=d}/\bar{g}_d$, this reduces to the system above. This is why the object of interest is a total derivative $d\bar{x}_f/dG_d$: when G_d changes, every equilibrium payline may move to keep all markets clearing simultaneously.

Proposition 3 (General Equilibrium Payline Adjustments)**.** *Suppose there exists an equilibrium $\bar{x}^*$ solving (10). The Jacobian evaluated at $\bar{x}^*$ is invertible and satisfies $J^{-1} \leq 0$ elementwise. The Implicit Function Theorem then implies the local comparative statics*

$$d\bar{x} = J^{-1}\mathrm{diag}(1/\bar{g}_f)\, dG. \tag{65}$$

Hence an increase in the budget of funded field d weakly lowers every field's payline:

$$\frac{\partial \bar{x}_f}{\partial G_d} = (J^{-1})_{fd}\frac{1}{\bar{g}_d} \leq 0 \quad \textit{for all } f. \tag{66}$$

Proof. By Lemma 2, the uniqueness proof establishes that $-J(\bar{x}^*)$ is a non-singular M-matrix. A non-singular M-matrix has a non-negative inverse, so $(-J(\bar{x}^*))^{-1} \geq 0$ elementwise. Since $J(\bar{x}^*)^{-1} = -(-J(\bar{x}^*))^{-1}$, it follows that $J(\bar{x}^*)^{-1} \leq 0$ elementwise. The

Implicit Function Theorem applied to $F(\bar{x}, G) = E(\bar{x}) - \text{diag}(1/\bar{g}_f)G$ gives

$$d\bar{x} = J(\bar{x}^*)^{-1}\text{diag}(1/\bar{g}_f)\, dG.$$

Because $\bar{g}_f > 0$ for every field, the sign of each payline response is inherited from $J(\bar{x}^*)^{-1}$. □

The expression above is local. It describes the slope of the equilibrium correspondence at the baseline allocation, not the exact response to a large discrete funding reform. For small budget changes, this linearisation is the correct object. For larger changes, one should re-solve the full nonlinear equilibrium system at the new budget vector rather than extrapolating from J^{-1}.

In partial equilibrium, holding $\bar{x}_f$ fixed for $f \neq d$, an increase in G_d lowers $\bar{x}_d$. This creates *cross-field crowding out*: applications to other fields f fall because Π_{of} declines. It also raises M_o, shifting the conditional quality distribution upwards.

In general equilibrium, the paylines $\bar{x}_f$ must also adjust to clear the remaining markets. Since applicant volume in field f falls while G_f is fixed, field f must lower $\bar{x}_f$ to fund a larger share of the remaining applicants. The response of applicant volumes depends on the full matrix J^{-1}, but the payline effect is weakly negative under the stated conditions.

Step 8: Assemble the elasticity. Steps 2–5 give $\frac{\partial Y_f}{\partial \bar{x}_{f'}}$ for a generic output field f. Relabelling that generic field as output field o gives $\frac{\partial Y_o}{\partial \bar{x}_f}$ in Step 1. Substituting the solution from Step 7 gives $\frac{d\bar{x}_f}{dG_d}$. The elasticity is then:

$$\text{CS1}_{do} = \frac{G_d}{Y_o} \sum_{f=1}^{C} \frac{\partial Y_o}{\partial \bar{x}_f} \cdot \frac{d\bar{x}_f}{dG_d}. \tag{67}$$

F. Field-Specific Knowledge Depreciation Rates

This section documents the construction of the cited-field-specific depreciation rates δ_d entering the knowledge stock in the production function.

F.1. Reference Age Distribution

To construct these rates, I begin with citation ages in the OpenAlex network. For each citing publication p with observed publication year t_p, I retain a reference r only when its publication year t_r is recoverable from OpenAlex metadata, it can be matched to an NIH grant through a unique PubMed identifier (PMID), and the matched grant is

attributed to a single cited field d.[16] For each retained publication–reference pair (p, r), the *reference age* is

$$a_{pr} = t_p - t_r, \quad a_{pr} \geq 1.$$

Restricting citing publications to those from 2002 onwards and aggregating the retained pairs gives, for each cited field d and age a, the citation count C_{da}: the number of times a paper attributed to field d was cited at age a.

F.2. Half-Life Estimation

For cited field d, the empirical cumulative citation distribution is

$$\hat{F}_d(a) = \frac{\sum_{a'=1}^{a} C_{da'}}{\sum_{a'=1}^{A} C_{da'}},$$

where A is the maximum observed age. Let a_d be the first age at which half of the references have accumulated. I linearly interpolate the empirical distribution between adjacent ages to obtain the *citation half-life* h_d:

$$a_d = \min\{a : \hat{F}_d(a) \geq 0.5\}, \qquad h_d = \begin{cases} a_d, & a_d = 1, \\ a_d - 1 + \dfrac{0.5 - \hat{F}_d(a_d - 1)}{\hat{F}_d(a_d) - \hat{F}_d(a_d - 1)}, & a_d > 1. \end{cases}$$

A short half-life indicates that citations to publications in cited field d are concentrated in the years immediately following publication (i.e. the field's stock of relevant knowledge turns over rapidly).

F.3. Implied Depreciation Rate

Under geometric decay, a knowledge unit remains in productive use after a periods with fraction $(1 - \delta_d)^a$. Setting this fraction to one half at the empirical half-life, $(1 - \delta_d)^{h_d} = 0.5$, gives

$$\delta_d = 1 - 0.5^{1/h_d}. \tag{68}$$

This rate governs knowledge accumulation. Starting from $K_{d,0} = 0$, the depreciated stock evolves as

$$K_{d,t} = (1 - \delta_d)\, K_{d,t-1} + P_{d,t},$$

where $P_{d,t}$ counts distinct R01-linked publications attributed to cited field d in year t, regardless of the outcome measure used in the production regression. Thus $K_{d,t}$ is the post-production stock generated by publications through year t and is available for

[16] References matched to multiple institutes are discarded to ensure unambiguous field attribution.

production from year $t + 1$.

G. GMM Implementation Details

Table A.3. Identifying variation used by the estimator

Parameter	Identifying variation	Primary moments
μ_{od} (full)	Bilateral scientist-flow shares vary with the mean idea quality of each origin–destination pair, conditional on paylines and pool sizes.	Flow-share moments m^F_{od}
σ	Cross-institute substitution patterns: a higher σ implies flatter, less responsive choice probabilities as paylines change over time, altering application volumes and the composition of scientist flows	Application moments $m^A_{d,t}$ and flow-share moments m^F_{od}
c_0, c_1	Outside-option intercept and time trend identified from the aggregate level and time path of total applications relative to pool size $N_{o,t}$, conditional on the idea-quality matrix and paylines	Application moments $m^A_{d,t}$
γ ($\eta_0 = 1$ fixed)	The distance gradient of evaluation noise changes the funding probabilities of origin–destination pairs. Its identifying variation is the joint response of aggregate applications and the composition of pooled scientist flows to scientific distance	Application moments $m^A_{d,t}$ and flow-share moments m^F_{od}
$\bar{x}_{d,t}$ (solved)	Determined from the market-clearing condition given $(\mu_{od}, \sigma, c_t, \eta_{od})$; not a free parameter	Market-clearing

Notes: Each row links a parameter block to the empirical variation that primarily identifies it and to the corresponding moment conditions. Identification is joint across blocks because each candidate parameter vector requires solving the market-clearing paylines $\bar{x}_{d,t}$. Paylines are equilibrium objects rather than free parameters.

This section documents the numerical implementation of the estimation procedure described in Section 5.

The structural parameter vector is

$$\theta = \big(\mathrm{vec}(\beta_d),\, \log\sigma,\, c_0,\, c_1,\, \gamma\big),$$

where $\beta_d \in \mathbb{R}^{C-1}$ and $\mu_{\cdot d} = H\beta_d$. We use the matrix $H \in \mathbb{R}^{C\times(C-1)}$ to enforce the parameter restriction that $\sum_o \mu_{od} = 0$, which is necessary for identification given the multinomial logit structure of Π_{od}. The matrix H is an orthonormal Helmert basis

satisfying $H^\top H = I$ and $\mathbf{1}^\top H = 0$. Baseline evaluation noise is fixed at $\eta_0 = 1$. The scale $\sigma > 0$ is reparametrised as $\log \sigma$, and γ is constrained to $[0, \infty)$. The total free parameter count is $C(C-1) + 4 = 346$ for $C = 19$.

G.1. Normalisation Map

Start with any set of parameter values, including one that has not yet been normalised. For each destination d, let $\bar{\mu}_{\cdot d} = C^{-1} \sum_o \mu_{od}$ be the average idea quality across origin fields. We subtract this average from the destination's idea-quality parameters and express the idea quality parameters relative to the baseline noise level η_0, which acts as the numeraire in that system. This gives the normalised representation:

$$\mu^*_{od} = \frac{\mu_{od} - \bar{\mu}_{\cdot d}}{\eta_0}, \quad \bar{x}^*_{d,t} = \frac{\bar{x}_{d,t} - \bar{\mu}_{\cdot d}}{\eta_0}, \quad \sigma^* = \frac{\sigma}{\eta_0}, \quad c^*_j = \frac{c_j}{\eta_0}, \quad \eta^*_0 = 1, \quad \gamma^* = \gamma. \tag{69}$$

These normalisations are necessary for the model to be identified.

The nonlinear system of market-clearing conditions is solved with SciPy's `root` routine using the hybrid Powell method `hybr`, trying multiple starting points to avoid spurious local solutions. Each year's solve is warm-started from the previous year's payline vector, exploiting temporal continuity.

The integrals S_{od} and $\widetilde{Z}_{od}(M_{o,t}; \eta_{od}) = \int z \, \Phi(z/\eta_{od}) \, f_Z(z; M_{o,t}, \sigma) \, dz$ are approximated by deterministic quadrature over the Gumbel$(0, \sigma)$ density, evaluating 100 quantile points with PDF weights.

The estimator starts from the preceding unrestricted estimate transformed by the normalisation map above. If no preceding estimate is supplied, μ_{od} is initialised from logarithmic bilateral scientist-flow shares, centred within destination, and projected onto the Helmert basis.

Both GMM stages use the limited-memory BFGS algorithm with box constraints (`L-BFGS-B`). I add a ridge penalty of $\lambda_1 = 10^{-4}$ in the first stage to avoid extreme parameter values. Non-convergence of the equilibrium solver for a given year incurs a smooth penalty proportional to the mean squared market-clearing residual, steering the optimiser away from parameter regions where the system has no solution.

To assess identification numerically, I evaluate the $1{,}037 \times 346$ Jacobian of the two-block moment vector by central finite differences. The implementation retains an active estimate only when both the unweighted and second-stage-weighted Jacobians have rank 346 at the declared relative tolerance 10^{-8}.

The unweighted and weighted Jacobian condition numbers are 3.16×10^7 and 2.23×10^7, respectively. At the stricter relative tolerance 10^{-6}, their numerical ranks are 342 and 343.

G.2. GMM Sandwich Inference

Let $m(\theta)$ denote the 1,037-dimensional unpenalised relative-moment vector, $D = \partial m(\theta)/\partial \theta^\top$ its numerical Jacobian, W the diagonal second-stage weight matrix, and Ω the sampling covariance of $m(\theta)$. The GMM covariance is

$$\widehat{V}_\theta = (D^\top W D)^{-1} D^\top W \widehat{\Omega} W D (D^\top W D)^{-1}. \tag{70}$$

The calculation uses the unpenalised moment map and excludes the first-stage ridge and equilibrium-failure penalties. The estimator retains the stage-one estimate and the exact diagonal stage-two matrix constructed from the stage-one residuals, so Equation (70) uses the same W as the second-stage objective.

The moment vector contains 266 application moments and 771 scientist-flow-share moments pooled separately over 2005–2009, 2010–2014, and 2015–2018. For the application block, I construct $A_{dt} = n_{dt}/\pi_{dt}$ from the reported funded count n_{dt} and success rate π_{dt}, condition on n_{dt}, and use the binomial variance $\pi_{dt}(1-\pi_{dt})/A_{dt}$ for the reported rate. Pooled bilateral flow shares use the conditional multinomial covariance within destination and epoch. The covariance conditions on the realised applicant-pool series, including its application-based scale adjustment, and does not propagate generated-pool uncertainty. Finally, the transformation from $\log \sigma$ to σ uses the delta method, while the fixed normalisation $\eta_0 = 1$ has no standard error.

The reported confidence intervals use a conditional covariance that excludes the three weak weighted-Jacobian directions. Write $\widetilde{D} = W^{1/2} D = U \operatorname{diag}(s) V^\top$ and set $\tau = 10^{-6} s_1$. Let $\mathcal{R} = \{j : s_j > \tau\}$; the active estimate has $|\mathcal{R}| = 343$. I construct the truncated left inverse

$$\widetilde{D}_\tau^+ = V_{\mathcal{R}} \operatorname{diag}(s_{\mathcal{R}}^{-1}) U_{\mathcal{R}}^\top, \qquad \widehat{V}_\theta^{(\tau)} = \widetilde{D}_\tau^+ \big(W^{1/2} \widehat{\Omega} W^{1/2}\big) \big(\widetilde{D}_\tau^+\big)^\top. \tag{71}$$

This operation does not change the GMM estimate, fitted moments, criterion, or equilibrium paylines. It holds the three discarded singular directions fixed when propagating uncertainty. The resulting intervals are therefore conditional on this discarding.

H. Computing Marginal Returns

The numerical comparative-statics tables, figures, and confidence intervals below use the GMM estimate reported in the main text and the production-function estimates.

Section 4 presents the analytical decomposition. I implement it numerically by finite differences around the final estimation year $t^* = T - 1$ (2018 in the current sample window). Let baseline available grants be $B_{d,t^*} \equiv G_{d,t^*}/\bar{g}_{d,t^*}$. I perturb one funded field

at a time by $H = 0.01$:

$$B'_{d,t^*} = (1 + H)B_{d,t^*}. \tag{72}$$

I then re-solve the equilibrium paylines and evaluate model-implied output at both the baseline and counterfactual equilibria.

Baseline output in output field o is evaluated from structural primitives as

$$\hat{Y}^{\text{base}}_{o,t^*} = \hat{f}_o \sum_{f=1}^{C} N_{f,t^*} \Pi_{fo,t^*} S_{fo,t^*} Q^{\hat{\lambda}}_{fo,t^*}, \tag{73}$$

where Q_{fo,t^*} is defined in (39). The extensive-margin elasticity matrix is then computed as

$$\varepsilon^{\text{ext}}_{do} \approx \frac{\hat{Y}_{o,t^*}(B'_{d,t^*}) - \hat{Y}_{o,t^*}(B_{d,t^*})}{H\, \hat{Y}_{o,t^*}(B_{d,t^*})}. \tag{74}$$

The knowledge channel propagates only the current output response of funded field d through Equation (24):

$$\varepsilon^{\text{know}}_{do} = \frac{q\hat{\alpha}_o \Gamma_{do}}{1 - q(1 - \hat{\delta}_d)} \frac{\hat{Y}^{\text{base}}_{d,t^*}}{K_{d,t^*}} \varepsilon^{\text{ext}}_{dd}, \qquad q = \frac{1}{1.05}. \tag{75}$$

This converts d's output response into a knowledge-stock change and discounts its effect on output field o from the following year; current output changes in other fields are not propagated. Under the approximation in Equation (24), I use the baseline ratio at t^* for the first future period. I assume a 5 per cent interest rate and estimate the field-specific depreciation rate $\hat{\delta}_d$ from the half-life of yearly citation counts for the average paper in field d. The total elasticities are

$$\varepsilon^{\text{tot}}_{do} = \varepsilon^{\text{ext}}_{do} + \varepsilon^{\text{know}}_{do}. \tag{76}$$

I compute $\varepsilon^{\text{tot}}_{do}$ from model-implied baseline output. To express the result as the pairwise output effect of a \$1 million increase in G_d, I scale the elasticity by observed baseline output and funding:

$$\Delta Y^{(\$1\text{M})}_{o \leftarrow d} = \varepsilon^{\text{tot}}_{do} \left(\frac{Y_{o,t^*}}{G_{d,t^*}} \right) 10^6. \tag{77}$$

Here Y_{o,t^*} denotes observed output in output field o at the baseline year, while the elasticity is evaluated at model-predicted output.

H.1. Marginal-Return Confidence Intervals

For each outcome q and funded field d, let $R_{qd}(\theta, \psi_q)$ denote the total 2018 marginal return obtained by summing the pairwise \$1 million effects over all output fields. Here θ is the structural GMM vector and $\psi_q = (\alpha_{q1}, \ldots, \alpha_{q19}, \lambda_q)^\top$ contains the destination-specific knowledge elasticities and the shared coefficient on the selection control. I evaluate $G_q = \partial R_q / \partial \theta^\top$ and $H_q = \partial R_q / \partial \psi_q^\top$ numerically by applying small parameter changes and recomputing the marginal returns. Every baseline and perturbed equilibrium must converge. Changes are one-sided for a knowledge elasticity at its non-negativity bound.

$$\widehat{V}_{R_q} = G_q \widehat{V}_\theta^{(\tau)} G_q^\top + H_q \widehat{V}_{\psi_q} H_q^\top. \tag{78}$$

The production covariance $\widehat{V}_{\psi_q}$ is the joint 20×20 working-weight covariance from the constrained negative-binomial production fit. I set the cross-stage covariance between θ and ψ_q to zero. This plug-in approximation does not propagate dependence induced by constructing the production-function selection control from the GMM estimates. I report pointwise 95 per cent intervals $\hat{R}_{qd} \pm 1.96\sqrt{[\widehat{V}_{R_q}]_{dd}}$.

I. Constructing the Output Measures

The production-function estimates use three measures of scientific output. Each observation i is a unique grant–investigator pair, so its grant is implicit in i. Let $\mathcal{P}_i$ denote the set of OpenAlex publications linked to that grant and let n_i^{PI} denote the number of investigators on it. I split each linked publication equally across the grant's investigators and retain each grant–publication link when a publication is linked to several grants. The output under measure m of observation i is

$$Y_i^m = \sum_{p \in \mathcal{P}_i} \frac{w_p^m}{n_i^{\mathrm{PI}}}, \tag{79}$$

where w_p^m is the weight attached to publication link p under measure m; the link embeds the relevant grant.

I.1. Academic Impact

The academic-impact measure weights publication quantity by OpenAlex's two-year mean-citedness statistic, `2yr_mean_citedness`, matched to each publication's journal source by ISSN-L (OpenAlex, n.d.). The data-cleaning code stores this statistic in the `impact_factor` field; here IF_p denotes that OpenAlex value, not the 2014 Journal

Citation Reports impact factor. The publication weight is

$$w_p^{\mathrm{IF}} = IF_p, \tag{80}$$

and the observation-level outcome is

$$Y_i^{\mathrm{IF}} = \sum_{p \in \mathcal{P}_i} \frac{IF_p}{n_i^{\mathrm{PI}}}. \tag{81}$$

This measure uses a journal-source statistic rather than publication-level citations. Publications without a matched journal-source statistic contribute zero to the weighted sum.

I.2. Commercial Potential

The commercial-potential measure weights publication quantity by commercial potential. For each publication p, I merge the DOI-level score from Masclans, Hasan and Cohen (2023) and let $c_p \in [0, 1]$ denote that score. The publication weight is

$$w_p^{\mathrm{CP}} = c_p, \tag{82}$$

and the observation-level outcome is

$$Y_i^{\mathrm{CP}} = \sum_{p \in \mathcal{P}_i} \frac{c_p}{n_i^{\mathrm{PI}}}. \tag{83}$$

The score is interpreted as the publication's commercial potential. Publications without a matched DOI-level score receive weight zero. In the analysis sample, the score is observed for 89 per cent of grant-publication observations.

I.3. Disease Burden

The NIH's 2017 RCDC–GBD crosswalk (National Institutes of Health, n.d.[c]) maps grant categories to burden values. I use its US DALY values, following the Global Burden of Disease 2017 DALY estimates (GBD 2017 DALYs and HALE Collaborators, 2018), to weight the funding grant rather than the publication.[17] Let T_g denote the RCDC tags of grant g, let C denote the set of RCDC tags matched to the burden workbook, and let B_r denote disease burden for matched tag r. For grants with $M_g = T_g \cap C \neq \emptyset$,

[17] The NIH crosswalk workbook contains 74 matched RCDC–GBD category pairs and corresponding US deaths, US DALYs, global deaths, and global DALYs for fiscal year 2017.

the matched burden is

$$\tilde{b}_g = \frac{1}{|M_g|} \sum_{r \in M_g} B_r. \tag{84}$$

If M_g is empty, I impute the burden with the mean matched burden among grants funded by the same institute. If that institute has no matched grant, I use the overall mean matched burden. Let b_g denote this observed-or-imputed raw grant burden. The normalised grant weight is

$$w_g^B = \frac{b_g}{|G|^{-1} \sum_{g' \in G} b_{g'}}, \tag{85}$$

so that the average grant has weight one. Let w_i^B denote the weight of the grant represented by observation i; every publication funded by that grant receives this weight. The disease-burden-weighted observation-level outcome is therefore

$$Y_i^{\mathrm{DB}} = \sum_{p \in \mathcal{P}_i} \frac{w_i^B}{n_i^{\mathrm{PI}}} = \frac{|\mathcal{P}_i| w_i^B}{n_i^{\mathrm{PI}}}. \tag{86}$$

This outcome measures publications weighted by the disease burden of the funding grant, not by disease content inferred from the publication.

Tables A.4 and A.5 report coverage. In the analysis sample, 45 per cent of grants have a direct DALY match, 33.4 per cent have no RCDC tag, and 21.6 per cent have RCDC tags without a non-missing workbook match. The grant-level disease-burden weight is weakly correlated with the impact-factor and commercial-potential measures (Table A.6); the corresponding scatter plots are reported in Figure 11.

Table A.4. RCDC–DALY coverage summary

Unit	Total	No RCDC	RCDC no DALY	Direct DALY	IC avg
Grants	65,349	21,859 (33.4)	14,088 (21.6)	29,402 (45.0)	35,947 (55.0)
Publication links	1,179,568	398,211 (33.8)	245,434 (20.8)	535,923 (45.4)	643,645 (54.6)

Notes: The table reports RCDC–DALY matching coverage in the 2002–2018 analysis sample. "No RCDC" denotes grants without an RCDC tag. "RCDC no DALY" denotes grants whose RCDC tags are absent from the US-DALY workbook. "Direct DALY" denotes at least one matched RCDC–DALY tag. "IC avg" identifies institute-average imputation when no direct match exists. Publication links are unique grant–publication pairs.

Table A.5. RCDC–DALY coverage

Institute	Grants	G direct	G no RCDC	G no DALY	Pub. links	P direct	P no RCDC	P no DALY
NIA	3,015	1,339	937	739	65,480	28,919	21,418	15,143
NIAAA	1,217	823	394	0	19,105	13,199	5,906	0
NIAID	6,477	2,245	2,116	2,116	105,208	36,557	34,344	34,307
NIAMS	1,847	520	633	694	35,800	10,173	12,659	12,968
NIBIB	1,223	371	481	371	30,151	8,862	12,477	8,812
NCI	9,366	6,113	3,249	4	162,875	107,286	55,513	76
NICHD	3,034	1,255	927	852	45,936	18,997	13,849	13,090
NIDCD	1,305	224	391	690	21,408	3,749	6,728	10,931
NIDCR	1,050	684	366	0	19,095	12,671	6,424	0
NIDDK	5,160	3,403	1,488	269	91,922	61,695	25,854	4,373
NIDA	2,688	1,683	1,004	1	44,024	26,174	17,807	43
NIEHS	1,089	512	336	241	21,279	10,456	6,047	4,776
NEI	1,961	364	553	1,044	35,899	6,923	11,078	17,898
NIGMS	7,198	1,243	2,763	3,192	125,779	20,710	50,530	54,539
NHLBI	8,350	4,478	2,714	1,158	174,637	96,800	55,385	22,452
NHGRI	457	41	136	280	9,087	850	2,055	6,182
NIMH	4,115	1,833	1,397	885	74,977	32,511	26,902	15,564
NINDS	5,215	2,011	1,771	1,433	89,921	36,221	31,183	22,517
NINR	582	260	203	119	6,985	3,170	2,052	1,763

Notes: Rows are NIH institutes in the 2002–2018 analysis sample. "G" denotes unique grants and "P" unique grant–publication links. "Direct" identifies observations with at least one matched RCDC–DALY tag; "No RCDC" and "No DALY" identify missing RCDC tags and unmatched tags, respectively. Column percentages use the corresponding grant or publication-link count as denominator.

Table A.6. Grant-level correlations between disease burden and publication outcomes

Pair	N	Pearson	Spearman	Log Pearson
Burden–IF	65,349	0.048	0.053	0.040
Burden–commercial	65,349	0.099	0.144	0.110
IF–commercial	65,349	0.846	0.819	0.809

Notes: The unit of observation is a unique grant in 2002–2018. For each pair, the table reports the common sample size, Pearson and Spearman correlations in levels, and the Pearson correlation between log disease-burden weight and $\log(1 + y)$, where y is impact-factor-weighted or commercial-potential-weighted publication output.

J. Production Function Estimation Details

This section documents aspects of the production function estimation not covered by the main text.

J.1. Estimator

I estimate the production function in equation (38) by constrained negative binomial maximum likelihood. This specification allows for over-dispersion relative to the Poisson benchmark, which is important given the heavy right tail of the impact-factor-weighted output distribution. I impose the economically required bounds $\alpha_d \geq 0$ and $\beta_d \geq 0$ throughout the optimisation. The constrained likelihood is maximised with L-BFGS-B, with gradients numerically approximated by SciPy for the negative-binomial objective.

J.2. Standard Errors

The standard errors in Table A.7 use a quasi-Newton GLM negative-binomial working-weight covariance proxy evaluated at the constrained maximum: $\widehat{\mathrm{Var}}(\hat{\theta}) = (X'WX)^{+}$, with weights $w_i = 1/(1/\hat{\mu}_i + a)$, where $\hat{\mu}_i$ is the fitted mean and the dispersion parameter is fixed at $a = 1$. At an interior solution, this proxy approximates the conventional observed-information standard error. At a boundary, such as $\hat{\beta}_{\mathrm{NIDCR}} = 0$, it reflects local curvature at the constraint.

J.3. Full Destination-Specific Estimates

The main text reports summary statistics across destinations. The full destination-specific production-function estimates for the three weighted outcomes are reported below.

Table A.7. Production-function estimates by output measure

Institute	Academic impact		Commercial potential		Disease burden	
	Knowledge elasticity $\hat{\alpha}_d$	Funding elasticity $\hat{\beta}_d$	Knowledge elasticity $\hat{\alpha}_d$	Funding elasticity $\hat{\beta}_d$	Knowledge elasticity $\hat{\alpha}_d$	Funding elasticity $\hat{\beta}_d$
Aging	0.244***	0.219***	0.382***	0.197***	0.055**	0.260***
	(0.022)	(0.024)	(0.025)	(0.027)	(0.023)	(0.026)
Alcoholism	0.553***	0.041	0.650***	0.000	0.376***	0.034
	(0.051)	(0.034)	(0.060)	(0.038)	(0.053)	(0.035)
Allergy and Infectious Diseases	0.219***	0.080***	0.286***	0.039**	0.232***	0.048***
	(0.017)	(0.015)	(0.019)	(0.016)	(0.019)	(0.016)
Arthritis and Musculoskeletal and Skin Diseases	0.392***	0.119***	0.397***	0.073**	0.270***	0.075**
	(0.032)	(0.029)	(0.035)	(0.030)	(0.034)	(0.030)
Biomedical Imaging and Bioengineering	0.134***	0.179***	0.116***	0.234***	0.068	0.234***
	(0.042)	(0.046)	(0.044)	(0.050)	(0.042)	(0.048)
Cancer	0.356***	0.106***	0.429***	0.068***	0.260***	0.102***
	(0.014)	(0.015)	(0.015)	(0.016)	(0.014)	(0.015)
Child Health and Human Development	0.390***	0.082***	0.424***	0.009	0.283***	0.083***
	(0.021)	(0.017)	(0.026)	(0.020)	(0.024)	(0.019)
Deafness and Communication Disorders	0.523***	0.061*	0.269***	0.044	0.195***	0.098***
	(0.055)	(0.035)	(0.060)	(0.037)	(0.060)	(0.038)
Dental and Craniofacial Research	0.402***	0.000†	0.462***	0.000†	0.425***	0.000†
	(0.042)	(0.034)	(0.046)	(0.036)	(0.048)	(0.037)
Diabetes and Digestive and Kidney Diseases	0.257***	0.051***	0.363***	0.020	0.143***	0.037***
	(0.016)	(0.013)	(0.018)	(0.013)	(0.017)	(0.014)
Drug Abuse	0.449***	0.082***	0.618***	0.000†	0.364***	0.094***
	(0.031)	(0.028)	(0.037)	(0.032)	(0.032)	(0.029)
Environmental Health	0.199***	0.220***	0.618***	0.091**	0.356***	0.153***
	(0.037)	(0.041)	(0.043)	(0.044)	(0.039)	(0.042)
Eye	0.172***	0.215***	0.107**	0.190***	0.186***	0.225***
	(0.043)	(0.037)	(0.046)	(0.040)	(0.047)	(0.041)
General Medicine	0.190***	0.040**	0.190***	0.015	0.110***	0.016
	(0.018)	(0.016)	(0.019)	(0.017)	(0.019)	(0.016)
Heart, Lung, and Blood	0.098***	0.124***	0.256***	0.095***	0.063***	0.124***
	(0.013)	(0.014)	(0.014)	(0.015)	(0.013)	(0.014)
Human Genome	0.542***	0.156**	0.561***	0.098	0.273***	0.109
	(0.056)	(0.076)	(0.064)	(0.082)	(0.058)	(0.078)
Mental Health	0.278***	0.145***	0.385***	0.074***	0.163***	0.083***
	(0.024)	(0.023)	(0.027)	(0.026)	(0.026)	(0.025)
Neurological Disorders and Stroke	0.237***	0.101***	0.234***	0.125***	0.116***	0.096***
	(0.022)	(0.020)	(0.024)	(0.023)	(0.024)	(0.022)
Nursing Research	0.200***	0.189***	0.383***	0.116*	0.093*	0.103**
	(0.048)	(0.048)	(0.061)	(0.061)	(0.050)	(0.050)
Selection-control coefficient ($\hat{\lambda}$) on $\ln Q_{od,t}$	-0.001		-0.045*		0.080***	
	(0.022)		(0.024)		(0.023)	
Year FE	×	×	×	×	×	×
Destination FE	×	×	×	×	×	×
Observations	55,710		55,710		55,710	

Notes: The three outcomes are impact-factor-weighted (Y_d^{IF}), commercial-potential-weighted (Y_d^{CP}), and disease-burden-weighted (Y_d^{DB}) publications per PI. The constrained negative-binomial estimates use 55,710 grant–PI observations from 2005–2018 and impose $\hat{\alpha}_d, \hat{\beta}_d \geq 0$. Knowledge input k_i aggregates field-specific depreciated stocks using citation-share weights; g_i is real grant funding per PI. All specifications include destination and year fixed effects and the structural selection control $\ln Q_{od,t}$ from Equation (39). Standard errors appear below the estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$; † denotes a binding non-negativity constraint.

K. Predicting Missing Disease-Burden Links

This section describes a text classifier used in one appendix robustness exercise. It predicts disease categories for projects whose RCDC tags do not match a category in the NIH–burden crosswalk. The main analysis continues to use the RCDC match and institute-mean imputation defined in Appendix I; the classifier does not replace a main-text estimate.

K.1. Data and Labels

The classifier predicts which of 74 disease categories are associated with a given NIH project. The category list is the set of Research, Condition, and Disease Categorization (RCDC) entries in the 2017 NIH–burden crosswalk. A project may receive several categories, and each category has an associated US disability-adjusted-life-year (DALY) value.

The corpus contains 114,230 unique projects. To obtain one record per project, I use the grant application corresponding to the year when the year is unique. Otherwise, I use the latest application with an abstract and then the latest application with a title. Every project has a title: 113,350 projects also have an abstract, while 880 are title-only. RCDC tags match at least one GBD category for 47,562 projects. I use these as the labelled projects. The remaining 66,668 projects have no matching category and require prediction. Existing category assignments are never replaced by model output.

K.2. Model and Training

I fine-tune BiomedBERT to read each project's title and abstract and predict the 74 categories. A classification layer with one output per category sits on the pretrained encoder, and training updates both the layer and the encoder. A project can receive more than one category.[18] Titles and abstracts are supplied together, with the input capped at 512 model tokens and truncation applied only to the abstract. During training, errors on rare categories receive greater weight; these weights are capped at 20. I use binary cross-entropy, four epochs, a learning rate of 0.00002, weight decay of 0.01, and an effective batch size of 32.

I divide the labelled projects into 33,250 training projects, 7,142 validation projects, and 7,170 test projects using a deterministic split designed to preserve the category mix in each sample. Projects with identical normalised title–abstract text remain in the same sample, so duplicate text cannot appear in both training and test data. The realised shares are 69.9, 15.0, and 15.1 per cent, respectively.

[18] The model identifier is `microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`. The immutable base-model revision is `e1354b7a3a09615f6aba48dfad4b7a613eef7062`.

I use the validation sample to choose the training epoch and a score cutoff for each category. For each category with enough observations, I choose the most inclusive cutoff that achieves at least 90 per cent precision and produces at least five assignments. In this setting, 90 per cent precision means that at least 90 per cent of projects assigned the category in validation have that observed category. Categories with fewer than ten validation positives use one common cutoff chosen across categories. If no cutoff meets the precision requirement, the category cannot be assigned under the cutoff rule. The test sample is not used to choose the model or the cutoffs. I fix and record the model, cutoffs, category order, and test-project identifiers before evaluation.

For an unlabelled project, I retain every category whose score reaches its validation cutoff, so a project may receive several categories. If no category crosses the threshold, the highest-scoring category is assigned to the project and the project is flagged. For the first robustness estimate, I assign that highest-scoring category and mark the case as a forced assignment. I report this estimate first. I then review the projects for which no category reached its cutoff.

K.3. Test Performance

Table A.8 compares the two assignment rules on the 7,170 held-out test projects. The threshold-only assignment rule attains micro precision of 0.893, micro recall of 0.795, and micro F1 of 0.841, assigning at least one category to 96.9 per cent of projects. Assigning the highest-scoring category when no cutoff is reached adds 224 assignments. It raises micro F1 to 0.847 and coverage to 100 per cent, while reducing micro precision to 0.890. The exact observed category set is recovered for 65.9 per cent of projects, and the Spearman correlation between observed and predicted project DALYs is 0.868.

Table A.9 shows how the threshold-plus-forced-assignment rule performs across administering institutes and centres. The administering organisation is recovered from the NIH project number and is not supplied to the model. Among institutes with substantial test support, performance is lowest for NIDA and NIAAA, whose F1 values are 0.605 and 0.644; recall is the main limitation. NHGRI has an F1 of 0.909, but its test sample contains only eight labelled projects, so the estimate says little about general-purpose genomic projects.

Table A.10 reports the same threshold-plus-forced-assignment statistics by the fiscal year of the selected source application. Performance is lowest in 2008, with an F1 of 0.720, and ranges from 0.828 to 0.889 in 2009–2023.

K.4. Performance for Rare Categories

Performance is worse for rare categories. Table A.11 groups categories by the number of positive projects in the training sample. The 17 rare categories have an F1 of

Table A.8. Test performance of the disease-burden classifier

Metric	Threshold-only assignment	Threshold plus forced assignment
Projects	7,170	7,170
Micro precision	0.893	0.890
Micro recall	0.795	0.808
Micro F1	0.841	0.847
Macro F1	0.638	0.648
Exact-set match	0.645	0.659
Mean predicted labels	1.521	1.552
DALY coverage	0.969	1.000
DALY Spearman correlation	0.875	0.868
Forced assignments	0	224

Notes: The unit of observation is a project in the held-out test partition. Micro statistics pool all project–category decisions. Macro F1 gives each of the 74 categories equal weight. Exact-set match equals one only when the complete predicted category set equals the observed RCDC-matched set. DALY coverage is the share of projects receiving a finite burden after taking the equal-weight mean across their assigned categories. The DALY rank correlation compares this value with the corresponding value constructed from observed categories. "Threshold-only assignment" applies the validation cutoffs. "Threshold plus forced assignment" additionally assigns the highest-scoring category when no cutoff is reached.

0.527, with precision of 0.906 but recall of 0.372, whereas the 17 common categories have an F1 of 0.864. Table A.12 lists the supported categories with test F1 below 0.70. Prescription Drug Abuse, Ovarian Cancer, and Preterm, Low Birth Weight and Health of the Newborn receive no final predictions because no cutoff for them meets the 90 per cent precision rule. Substance Abuse has an F1 of 0.240. The weak performance of overlapping substance-use and broad mental-health categories also explains part of the low NIDA, NIAAA, and NIMH recall.

K.5. Limits for Unmatched Projects

The test data contain only projects with an observed crosswalk-matched category. The unmatched data also contain projects whose RCDC tags are missing or do not match the burden vocabulary. Many of these projects develop general methods, infrastructure, or ethical frameworks rather than target one disease. Test performance therefore does not tell us how well the model works for these unmatched projects; deciding whether any of the 74 categories applies is a different task.

From the test data, only 224 projects, or 3.1 per cent, require a highest-score assignment, compared with 12,535 of the 66,668 projects from the unmatched data, or 18.8 per cent. This reflects that many projects with no disease match are more likely to have no obvious target disease.

K.6. Marginal Returns with Predicted Categories

To test the sensitivity of marginal-return estimates to missing disease labels, I replace the institute-mean imputation with the model's category assignments. Among the 85,290 unique projects in the estimation sample, I preserve the observed RCDC–DALY linkage for 39,145 projects and use the model predictions for the remaining 46,145, including 8,830 forced assignments. For each project, I average the DALY values of its assigned disease and re-normalise the resulting burden weights to mean one across unique projects. I then re-estimate the disease-burden production function on the same 55,710 grant–PI observations, holding the application model, funding allocation, knowledge stocks, depreciation rates, and comparative-statics procedure fixed.

Table 11 compares total marginal returns at the 2018 allocation under the institute-mean imputation and the model predictions. The two sets of estimates have a correlation of 0.986. The mean absolute percentage change is 9.3 per cent and the median is 7.4 per cent. Returns rise for 14 of the 19 institutes; Human Genome remains first and Nursing Research remains last. Eye, Human Genome, and Deafness and Communication Disorders show the largest proportional increases, at 29.6, 25.1, and 21.0 per cent, respectively, while Alcoholism shows the largest decrease, at 6.2 per cent. Figure A.7 shows that most changes are small relative to cross-institute dispersion, although the Human Genome and Eye estimates increase materially. Crucially, the overall ranking of institutes remains largely unchanged.

Table A.9. Prediction performance by institute

Administering IC	Projects	Precision	Recall	F1	Exact-set match	Forced share
FIC	7	0.900	0.818	0.857	0.714	0.000
NCCIH	29	0.800	0.720	0.758	0.448	0.000
NCI	1,355	0.933	0.899	0.915	0.731	0.002
NCRR	2	1.000	0.600	0.750	0.500	0.000
NEI	94	0.894	0.853	0.873	0.809	0.064
NHGRI	8	0.909	0.909	0.909	0.875	0.125
NHLBI	1,022	0.903	0.872	0.887	0.721	0.001
NIA	435	0.915	0.892	0.904	0.830	0.005
NIAAA	175	0.888	0.505	0.644	0.331	0.326
NIAID	627	0.861	0.850	0.855	0.715	0.014
NIAMS	130	0.884	0.823	0.852	0.762	0.015
NIBIB	115	0.832	0.841	0.837	0.661	0.026
NICHD	313	0.842	0.691	0.759	0.565	0.058
NIDA	387	0.817	0.480	0.605	0.119	0.070
NIDCD	43	0.788	0.743	0.765	0.605	0.023
NIDCR	143	0.888	0.855	0.871	0.727	0.007
NIDDK	828	0.878	0.856	0.867	0.634	0.002
NIEHS	131	0.845	0.812	0.828	0.626	0.008
NIGMS	251	0.811	0.829	0.820	0.673	0.028
NIMH	460	0.876	0.660	0.753	0.539	0.143
NIMHD	45	0.831	0.728	0.776	0.533	0.111
NINDS	496	0.915	0.876	0.895	0.804	0.022
NINR	53	0.909	0.843	0.875	0.755	0.019
NLM	17	0.917	0.846	0.880	0.647	0.000
OD	4	0.833	0.833	0.833	0.750	0.000

Notes: Each row applies the threshold-plus-forced-assignment rule to test projects administered by the indicated NIH institute or centre. The table reports subgroup size. Precision, recall, and F1 are micro statistics over all project–category decisions in the subgroup. Exact-set match is the share for which the full predicted category set equals the observed set. Forced share is the fraction receiving the highest-scoring category because no score reached its validation cutoff. Estimates based on small subgroups, including FIC, NCRR, NHGRI, NLM, and OD, are descriptive and imprecise.

Table A.10. Prediction performance by source fiscal year

Fiscal year	Projects	Precision	Recall	F1	Exact-set match	Forced share
2008	298	0.692	0.751	0.720	0.436	0.054
2009	529	0.878	0.784	0.828	0.645	0.062
2010	437	0.880	0.813	0.845	0.657	0.041
2011	356	0.905	0.837	0.869	0.722	0.022
2012	375	0.908	0.810	0.856	0.691	0.040
2013	329	0.890	0.822	0.855	0.693	0.018
2014	346	0.906	0.832	0.867	0.720	0.038
2015	422	0.886	0.824	0.854	0.661	0.021
2016	433	0.917	0.798	0.853	0.667	0.023
2017	526	0.896	0.769	0.828	0.622	0.034
2018	502	0.901	0.787	0.840	0.624	0.030
2019	487	0.886	0.779	0.829	0.626	0.014
2020	533	0.896	0.801	0.846	0.634	0.032
2021	506	0.903	0.837	0.868	0.698	0.024
2022	537	0.900	0.830	0.864	0.687	0.024
2023	554	0.929	0.853	0.889	0.726	0.025

Notes: Rows partition the held-out test projects by the fiscal year of the NIH application record supplying the title, abstract, and observed RCDC category set. The table reports subgroup size. Precision, recall, and F1 are micro statistics. Exact-set match requires equality of the complete category sets. Forced share records use of the highest-scoring category when no cutoff is reached.

Table A.11. Prediction performance by category frequency

Training-label frequency	Labels	Precision	Recall	F1
Rare: 0–99 training positives	17	0.906	0.372	0.527
Medium: 100–999 training positives	40	0.888	0.744	0.809
Common: 1,000 or more training positives	17	0.890	0.840	0.864

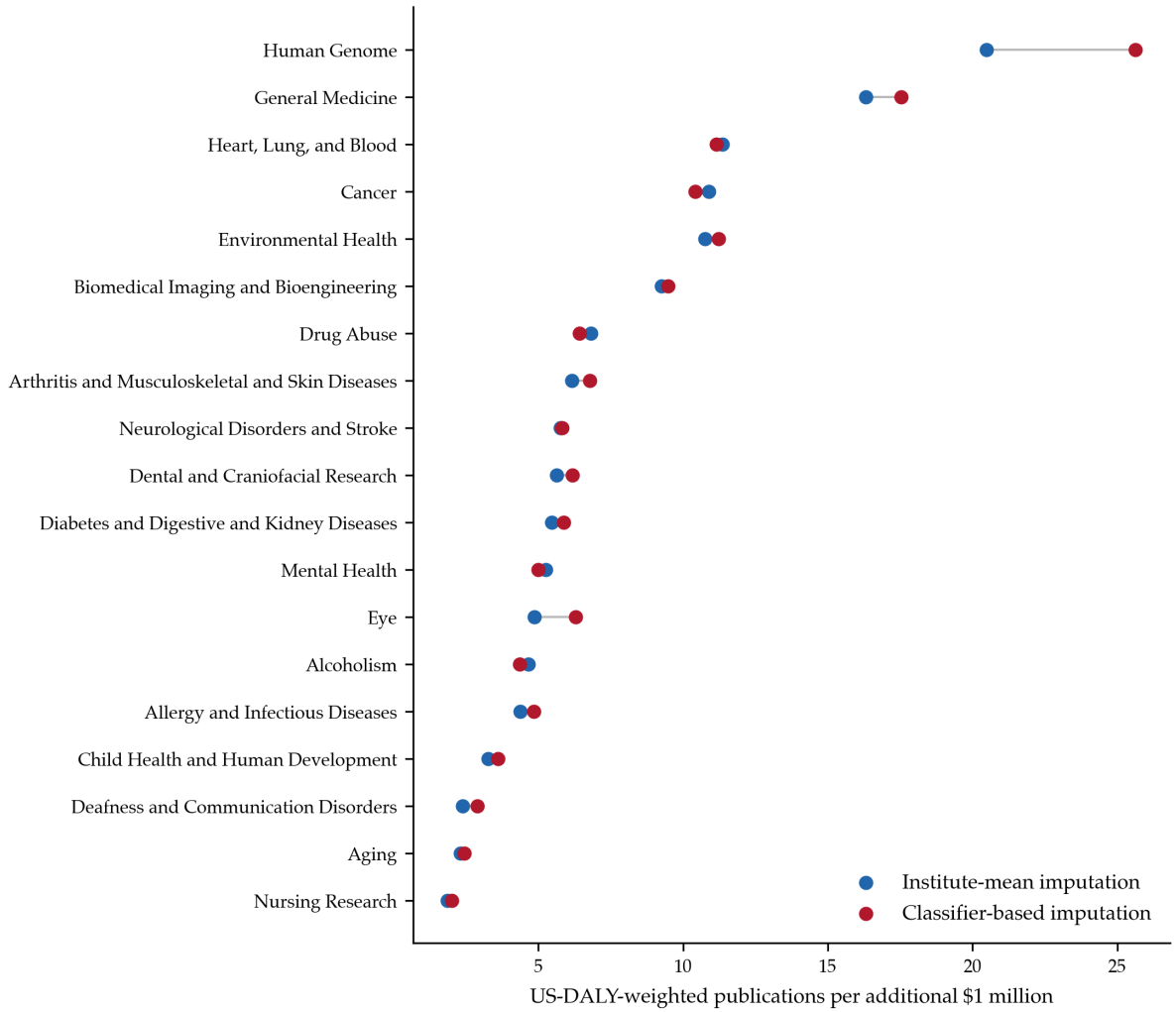
Notes: Categories are partitioned by their number of positive projects in the training sample. Label counts are the numbers of categories in each frequency group. Precision, recall, and F1 pool all test project–category decisions within the indicated group. Frequency groups and cut-offs are fixed independently of test outcomes.

Table A.12. Supported disease-burden categories with low prediction performance

Burden category	Test positives	Predicted positives	Precision	Recall	F1
Prescription Drug Abuse	24	0	0.000	0.000	0.000
Ovarian Cancer	54	0	0.000	0.000	0.000
Preterm, Low Birth Weight and Health of the Newborn	87	0	0.000	0.000	0.000
Substance Abuse	514	85	0.847	0.140	0.240
Mental Illness	226	77	0.662	0.226	0.337
Sexually Transmitted Infections	45	22	0.682	0.333	0.448
Anxiety Disorders	54	25	0.800	0.370	0.506
Chronic Liver Disease and Cirrhosis	166	87	0.828	0.434	0.569
Alcoholism, Alcohol Use and Health	126	71	0.887	0.500	0.640
Pneumonia	57	31	0.968	0.526	0.682
Injury (total) Accidents/Adverse Effects	180	132	0.811	0.594	0.686

Notes: The table reports every category with at least 20 observed positives in the held-out test partition and threshold-plus-forced-assignment F1 below 0.70. Predicted positives count projects assigned the category. Precision, recall, and F1 are category-specific.

Figure A.7. Disease-burden marginal returns: institute-mean imputation versus classifier-based imputation



Notes: Dots reproduce the Institute-mean imputation and Classifier-based imputation columns of Table 11. Institutes are ordered by the institute-mean imputation estimate. Horizontal segments show the change induced by replacing the institute-mean imputation with classifier-based imputation for projects without an observed RCDC–DALY match.

L. Additional Results

L.1. Detailed Return Decompositions

Tables A.13–A.15 report the full institute-level decomposition of the measure-specific marginal returns presented in Section 7.2. To keep the layout readable, each table is split into an own-field panel and a cross-field panel.

L.2. Pairwise Knowledge-Spillover Heatmaps

Appendix Figures A.8–A.10 report the full pairwise knowledge-spillover matrices for the three measures presented in Section 7.1.

L.3. Allocation Rules Excluding Human Genome

Human Genome ranks first under all three marginal-return criteria in Section 7.2, so it also receives the largest proportional share under every marginal-return rule in Section 7. Table A.16 repeats the local $1 million allocation-rule comparison of Table 12 after removing Human Genome from the set of eligible recipients. Shares under every rule, including the current-share benchmark, are re-normalised over the remaining 18 institutes; output gains use only these institutes' marginal returns.

The average disease-burden rule still raises output relative to current shares under all three evaluation measures, and each marginal-return rule still dominates it under its own evaluation measure. Excluding Human Genome narrows the gaps: relative to current shares, the average disease-burden rule's gain falls from 22.3 to 3.5 per cent for academic impact, from 21.9 to 5.7 per cent for commercial potential, and from 25.7 to 13.2 per cent for disease-burden-weighted publications. The corresponding marginal-return rules still add a further 9.7, 10.5, and 7.4 per cent, respectively, over the average disease-burden rule. Human Genome's outsized marginal returns amplify the gains in the 19-institute comparison.

Table A.13. Marginal returns: academic impact

Institute	Current own-field return (PE)	Current own-field return (GE)	Own-field knowledge spillovers	Own-field total
Human Genome	35.872	35.872	22.538	58.410
General Medicine	45.481	45.481	17.297	62.778
Environmental Health	25.218	25.218	7.529	32.748
Diabetes and Digestive and Kidney Diseases	26.765	26.765	10.851	37.616
Cancer	25.486	25.486	18.954	44.440
Dental and Craniofacial Research	28.588	28.588	12.007	40.595
Mental Health	24.032	24.032	11.394	35.426
Neurological Disorders and Stroke	25.320	25.320	9.126	34.446
Biomedical Imaging and Bioengineering	23.348	23.348	2.348	25.696
Arthritis and Musculoskeletal and Skin Diseases	22.606	22.606	9.349	31.955
Eye	24.416	24.416	8.760	33.175
Drug Abuse	16.348	16.348	11.350	27.698
Allergy and Infectious Diseases	21.766	21.766	9.704	31.470
Heart, Lung, and Blood	23.952	23.952	4.374	28.326
Child Health and Human Development	11.125	11.125	5.434	16.558
Alcoholism	12.449	12.449	9.030	21.480
Deafness and Communication Disorders	11.983	11.983	11.812	23.795
Aging	12.321	12.321	4.755	17.076
Nursing Research	3.916	3.916	0.494	4.410

Institute	Talent-flow externality (PE)	Talent-flow externality (GE)	Cross-field knowledge spillovers	Total Return
Human Genome	-23.400	0.004	137.756	196.170
General Medicine	-25.846	-0.003	53.181	115.957
Environmental Health	-17.403	-0.002	33.555	66.301
Diabetes and Digestive and Kidney Diseases	-13.645	-0.002	27.931	65.545
Cancer	-11.116	-0.001	18.865	63.304
Dental and Craniofacial Research	-18.123	-0.003	21.791	62.383
Mental Health	-12.797	-0.002	26.375	61.799
Neurological Disorders and Stroke	-13.616	-0.001	23.381	57.826
Biomedical Imaging and Bioengineering	-21.753	-0.001	30.826	56.521
Arthritis and Musculoskeletal and Skin Diseases	-16.696	-0.002	21.723	53.677
Eye	-16.541	-0.002	20.219	53.392
Drug Abuse	-12.488	-0.002	19.171	46.867
Allergy and Infectious Diseases	-11.023	-0.001	14.229	45.698
Heart, Lung, and Blood	-9.360	-0.001	17.287	45.612
Child Health and Human Development	-13.310	-0.001	21.030	37.587
Alcoholism	-12.624	-0.002	15.956	37.433
Deafness and Communication Disorders	-13.085	-0.001	9.309	33.102
Aging	-5.986	-0.000	8.608	25.684
Nursing Research	-9.954	-0.002	6.123	10.531

Notes: Each row reports the OpenAlex two-year mean-citedness-weighted publication response to a $1 million increment at the indicated institute, evaluated at the 2018 R01 allocation. The first panel decomposes own-field responses. The second reports cross-field channels and the total marginal return. "Own-field total" is the GE current own-field return plus own-field knowledge spillovers. Institutes are ordered by the total return.

Table A.14. Marginal returns: commercial potential

Institute	Current own-field return (PE)	Current own-field return (GE)	Own-field knowledge spillovers	Own-field total
Human Genome	2.022	2.022	1.347	3.369
General Medicine	3.634	3.634	1.313	4.947
Biomedical Imaging and Bioengineering	2.668	2.668	0.279	2.947
Dental and Craniofacial Research	2.642	2.642	1.330	3.971
Diabetes and Digestive and Kidney Diseases	2.003	2.002	1.175	3.178
Environmental Health	1.433	1.433	1.062	2.495
Cancer	1.891	1.891	1.714	3.605
Neurological Disorders and Stroke	2.159	2.159	0.802	2.960
Arthritis and Musculoskeletal and Skin Diseases	1.965	1.965	0.887	2.852
Eye	2.712	2.712	0.652	3.364
Heart, Lung, and Blood	2.182	2.181	1.099	3.280
Allergy and Infectious Diseases	1.673	1.673	0.986	2.659
Mental Health	1.168	1.168	0.675	1.843
Drug Abuse	0.944	0.944	0.865	1.809
Child Health and Human Development	0.826	0.826	0.427	1.253
Alcoholism	0.858	0.858	0.738	1.596
Deafness and Communication Disorders	0.960	0.960	0.579	1.539
Aging	0.818	0.818	0.471	1.289
Nursing Research	0.206	0.206	0.046	0.253

Institute	Talent-flow externality (PE)	Talent-flow externality (GE)	Cross-field knowledge spillovers	Total Return
Human Genome	-1.753	0.008	12.403	15.780
General Medicine	-2.016	-0.006	5.442	10.383
Biomedical Imaging and Bioengineering	-1.702	-0.003	3.349	6.293
Dental and Craniofacial Research	-1.319	-0.006	2.220	6.185
Diabetes and Digestive and Kidney Diseases	-1.083	-0.005	2.878	6.052
Environmental Health	-1.345	-0.005	3.216	5.706
Cancer	-0.888	-0.003	1.989	5.590
Neurological Disorders and Stroke	-1.017	-0.003	2.294	5.252
Arthritis and Musculoskeletal and Skin Diseases	-1.327	-0.005	2.335	5.182
Eye	-1.279	-0.005	1.728	5.088
Heart, Lung, and Blood	-0.717	-0.003	1.643	4.920
Allergy and Infectious Diseases	-0.873	-0.003	1.382	4.038
Mental Health	-0.997	-0.004	1.999	3.837
Drug Abuse	-0.907	-0.005	1.450	3.254
Child Health and Human Development	-1.000	-0.003	1.812	3.062
Alcoholism	-0.915	-0.006	1.327	2.917
Deafness and Communication Disorders	-1.034	-0.003	0.753	2.289
Aging	-0.463	-0.001	0.794	2.083
Nursing Research	-0.748	-0.004	0.497	0.746

Notes: Each row reports the commercial-potential-weighted publication response to a $1 million increment at the indicated institute, evaluated at the 2018 R01 allocation. The first panel decomposes own-field responses. The second reports cross-field channels and the total marginal return. "Own-field total" is the GE current own-field return plus own-field knowledge spillovers. Institutes are ordered by the total return.

Table A.15. Marginal returns: disease burden

Institute	Current own-field return (PE)	Current own-field return (GE)	Own-field knowledge spillovers	Own-field total
Human Genome	4.915	4.914	1.616	6.530
General Medicine	8.895	8.897	1.849	10.746
Heart, Lung, and Blood	8.407	8.409	1.104	9.513
Cancer	5.801	5.803	3.179	8.982
Environmental Health	4.877	4.877	2.285	7.163
Biomedical Imaging and Bioengineering	5.460	5.460	0.321	5.781
Drug Abuse	3.093	3.093	2.016	5.109
Arthritis and Musculoskeletal and Skin Diseases	2.885	2.886	0.973	3.858
Neurological Disorders and Stroke	3.189	3.189	0.578	3.767
Dental and Craniofacial Research	2.585	2.585	0.768	3.353
Diabetes and Digestive and Kidney Diseases	1.933	1.934	0.447	2.381
Mental Health	2.212	2.212	0.716	2.928
Eye	2.225	2.225	0.985	3.211
Alcoholism	1.787	1.787	1.023	2.810
Allergy and Infectious Diseases	1.945	1.946	0.931	2.876
Child Health and Human Development	0.906	0.907	0.325	1.232
Deafness and Communication Disorders	1.137	1.137	0.485	1.622
Aging	1.265	1.265	0.116	1.380
Nursing Research	1.030	1.030	0.085	1.115

Institute	Talent-flow externality (PE)	Talent-flow externality (GE)	Cross-field knowledge spillovers	Total Return
Human Genome	-3.542	-0.025	13.979	20.484
General Medicine	-4.315	0.023	5.551	16.319
Heart, Lung, and Blood	-1.296	0.010	1.832	11.354
Cancer	-1.845	0.010	1.898	10.890
Environmental Health	-3.050	0.020	3.569	10.752
Biomedical Imaging and Bioengineering	-3.996	0.008	3.460	9.249
Drug Abuse	-1.715	0.019	1.693	6.820
Arthritis and Musculoskeletal and Skin Diseases	-2.646	0.021	2.283	6.162
Neurological Disorders and Stroke	-2.224	0.011	1.993	5.771
Dental and Craniofacial Research	-2.707	0.024	2.262	5.639
Diabetes and Digestive and Kidney Diseases	-2.696	0.017	3.076	5.474
Mental Health	-1.878	0.018	2.323	5.268
Eye	-2.572	0.019	1.627	4.857
Alcoholism	-1.925	0.022	1.825	4.657
Allergy and Infectious Diseases	-2.100	0.010	1.500	4.386
Child Health and Human Development	-2.141	0.012	2.022	3.266
Deafness and Communication Disorders	-2.009	0.012	0.752	2.387
Aging	-1.017	0.004	0.934	2.318
Nursing Research	-1.631	0.014	0.733	1.862

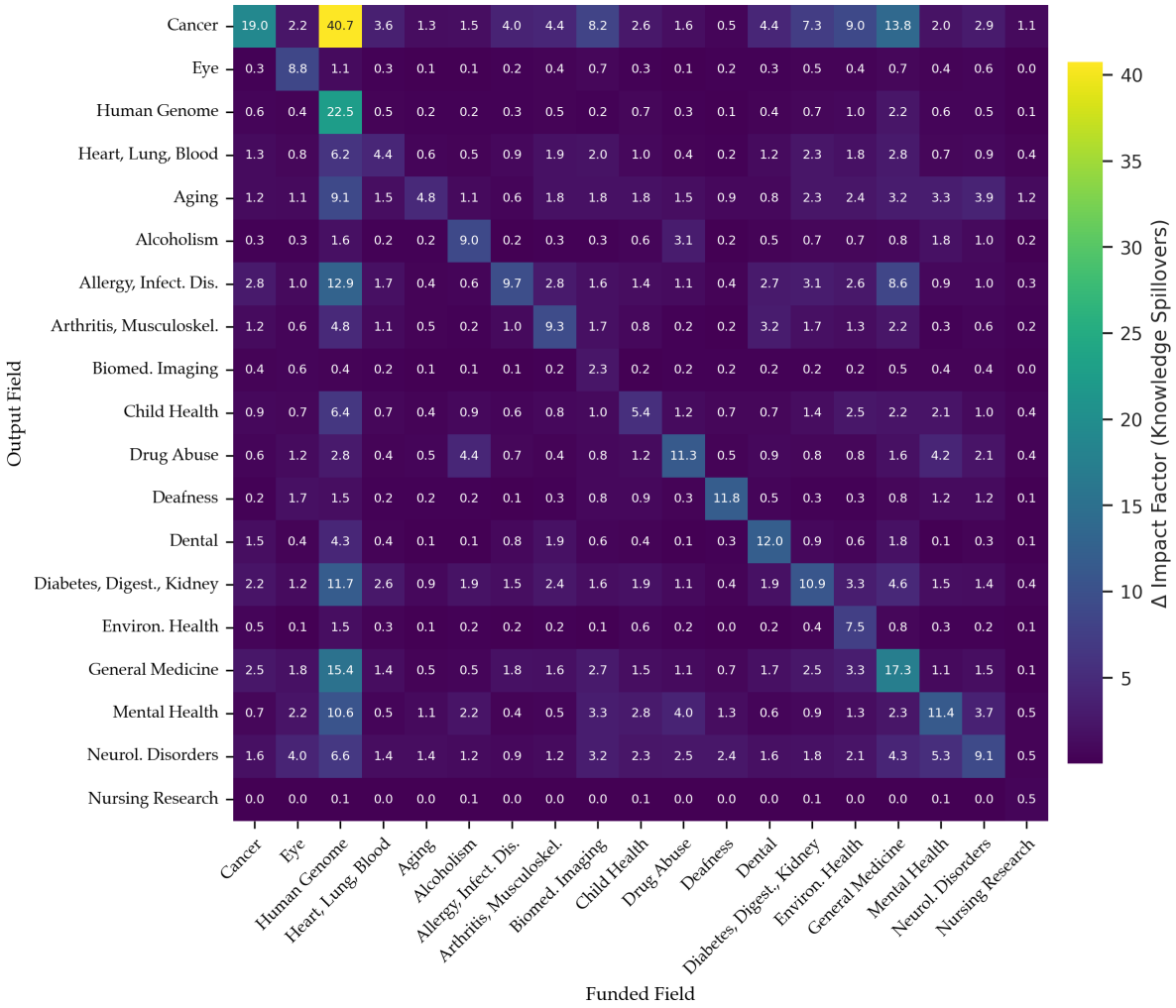
Notes: Each row reports the disease-burden-weighted publication response to a $1 million increment at the indicated institute, evaluated at the 2018 R01 allocation. The first panel decomposes own-field responses. The second reports cross-field channels and the total marginal return. "Own-field total" is the GE current own-field return plus own-field knowledge spillovers. Institutes are ordered by the total return.

Table A.16. Output gains by allocation rule, excluding Human Genome

Allocation rule	Academic impact	Commercial potential	Disease burden
Current allocation	53.60 (1.000)	4.75 (1.000)	7.09 (1.000)
Average disease burden	55.47 (1.035)	5.03 (1.057)	8.03 (1.132)
Academic-impact MR	60.87 (1.136)	5.42 (1.139)	7.74 (1.092)
Commercial-potential MR	61.39 (1.145)	5.55 (1.168)	7.92 (1.117)
Disease-burden MR	61.91 (1.155)	5.59 (1.175)	8.62 (1.216)

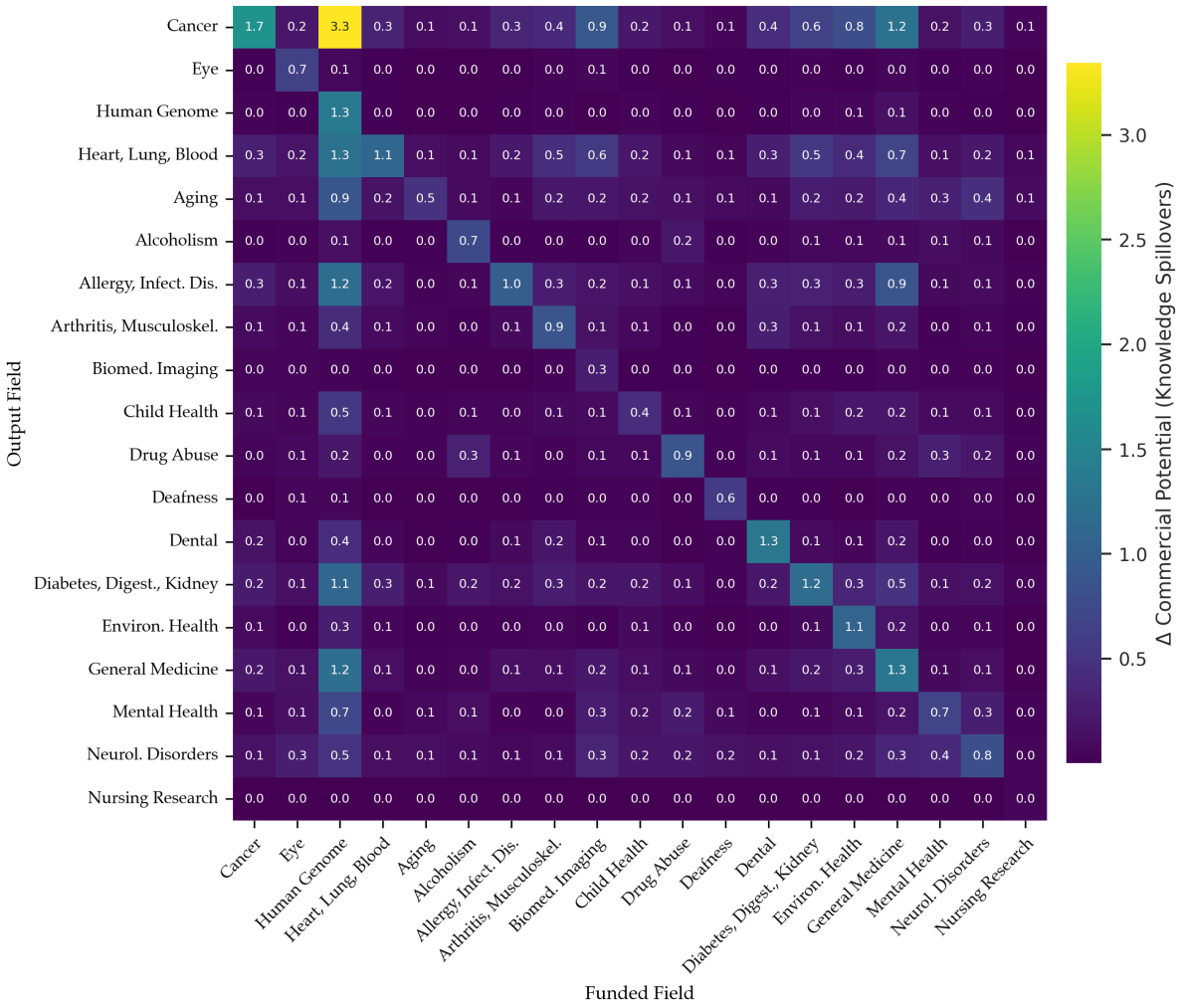
Notes: Repeats the exercise in Table 12 after removing Human Genome, the top-ranked institute under all three marginal-return criteria, from the set of eligible recipients of the local $1 million increment. Allocation shares under every rule are re-normalised over the remaining 18 institutes, and output gains $\Delta Y^{r,m}$ use only these institutes' marginal returns. Parentheses report the ratio to the current-share benchmark, recomputed on the same 18-institute set.

Figure A.8. Pairwise knowledge spillovers: academic impact



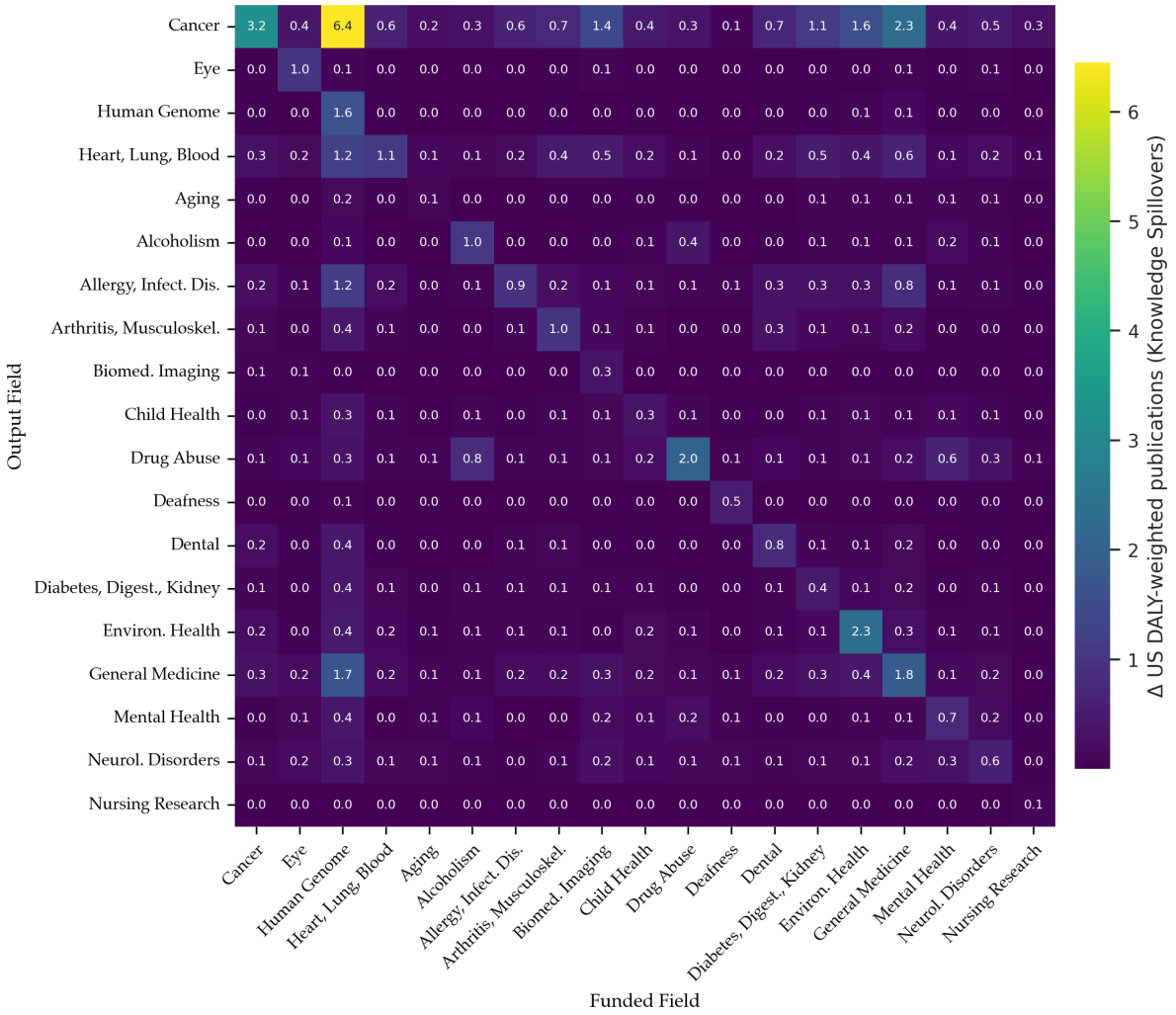
Notes: Rows are output fields o and columns are funded fields d. Each cell reports the change in impact-factor-weighted publication output in o caused by the knowledge-spillover channel of a $1 million budget increment in d, evaluated under the local log-output approximation at the 2018 R01 allocation. Cell labels and the colour bar are in impact-factor-weighted publication units. Lighter cells indicate larger positive spillovers.

Figure A.9. Pairwise knowledge spillovers: commercial potential



Notes: Rows are output fields o and columns are funded fields d. Each cell reports the change in commercial-potential-weighted publication output in o caused by the knowledge-spillover channel of a \$1 million budget increment in d, evaluated under the local log-output approximation at the 2018 R01 allocation. Cell labels and the colour bar are in commercial-potential-weighted publication units. Lighter cells indicate larger positive spillovers.

Figure A.10. Pairwise knowledge spillovers: disease burden



Notes: Rows are output fields o and columns are funded fields d. Each cell reports the change in disease-burden-weighted publication output in o caused by the knowledge-spillover channel of a $1 million budget increment in d, evaluated under the local log-output approximation at the 2018 R01 allocation. Cell labels and the colour bar are in disease-burden-weighted publication units. Lighter cells indicate larger positive spillovers.